

LEARNING ABOUT WOMEN’S COMPETENCE: THE DYNAMIC RESPONSE OF POLITICAL PARTIES TO GENDER QUOTAS IN SOUTH KOREA*

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June 30, 2026

We study the dynamic response of political parties to gender quotas in South Korean municipal councils, a setting with nearly zero women pre-quota. We exploit two unique institutional features: quota intensity is discontinuous in council size; the quota regulates only one of two election arms. Political parties initially counteract the quota by lowering women’s share among candidates in the unregulated arm, but this response gradually reverses over time. We provide evidence consistent with parties learning about women’s competence through exposure to female politicians. Quotas can thus generate effects beyond direct compliance, acting as information interventions that reshape future nomination strategies.

Keywords: Gender quota, political parties, discrimination, biased beliefs, learning

JEL codes: D72, D83, J16, J71, J78

*This research is supported by the STICERD Research Grant and a grant of the French National Research Agency (ANR), “Investissements d’Avenir” (LabEx Ecodec/ANR-11-LABX-0047). We are grateful to Oriana Bandiera, Manuel Bagues, Pamela Campa, Katherine Casey, Rachel Cassidy, Luca Citino, Nicola Fontana, Martina Kirchberger, Alan Manning, Nicola Mastroiocco, Guy Michaels, Gaia Narciso, Torsten Persson, Steve Pischke, Anna Raute, Viola Salvestrini, Vincenzo Scrutinio, Horng Chern Wong, and numerous seminar and conference participants, for their extremely helpful comments. The use of the data was approved by the LSE Ethics Committee in December 2018.

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I INTRODUCTION

Gender quotas in politics are currently used by 130 countries. Proponents view quotas as a tool to dismantle negative biases against female politicians. However, it is precisely where these biases are the greatest, and therefore quotas are most needed, that quotas also run the highest risk of being ineffective. The critical risk is strategic counteraction by political parties. Voters, too, may resist quotas (Clayton, 2015), but political parties are the first gatekeepers who control the set of candidates available for voter selection (Norris and Lovenduski, 1995; Dahlerup, 1998; Esteve-Volart and Bagues, 2012; Fujiwara et al., 2025).

Strategic counteraction against quotas by parties has been documented, as part of a large literature evaluating the effect of gender quotas on female representation and policy outcomes. Parties often nominally comply with quotas while undermining them in practice—for instance, by placing women far down the party list or in less-winnable constituencies (Lippmann, 2021; Bagues and Campa, 2021; Casas-Arce and Saiz, 2015; Esteve-Volart and Bagues, 2012; Murray, 2008).¹ Lippmann (2021) also shows that counteraction can decrease over time. However, little is known about *why* parties’ resistance to quotas might weaken.

In this paper, we study how highly male-dominated political parties react to quotas in their nomination of candidates for municipal councils in South Korea, where the female share of councilors was as low as 2%. Our key contribution is to shed light on how exposure to women in office changes parties’ later nomination strategies by shifting their beliefs about women’s competence as politicians. We take advantage of a unique institutional feature: the gender quota regulates only one of two separate arms through which councilors get elected. In South Korea’s mixed electoral system, 80-90% of councilors are elected by plurality vote in the municipality’s constituent wards (“ward arm”) while the rest are elected by closed-list proportional representation (“PR arm”).² The gender quota regulates only the PR arm, stipulating that all odd-numbered candidates in the party list be women. The unregulated ward arm, therefore, provides a rare window into a rich set of endogenous party responses beyond mechanical quota compliance.

We assemble two novel datasets. The first covers the universe of candidates on the ballot and elected councilors over seven election cycles, including detailed candidate characteristics that are often unavailable in empirical studies of quotas. The second dataset records the number of bills each elected councilor proposes, which we use as a proxy for performance in office. Together, the data trace the chain from nomination to election to observed performance in office.

¹Krook (2016) reviews various tactics of counteraction employed by parties.

²Parties predominantly determine the set of candidates running for election. The case is obvious for the PR arm, where parties compete for seats. Even in the ward arm, 70% of candidates run with party affiliation.

Our identification strategy is a regression discontinuity design exploiting variation in quota intensity across municipalities. The number of PR seats increases as a step function of a municipality’s council size, creating discontinuities in quota intensity at certain council-size cutoffs. Two rules interact: the quota requires all odd-numbered candidates on PR party lists to be female, while the seat-allocation rule makes the additional PR seat more likely to go to a candidate in an odd-numbered position than to one in an even-numbered position.³ Therefore, comparing municipalities just above (treatment) and just below (control) each cutoff isolates exogenous variation in quota intensity, which translates into greater exposure to female councilors. Treatment municipalities elect an estimated 0.72 more female PR councilors than control municipalities.

Consistent with evidence from other contexts, we find that parties in treatment municipalities initially counter the quota by shifting the unregulated ward arm toward men: they nominate more male ward candidates, thereby reducing the female share, especially in ballot positions with higher chances of election. Hence, although the quota successfully increases the number of women elected through the regulated PR arm, the overall increase in female councilors is dampened by counteraction in the ward arm.

This pattern reverses over time. Over the following three election cycles, parties in treated municipalities gradually increase the number of female candidates in the ward arm. Remarkably, by the final election cycle in our data, treated municipalities nominate 1.17 *more* female ward candidates than control municipalities. The same policy that initially induces strategic counteraction eventually leads parties to flip the direction of their response.

What drives this reversal? One mechanism is incumbency. The women elected through the quota gain an incumbency advantage and, given the no-reelection norm in the PR arm (Shin, 2014), are often renominated in the ward arm in later elections.⁴ Critically, however, the quota’s effect extends beyond incumbent women. The gradual increase in female ward candidates is also observed among *rookie* women with no prior council experience.⁵

We hypothesize that these positive spillovers to rookies reflect a change in how parties appraise women as a group, not only how they evaluate specific female incumbents. We formalize this interpretation using a dynamic model of electoral competition with statistical discrimination, in which candidate ability matters for electoral success (Galasso and Nannicini, 2011; Le Barbanchon and Sauvagnat, 2022; Aigner and Cain, 1977).⁶ When parties have limited information about

³For instance, in municipalities with two PR seats, the second PR seat is far more likely to go to the number-1 candidate of the second-ranked party than to the number-2 candidate of the first-ranked party.

⁴As PR councilors get elected through the party rather than individual merit, it is seen as a one-time “bonus” affair.

⁵We find similar results in a complementary party-level analysis of close races in the PR arm: parties that marginally won a female PR councilor in cycle 4 nominated statistically significantly more rookie female candidates in cycles 6 and 7 than parties that marginally lost.

⁶We verify the model’s key assumptions in the data. Candidates with higher education and prior political experi-

women as politicians, they may hold downward-biased beliefs about women’s average competence and nominate few women. Each new woman provides a signal that helps parties update their beliefs and accelerates learning in municipalities more strongly affected by the quota.

Several pieces of evidence support this learning interpretation. First, the gradual increase in female candidates is concentrated in competitive wards, where candidate quality matters most, including among rookies. This pattern is consistent with an increase in parties’ perceived competence of women. By contrast, if exposure to women only weakened parties’ distaste for female candidates, parties would become more willing to nominate women without any change in their beliefs about their competence. The newly added women are therefore expected to be less competent than existing female candidates and thus would be placed in less competitive wards.

Second, consistent with statistical discrimination, changes in candidate selection are related to information acquisition. Treated municipalities elect more rookie female councilors starting immediately after the quota reform and continuing thereafter, consistent with these rookies providing new signals about women’s competence that trigger a process of party learning. This learning process appears concentrated where information was scarce to begin with: both the initial counteraction and the subsequent reversal are driven by municipalities with no female councilors before the quota. Furthermore, the content of this information matters. The shift toward female candidates in cycles 5-7 is larger where the legislative performance of the first female PR councilors elected in cycle 4 is stronger than expected, as measured by the number of bills they propose. This finding speaks directly to beliefs about competence, as the model predicts faster updating when observed women generate more positive surprises about their ability.

Last, we explore alternative mechanisms that could generate the same pattern of initial counteraction followed by gradual reversal. These include differential changes, in treatment municipalities relative to control, in voter preferences, the supply of potential female candidates, and elected women’s power within the party. We do not find evidence that these mechanisms drive the treatment effect at the quota-intensity discontinuities.⁷

Taken together, the evidence provides coherent support for the view that the quota changed parties’ long-term nomination strategies through two mechanisms: incumbency advantages among women and learning from exposure to female councilors. We close the empirical analysis with an accounting decomposition. The mechanical growth in the stock of previously elected women

ence receive higher vote shares, consistent with ability mattering for electoral success. Moreover, candidate education exhibits a clear inverted V-shaped relationship with party popularity in the ward, peaking in the most competitive wards for both main parties, consistent with ability mattering especially in competitive races.

⁷This does not imply that these factors are unaffected by the increase in women in politics. These margins may well respond to the nationwide quota system, which brought in women also in control municipalities. Our point is that they do not change discontinuously at the quota-intensity cutoffs.

explains a substantial, but incomplete, share of the total increase in female ward candidates: 27% in cycle 5, 56% in cycle 6, and 50% in cycle 7. The increase in the female share among rookies—a margin closely tied to learning about women’s average competence—accounts for 17% of the total effect in cycle 5, 24% in cycle 6, and 34% in cycle 7. Hence, the learning-consistent margin is economically meaningful and becomes increasingly important over time.

From a policy perspective, our findings imply that quotas can be effective in the long run when designed appropriately, even when resisted by the very gatekeepers whose behavior they are intended to change. In South Korea, where more than 60 percent of respondents agreed that men make better political leaders than women (Appendix Figure A.1), parties likely operated in an environment of strongly negative prior beliefs about women in politics. Yet, the quota design i) incentivized parties to nominate competent women, and ii) ensured those women got elected, thereby paving the path for learning to take place.⁸ The broader lesson is not specific to South Korea or to politics, but extends to domains with similarly entrenched incumbents and limited prior information about underrepresented groups.

We contribute to the literature on gender quotas and political selection in two ways. First, we document a novel dynamic response: strategic counteraction to quotas can reverse, not merely attenuate over time as in Lippmann (2021). In our setting, parties initially reduce female nominations in the unregulated ward arm, but later nominate more women in that same arm. This reversal suggests that quotas can operate not only as a compliance device but also as an information intervention that corrects existing inefficiencies. Once the process is set in motion, the policy itself may no longer be necessary. Second, we provide evidence on a less-studied source of female disadvantage in party selection. Existing work points to other sources, such as women threatening leaders’ survival (Besley et al., 2017), lack of incumbency advantage (Lippmann, 2021), or expectations of voter bias (Le Barbanchon and Sauvagnat, 2022). There is little evidence on whether parties’ preferences or beliefs may play a role in explaining this strategic counteraction (Dahlerup, 2021). Leveraging our unique setting and rich data, we make progress by assembling evidence that collectively points toward statistical discrimination rooted in biased beliefs and a lack of information about women’s competence.

We also contribute to the literature on exposure and belief updating. The closest conceptual predecessor is Beaman et al. (2009), who show that exposure to women politicians improves voters’ perceptions of women as effective leaders. We study a different set of learners: political

⁸Downward-biased beliefs might have persisted if the quota had instead brought in unqualified women, such as female family members of incumbent males—a widespread tactic observed in Argentina, India, and Mexico (Krook, 2016).

parties.⁹ This distinction is important because parties are the gatekeepers that decide who runs and how viable their races are. Recent work suggests that biased nomination strategies remain an important source of women’s underrepresentation, even when voter bias is limited (Däubler et al., 2025; Fujiwara et al., 2025; Schwarz and Coppock, 2022; Teele et al., 2018; Hayes et al., 2014).

We also speak to the literature on the dynamics of statistical discrimination. Theoretical frameworks have focused on how discrimination evolves in response to repeated signals about an individual’s ability (Fryer, 2007; Bohren et al., 2019). Our model shifts this logic to the group level: the revealed abilities of elected women provide repeated signals that parties aggregate to update their beliefs about women’s average ability. Empirically, we find evidence consistent with statistical discrimination based on biased beliefs. When information frictions exist, quotas need not increase inclusion at the expense of competence, but can instead help correct a persistent misallocation of talent. Our study, therefore, shows how quotas can help societies escape “learning trap” discrimination dynamics (Komiya and Noda, 2024; Lepage, 2024), where limited representation prevents learning about the quality of members of a disadvantaged group.

Lastly, our paper contributes to the literature on women in South Korean municipal councils. We provide causal evidence that quantitatively supports arguments in qualitative studies (Shin, 2014; Yoon and Shin, 2017): i) parties, not voters, dictate electoral success; ii) the quota was resisted by parties; iii) the PR women who demonstrated their caliber during their term allowed parties and voters to revise their perceptions on women’s capabilities. Joo and Lee (2018) do causally estimate the effect of electing women on female candidate nomination, using an instrumental variables strategy based on an arbitrary name-order advantage on the ballot. Their finding of null spillover effects on other females in the following election cycles contrasts with ours. This highlights two points we make: the importance of electing *competent* women to propel the process of learning,¹⁰ and the importance of examining effects over the long term.

The remainder of the paper is organized as follows. Section II describes the institutional setting and data. Section III lays out our empirical strategy, and Section IV discusses the results. In Section V, we present a model and discuss the pieces of evidence that point towards learning as an explanation for the results. Finally, Section VI concludes.

⁹Party beliefs are peripherally investigated by Bhavnani (2009). He presents “weak evidence” (p.32) that reserving seats for women helped parties learn that women can win elections: parties with higher success rates of female candidates nominate a higher share of female candidates in the next election. However, the correlational evidence is limited by the small sample size (four observations; one per party) and could be confounded by a higher number of women benefiting from incumbency advantage.

¹⁰The compliers in their IV method are females who only got elected thanks to the name-order advantage, and therefore may not be of high competence.

II INSTITUTIONAL SETTING AND DATA

II.A The role of municipal councils, electoral rules, and gender quotas

There are 226 municipal councils in South Korea. Municipal councils represent the legislative branch that works with municipal governments—the executive branch—to oversee local matters. Councils have several legally defined responsibilities, such as reviewing and approving the spending of municipal governments, adopting and revising local bills, monitoring municipal governments, and addressing petitions submitted by residents. Municipal governments administer around a third of South Korea’s total public expenditure.

Municipal councils were established in the mid-1990s, and since then, elections have taken place every four years. Our sample covers seven elections, with 2018 as the last election year. Up to the third election in 2002, all councilors were directly elected through plurality vote in single-member constituent villages. All candidates ran as independents with no party affiliation.

However, major reforms were made to the electoral rules from the fourth election in 2006. We describe below the two main reforms of interest for our analysis. While there were other reforms introduced simultaneously, discussed in Appendix C.1, none are related to the council-size thresholds we exploit for identification.

First, a double-arm voting system was introduced. Candidates could be elected through two alternative arms¹¹: a closed-list proportional representation arm (“PR arm”), and a multi-member plurality vote arm in constituent wards (“ward arm”). Appendix Figure A.2 illustrates what the ballot papers look like for the two arms and explains in detail how each arm operates. Ward candidates could now run with party affiliation.¹² Each ward elected between 2 and 4 councilors, so multiple candidates from one party could run in the same ward.¹³

Second, a gender quota was put in place: all odd-numbered candidates in the PR party list needed to be female. As the number of PR seats is small, e.g. 1 out of 10 total council seats, around 90% of elected PR councilors turned out to be the first candidates on the lists and therefore female. Consequently, the introduction of quotas sharply increased the share of female councilors.

Municipal councils were severely male-dominated prior to the reform, with only 2% of councilors being female. Since the introduction of quotas in 2006, however, the female share of coun-

¹¹The same person could run only on one arm.

¹²A party holds the authority to solicit, vet, and determine candidate placement across wards, with electability and ideological alignment being key criteria. Formally, candidate nominations for all regional elections—spanning from provincial governors to municipal councilors—are managed by the province-level Nomination Management Committee. However, for municipal councilor nominations, the recommendations of the head of the municipal party chapter have been given particular priority. (Park et al., 2014).

¹³A party could nominate candidates for a ward up to the preset number of seats allotted for that ward.

cilors sharply increased to 15% and continued to grow to 30% by 2018 (Table I). The female share rose beyond the stipulations of the quota: it rose among ward candidates, whose gender is not regulated, and even among “rank 1” ward candidates, who have a higher likelihood of election.

Table I - Descriptive Statistics on Councilors and Candidates

Election cycle:	Quota on PR arm (passed in 2005)						
	1 (1995)	2 (1998)	3 (2002)	↓ 4 (2006)	5 (2010)	6 (2014)	7 (2018)
Number of...							
councilors	19.3	14.9	14.9	12.4	12.4	12.6	12.7
PR councilors	-	-	-	1.61	1.61	1.65	1.68
ward candidates	50.8	33.0	35.8	34.3	25.0	23.3	23.1
PR candidates	-	-	-	4.39	3.88	3.23	3.84
Female share of...							
councilors	0.01	0.01	0.02	0.15	0.21	0.25	0.30
PR councilors	-	-	-	0.87	0.96	0.97	0.98
ward candidates	0.01	0.01	0.02	0.04	0.08	0.13	0.18
ward candidates, rank 1	-	-	-	0.04	0.13	0.18	0.23
PR candidates	-	-	-	0.76	0.83	0.91	0.93
PR candidates, rank 1	-	-	-	0.91	1	1	1

Note: The table reports the average number and female share of councilors or candidates in each municipality, by election cycle. The PR arm was introduced in cycle 4. “Rank 1” ward candidates refer to the first or only candidate of a party in a ward, relevant from cycle 4 when party affiliation started. “Rank 1” PR candidate refers to the first candidate of a party in the PR party list. The sample includes the municipalities in the analysis, excluding five municipalities that merged over time. Statistics, including all municipalities, are extremely similar. Due to imperfect compliance, the rank-1 PR candidates are not all women in 2006, but there is no difference between treated and control municipalities (p-value=0.65). For descriptive statistics by party, refer to Appendix Table B.1.

Political landscape Over the sample period, South Korean politics was dominated by two blocs (Progressives and Conservatives), reflecting contrasting views on economic policy, welfare programs, and North Korea relations. Deep-set regional loyalties played a pivotal role in party support. Small parties periodically emerged to challenge the duopoly but struggled to gain traction.

Background behind the adoption of gender quotas If some parties had a pro-female ideology and led the movement for the reform against opposition from other parties, then we might expect parties’ strategic responses to the quota to be very heterogeneous in nature. Yet, there was consensus on the need to address South Korea’s low levels of women’s political representation, and both major parties led the passage of the gender quota. It has also been argued that the quota was merely a political tactic to expand the number of politicians (Jeon, 2013) and to strengthen the power of parties¹⁴ (Kim, 2005), unrelated to any genuine interest in female representation. Appendix C.2 elaborates on the background of the quota’s adoption. Parties were not very divided in their support of the quota.

¹⁴The quota was tied to party-list proportional representation, after all.

II.B Data

Our data were collected by scraping the website of the National Election Commission (2018). The website posts comprehensive information on all past elections, including candidate details, vote counts, and elected councilor lists.

The candidate characteristics available are election arm (ward or PR) classification, ward name, candidate number, party affiliation, name, gender, date of birth, age, occupation, education, and pertinent work experience. We use personal information to track incumbency status, and use candidate numbers (order on the ballot) to distinguish candidate positions with high likelihoods of election. We also use occupation and education as signals of competence observed by parties at candidate selection.

Vote counts are available by ward. For each party, we categorize wards into stronghold, competitive, and weakhold wards, based on the vote count the party obtained in the PR arm of the immediately preceding election. Moreover, the gap in vote shares of male and female ward candidates informs us of voter preferences for women.

Because ward divisions are centrally determined based on population size, the NEC also publishes population data, such as the number of residents by ward, voting eligibility, and gender. These data are used in balance tests.

We also scrape the “Municipal Council Portal” of the National Assembly Library (2026) to collect data on the bills proposed by elected councilors, which we use as a proxy for performance in office that is not revealed before election.

Other supplementary data, such as municipal budgets, used in balance tests (Appendix Table B.2) and descriptive figures (Appendix Figures A.1 and A.3), are explained in the corresponding notes.

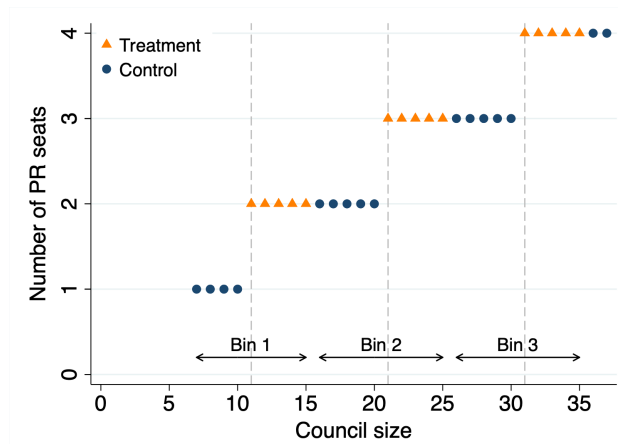
III EMPIRICAL STRATEGY

III.A Regression discontinuity design around the number of PR seats

To estimate the causal effect of stronger quota requirements, we use the fact that quota intensity varies across municipalities with the number of PR seats in the council. The number of PR seats is important as the gender quota only applies to the PR arm, and the quota stipulates that all odd-numbered candidates in the party list be female. The number of PR seats increases as a step function of the total council size, which is pre-determined centrally by the National Election Commission loosely based on population size and regional representativeness. The step function,

depicted in Figure I, provides discontinuities in the number of PR seats at given thresholds of council size. For all councils with up to 10 total seats, one councilor must be elected through PR. For councils with 11-20 total seats, two councilors must be elected through PR, etc.

Figure I - The Number of PR Seats by Council Size



Note: This figure depicts how the number of seats reserved for the PR arm increases as a step function of the total number of councilors in a municipality. Each council is categorized into a bin based on its most proximate threshold: bin 1 = {7, 8, ..., 15}, bin 2 = {16, ..., 25}, and bin 3 = {26, ..., 35}. Councils above and below the threshold in each bin are assigned to treatment (orange triangles) and control (navy circles) groups, respectively.

We exploit the discontinuous change in the number of PR seats in a regression discontinuity design: we compare candidates in municipalities on each side of the threshold, controlling for council size. To account for the fact that there are multiple thresholds (11, 21, 31), we categorize councils into bins based on the proximity to thresholds, as illustrated in Figure I. We compare treated municipalities above each threshold to control municipalities below it, within bins.

Importantly, the thresholds at 11, 21, 31, etc., do not appear to have fiscal, administrative, or political significance beyond the quota. They are not referenced in any fiscal or electoral regulation, including the quota provision itself, but arise mechanically from the rule that the number of PR seats should equal 10% of total council size, rounded up. The absence of other institutional features tied to these cutoffs alleviates concerns that other incentives change discontinuously at the thresholds and confound our identification strategy. We perform tests that validate our strategy in Section III.C.

We define treatment status based on council size in election cycle 4, instead of contemporaneous council size. This way, we can compare the estimated treatment effects across election cycles and identify the long-term effect of quotas. Since the composition of treated municipalities is held fixed across cycles, differences in treatment effects over time reflect immediate versus follow-up

effects for the same councils, not councils switching treatment status.¹⁵

The regression discontinuity specification is given by:

$$Y_{cbt} = \alpha_b + \alpha_t + \sum_{s=4}^7 \beta_s \cdot (\text{Treat at cycle 4})_{cb} \times \mathbb{1}(t = s) + f(x_{cb}) + X'_{cbt}\gamma + \epsilon_{cbt} \quad (1)$$

where Y_{cbt} denotes the outcome variable for council c in bin b in election cycle t . Since we study parties' candidate nomination strategies, the outcomes include the number of ward and PR candidates and councilors by gender. The baseline running variable is $x_{cb} \equiv (\text{council size})_{cb} - \text{threshold}_b$ in cycle 4, with $\text{threshold}_b \in \{11, 21, 31\}$.¹⁶ For outcomes related to ward elections, we change the running variable to $\tilde{x}_{cb} \equiv (\text{N. of ward seats})_{cb} - (\text{N. of ward seats at the threshold})_b$ in cycle 4, to facilitate the interpretation of the coefficients.¹⁷

$(\text{Treat at cycle 4})_{cb} \equiv \mathbb{1}(x_{cb} \geq 0)$ signifies an additional PR councilor in cycle 4. We interact it with cycle dummies to estimate β_s , the effect of an additional PR councilor in cycle 4 on the outcome in cycle s , pooling all bins. We focus on the post-reform period (cycles 4-7), since the PR arm was introduced only in cycle 4. Moreover, the baseline functional form of f is linear and fixed across both sides of the threshold. Making f quadratic or allowing for differential trends on either side of the threshold barely makes a difference, as shown in Appendix Tables B.4- B.6. X_{cbt} denotes control variables, such as contemporaneous council size or number of ward seats. The rationale behind the control is explained in Section IV.A.

Standard errors are clustered by municipality for two reasons. First, treatment assignment varies at the municipality level. Second, parties formulate strategies chiefly within a municipality and very rarely move candidates across municipalities. Several factors bind a candidate to a municipality, such as a legal residency requirement for nomination, and the electoral advantages of familiarity with local matters and voters.¹⁸

¹⁵Treatment status changes for only 5.4% of council $\times$ election cycle observations during the analysis period.

¹⁶The running variable is discrete (low degree of variability), so equation (1) is not suitable for the RD estimation method using optimal bandwidths (Calonico et al., 2014) common in regression discontinuity designs. However, we check that the estimated treatment effect is robust to different bandwidths, i.e. number of seats around the threshold (Appendix Table B.3). We use optimal bandwidths and graph the accompanying RD plots in another regression specification we introduce later, where the method is suitable (Appendix Table F.1, Appendix Figure F.1).

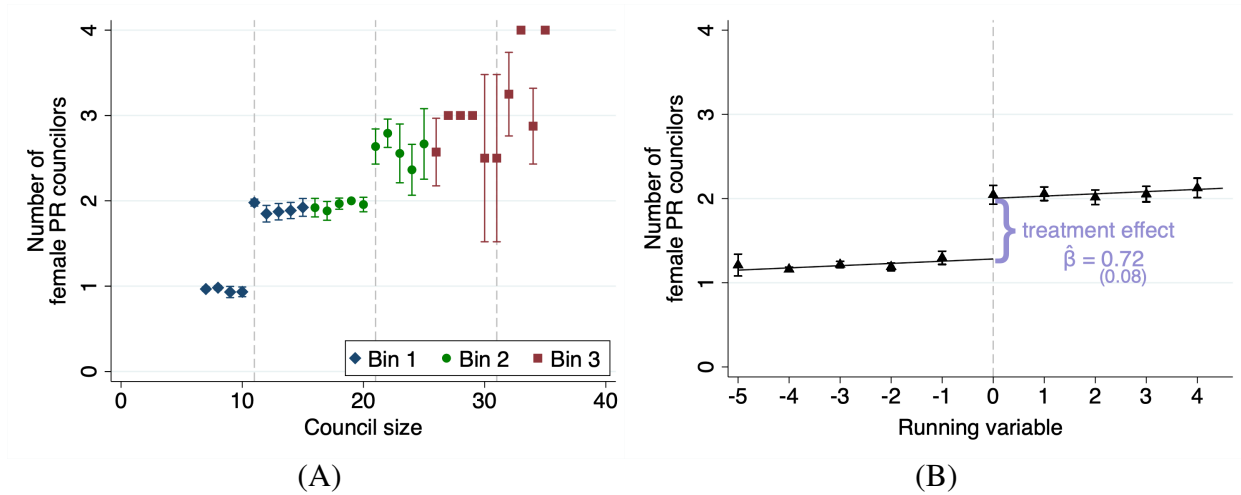
¹⁷The distinction between x_{cb} and $\tilde{x}_{cb}$ is best explained graphically; see Appendix D.1. Briefly, the difference arises because the total council size and the number of ward seats do not move together at the threshold. For example, compare a treated council with 11 total seats to a control council with 10 total seats: both have 9 ward seats, because the additional seat is a PR seat. The council-size running variable (x_{cb}) treats these councils as one unit apart. For ward-arm outcomes, controlling for it induces a mechanical discrepancy in ward-seat capacity: relative to the control-side benchmark at the cutoff, the treatment council effectively has *one fewer* ward seat. As a result, for outcomes such as the number of female ward councilors, the estimated treatment effect can be negative even if treatment increases the number of female ward councilors. Redefining the running variable as $\tilde{x}_{cb}$ removes the mechanical discrepancy by aligning the number of ward seats in treated and control councils at the cutoff. Changing the running variable this way does not change much else; $\hat{\alpha}_b$, $\hat{\alpha}_t$, $\hat{\gamma}$, the R^2 stay the same.

¹⁸Appendix D.2 elaborates on these factors.

III.B Did the quota bite?

Our strategy identifies the effect of an additional PR councilor, rather than an additional *female* PR councilor. However, an additional PR councilor strongly implies an increase in female PR councilors. For instance, in municipalities with two PR seats, the seat allocation rule¹⁹ implies that the second PR seat is far more likely to go to the number-1 candidate of the second-ranked party than to the number-2 candidate of the first-ranked party. Since the quota restricts odd-numbered PR candidates to be female, while leaving even-numbered candidates unrestricted, the additional seat is disproportionately likely to be filled by a woman.

Figure II - Treatment Effect on the Number of Female PR councilors



Note: Panels A plots the average raw number of female PR councilors by council size, pooling all post-quota cycles. Error bars represent 95% confidence intervals; where missing, only one observation exists. Panel B contains the analysis sample (bins 1 and 2). It illustrates the results of equation (1), except with a constant treatment effect for presentational clarity. The triangles plot the average number of female PR councilors, residualized and re-centered for bin and cycle effects (α_b and α_t), for each value of the running variable (x_{cb}). The lines represent $f(x_{cb})$.

Figure II illustrates this empirically in two ways. Panel A plots the average of the raw number of female PR councilors for each value of council size, pooling all post-quota cycles. The sharp rise in female PR councilors at the threshold is clear in bins 1 and 2. In bin 3, the pattern is less apparent, partly reflecting the very small number of observations.²⁰ It is also consistent with the seat allocation rule: with more PR seats, the link between an additional PR seat and an additional female PR councilor becomes less mechanical, as party vote shares play a larger role in determining

¹⁹PR seats are allocated using the largest remainder method (Hare-Niemeyer method) among parties receiving more than 5% of total votes. Each qualifying party first receives the integer part of its allocation score, defined as the number of PR seats multiplied by its vote share among qualifying parties. Remaining seats are then allocated to parties with the largest fractional remainders. Appendix D.3 provides details on the implications of the allocation rule for the gender of the additional PR seat in our empirical setting.

²⁰Bin 3 contains 9 municipalities and 30 municipality $\times$ election cycle observations.

whether the marginal seat goes to an odd- or even-numbered candidate. We therefore restrict the reduced-form analysis to bins 1 and 2, where the sample is larger and the exogenous discontinuity in PR seats most cleanly translates into variation in female PR councilors.²¹

Panel B contains exactly this analysis sample. It illustrates the results of equation (1), except with a constant treatment effect for presentational clarity: $Y_{cbt} = \alpha_b + \alpha_t + \beta \cdot (\text{Treat in cycle 4})_{cb} + \pi x_{cb} + \epsilon_{cbt}$. The triangles plot the average number of female PR councilors, residualized and re-centered for bin and cycle effects (α_b and α_t), for each value of the running variable (x_{cb}). Again, the (residualized) number of female PR councilors sharply increases at $x_{cb} = 0$, and $\hat{\beta} = 0.72$ indicates that there are 0.72 more female PR councilors at the threshold. Note that the number of female PR councilors slightly increases in the running variable (the slope of the fitted line, also plotted, is $\hat{\pi} = 0.026$), unlike the “flat” raw averages in Panel A, because the running variable is based on cycle-4 council size and council sizes grow slightly over time.

If we estimate equation (1) with heterogeneous treatment effects by election cycle,²² we find that the treatment increases female PR councilors by a similar magnitude each cycle. The constancy in the direct effect confirms that the nature and magnitude of the treatment, as well as the composition of the treatment municipalities, are stable. Thus, any change in the treatment effect on other outcomes over time can be safely attributed to the initial treatment setting treatment and control groups on different paths.

Finally, to formally test the regression discontinuity design, we check in Appendix D.4 that the number of female PR councilors increases only at the threshold and at no other point.

Our empirical strategy leverages variation in quota intensity, which leads treatment municipalities to experience an estimated 0.72 more PR women than control municipalities on average. As explained in Section V.B, the number of women experienced matters because each additional female councilor provides parties with another signal about women’s competence in office.²³

III.C Validity of the regression discontinuity design

Balance Tests The critical identifying assumption is that the relationship between council size and potential outcomes is smooth around the threshold, so that any discontinuity in observed out-

²¹We also exclude five municipalities that merge into two larger municipalities over time. Our results are robust to alternative sample criteria: including bin-3 municipalities (Appendix Table B.7), including the five aforementioned municipalities (Appendix Table B.8), and additionally excluding one municipality that becomes a provincial council in election cycle 6 (Appendix Table B.9).

²²Results can be found in Table II, column 4.

²³As such, our empirical setup bears some resemblance to a fuzzy-RD design: the discontinuity in quota intensity increases exposure to women elected through the PR arm (akin to a first stage), potentially shaping parties’ ward-arm nominations in the following election cycles (akin to a second stage).

comes at the cutoff can be attributed to treatment. We show there are no discontinuities in *pre-determined* characteristics at the threshold in cycle 4. Appendix Table B.2 reports treatment effects on these characteristics from estimating equation (1); the estimates are all statistically insignificant and small in magnitude relative to the mean of the dependent variable. Appendix Figure A.4 provides an RD plot for population size (Panel A, column 1) as an example. Population increases with council size—as expected, since it is one of the determinants of council size—but, importantly, there is no discontinuity at the threshold.

Panels A, B, and C of Appendix Table B.2 show that various population characteristics are smooth through the threshold. In particular, there is no discontinuity in the voting-age population by gender, alleviating concerns that voter preferences for female councilors differ systematically between treated and control municipalities around the cutoff. Likewise, there is no discontinuity in education or labor force participation by gender.²⁴ Panel D shows that there are also no discontinuities in ideological leaning, economic prosperity, or council performance around the threshold (columns 1-4), measured using the vote shares of the two main parties in the PR arm of the 2004 National Assembly Election, the municipal government budget, and the share of municipal spending devoted to council operations, respectively. The structure of the ward election arm—the number and size of wards—is also smooth through the threshold (columns 5-6).

Bunching Secondly, we check for bunching at the council size threshold. The concern is that councils might be able to manipulate their constituent areas to affect their council’s size and, therefore, treatment status.

The evidence does not point to clear bunching around the thresholds, although bunching is difficult to assess in our setting. Visually, it is hard to say there is bunching around the thresholds of 11 and 21.²⁵ Formally, because of the coarseness of the council size variable, even the discrete version of the McCrary (2008) density test proposed by Frandsen (2017) does not perform well: no bunching is rejected at the threshold of 11, but it is also rejected with similar p-values at randomly selected cutoffs of council size.²⁶

Nonetheless, the balance tests above and the electoral rules support the hypothesis that municipalities do not manipulate their council size around the threshold. Municipalities are not found to be systematically different above and below the threshold, which we would expect if manipulation were possible. Additionally, strict electoral rules make gerrymandering difficult. The division of

²⁴The data for Panels B and C are at the province level (16 provinces), so those results should be interpreted with more caution.

²⁵Appendix Figure A.5 displays the histogram of the frequency of municipalities by council size.

²⁶The p-value is 0.000 in the Frandsen (2017) test for the thresholds of 9, 10, 11, 18, 19, 22, 23, 25, 26, 27, for all values of the bound coefficient $k \in \{0, 0.01, 0.02, 0.05\}$.

election constituencies is determined by the Municipal Council Election Committee, which municipal councilors or party members cannot join.²⁷ The committee sets the council size, considering population, administrative districts, topography, transportation, and other conditions. It also cannot split the smallest administrative district to add it to another ward.

Placebo Test To verify that the discontinuity thresholds become meaningful for candidacy patterns only after the electoral reform and not before, we conduct two placebo tests using the three election cycles preceding the introduction of the quota. In both, the outcome is the number of male ward candidates. We focus on this outcome because (i) there was no PR arm before the reform, so all candidates ran in the ward arm; and (ii) only 1-2% of candidates were women before the reform (Table I), limiting the scope to detect meaningful pre-reform discontinuities for female candidates.

The first placebo test (column 1, Appendix Table B.10) uses our main treatment variable, defined by whether a municipality would receive an additional PR seat in cycle 4. This specification tests whether municipalities that later become treated already differ from control municipalities in ways that affected candidate selection before the reform.

The second placebo test (column 2, Appendix Table B.10) defines treatment contemporaneously in each pre-reform cycle. It tests whether the discontinuity thresholds themselves (e.g. 10 vs. 11, or 20 vs. 21) have any independent significance before the reform. If the cut-offs were tied to other long-standing fiscal, administrative, or political incentives unrelated to the quota, we would expect to detect discontinuities in the pre-reform period.

Under both approaches, the estimated treatment effects in cycles 1-3 are statistically indistinguishable from zero. These findings support our interpretation that the discontinuities become meaningful only with the introduction of the quota, rather than reflecting pre-existing differences between treatment and control municipalities, any inherent importance of the thresholds themselves, or other long-standing discontinuous incentives at those cutoffs.

Discussion on SUTVA The Stable Unit Treatment Value Assumption (SUTVA) requires no spillovers across treatment groups. While we are less worried about the violation of SUTVA in the first election after the quota, it is possible that party learning in treatment municipalities spills over to control municipalities, e.g. through communication with party leaders at the district or national level. If so, our treatment effects would be underestimated, indicating that our long-term results represent conservative estimates.

²⁷The committee consists of up to 11 members appointed by the provincial mayor among individuals nominated by the media, legal and academic communities, civic groups, the provincial council, and the Provincial Election Committee.

IV MAIN RESULTS

IV.A The evolution of councils' gender composition

Did the reform have the intended effect? Table II reports the results of equation (1) on the number of male and female councilors elected in each municipality, by election arm.

Table II - Treatment Effect on the Number of Councilors

Election arm:	All political parties					
	Ward		PR		All	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)
Treat at cycle 4 $\times$ Cycle 4	0.45* (0.26)	-0.33 (0.22)	0.09 (0.08)	0.76*** (0.09)	-0.29 (0.33)	0.29 (0.28)
Treat at cycle 4 $\times$ Cycle 5	-0.13 (0.29)	0.32 (0.23)	0.10 (0.06)	0.70*** (0.08)	-0.87** (0.35)	0.88*** (0.29)
Treat at cycle 4 $\times$ Cycle 6	-0.23 (0.33)	0.53* (0.27)	0.09 (0.06)	0.70*** (0.09)	-0.98** (0.38)	1.10*** (0.32)
Treat at cycle 4 $\times$ Cycle 7	-0.28 (0.36)	0.82*** (0.29)	0.04 (0.06)	0.72*** (0.08)	-1.08*** (0.41)	1.41*** (0.35)
Running variable form	ward	ward	council	council	council	council
<i>N</i>	865	865	865	865	865	865
<i>p</i> -value: Cycle 4=Cycle 5	0.002	0.000	0.919	0.418	0.005	0.001
<i>p</i> -value: Cycle 4=Cycle 6	0.007	0.000	0.882	0.417	0.009	0.000
<i>p</i> -value: Cycle 4=Cycle 7	0.016	0.000	0.447	0.593	0.014	0.000

Note: This table reports the effect of being above the threshold on the gender composition of councilors across election cycles. The regression specification is given by equation (1). The sample includes bins 1 and 2, and all parties participating in municipal elections. The outcome variable is the number of councilors elected overall and separately through the two arms—ward and PR—by gender in each municipality. Note that the coefficients for the ward and PR arms do not sum to the coefficient for all arms (e.g., columns 1 and 3 do not sum to column 5), because the “Running variable form” for the ward arm regressions is different, as explained in Section III.A. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

In the first cycle after the introduction of the quota (cycle 4), treated municipalities display an overall higher number of elected female councilors (column 6), although the increase is not statistically significant. Column 4 shows that a higher number of women do get elected in treated councils through the PR arm—0.76 for every additional PR seat. However, this effect is partially offset by 0.33 fewer women (not statistically significant) and 0.45 more men getting elected through the unregulated ward arm (columns 1-2).

Yet the initial treatment effect does not persist across election cycles. From cycle 5 onward, treated municipalities elect significantly more female councilors and significantly fewer male councilors (columns 5-6). Since columns 3-4 confirm that a constant number of additional women is elected through the PR arm, this reversal cannot be due to the intensity of the quota effect changing

over time. Instead, it traces back to fewer men and more women being elected through the ward arm starting in cycle 5 (columns 1-2). By cycle 7, 0.82 more female ward councilors are elected in the treated municipalities than control municipalities.

One factor to note is that councils treated in cycle 4 grow slightly larger over time relative to control, as reflected in the sum of male and female councilors in columns 5-6.²⁸ To account for this, we control for contemporaneous council size in all subsequent tables estimating equation (1) (as with the running variable, for ward-arm outcomes we instead control for the contemporaneous number of ward seats).

IV.B Party strategies and candidate selection

As election outcomes, the results in Table II could arise from either the voter’s side or the party’s side: they may be driven by voters expressing gender preferences among a given set of candidates, or by parties expressing gender preferences in candidate selection. Given the strong tendency of voters to vote for candidates of their preferred party,²⁹ we focus on parties’ candidate selection. We return to the discussion of voter preferences in Section V.E.

We find that the treatment effects on candidates closely track the previous results on elected councilors, pointing to candidate selection as the main driver of the effects on electoral outcomes. Two patterns stand out. First, parties initially counteract the quota by increasing the number of male candidates in the unregulated ward arm, which reduces the female share for a given set of candidate positions. Second, this response reverses over time, with treated municipalities gradually nominating more female ward candidates.

To show these patterns, Table III reports the effects on the gender composition of candidates separately by election arm and ballot position. Columns 1-6 consider candidates of all parties participating in municipal elections. We exclude independent candidates who run alone without party affiliation and are thus unaffected by PR-arm regulations. Columns 7-15 restrict the sample to the two main parties—the Progressive Party and the Conservative Party. Their nomination strategies are the most consequential, as these two parties alone produce at least 74% of ward councilors and 82% of PR councilors in every election (Appendix Table B.1). Lastly, columns 10-15 further focus on candidates in electorally viable ballot positions: “useful” positions—position 1 in wards electing 1-2 councilors and positions 1-2 in wards electing 3-4 councilors—and “rank 1” positions, defined as the party’s first or only candidate in the ward. Changes in the composition

²⁸If we defined treatment status contemporaneously, the coefficients in columns 1 and 2 would have summed to 0, those in columns 3 and 4 to 1, and those in columns 5 and 6 to 0, in each election cycle.

²⁹The predicted vote share of a candidate in the ward arm, based on (i) the popularity of their party—measured by the party’s vote share in the PR arm—and (ii) the historical tendency of voters to favor candidates higher up on the ballot, has a correlation coefficient of 0.90 with the actual vote share.

of ward candidates may not be consequential if they are driven by candidates in positions that have no hope of getting elected.

In columns 3, 6, 9, 12, and 15, we control for the total number (male+female) of relevant candidates (either ward or PR) in the relevant positions (either all, useful, or rank 1). This is an accounting exercise: holding the male+female total fixed, the treatment effects on the number of male and female candidates are mechanically opposite in sign and equal in magnitude. The coefficient then represents the number of additional female candidates (=fewer male candidates) in councils above the threshold relative to below, holding the total number of candidate positions fixed. A positive coefficient indicates that a woman is partially substituting for a man in a given candidate position, and thus corresponds to a higher female share of candidates.

The key findings are in the ward arm. In response to treatment, 3.44 more male candidates are nominated in the ward arm in treatment municipalities in cycle 4 (column 4). However, this number gradually decreases, leading to no significant difference in male ward candidates by cycle 7. The opposite trajectory holds for female ward candidates: the coefficient in column 5 changes from negative (though statistically insignificant) in cycle 4 to positive from cycle 5 onward. Combining both genders, column 6 shows that treated municipalities nominate 0.86 fewer women (0.86 more men) than control municipalities in cycle 4, turning into 1.00 more women (1.00 fewer men) by the last election cycle. These patterns also hold when we restrict attention to the main parties (columns 7-9) and to candidates in pivotal ballot positions (columns 10-15).³⁰

The increase in female candidates also extends to the PR arm (columns 1-3). The negative coefficient in cycle 4 in column 3—indicating a lower female share of PR candidates in treated municipalities—does not necessarily reflect an intentional strategy against women. Rather, it reflects that treated councils have one additional PR seat, which leads parties to submit longer PR lists and mechanically creates more opportunities to include men.³¹ By contrast, the weak growth over time in the number of female PR candidates in column 2 and the positive coefficient in cycle 7 in column 3 are more informative about nomination behavior. They imply that, over time, women increasingly replace men in even-numbered positions on PR lists in treated municipalities.

³⁰As an example of an RD plot for the ward arm, Appendix Figure A.6 visualizes the results in column 12.

³¹For example, even if all parties—treated and control—prefer not to nominate women unless required, control parties in bin 1 nominate one woman, while treated parties in bin 1 nominate one woman and one man, resulting in a lower female share in treated municipalities.

Table III - Treatment Effect on the Number of Candidates

All political parties										Main political parties									
Election arm: Position on ballot:	PR			Ward			Ward			Ward			Ward						
	All candidates			All candidates			All candidates			Useful positions			Rank 1						
	Male (1)	Female (2)	Female (3)	Male (4)	Female (5)	Female (6)	Male (7)	Female (8)	Female (9)	Male (10)	Female (11)	Female (12)	Male (13)	Female (14)	Female (15)				
Treat at cycle 4 × Cycle 4	0.95*** (0.22)	0.97*** (0.22)	-0.32* (0.18)	3.44*** (0.94)	-0.30 (0.33)	-0.86*** (0.29)	1.31* (0.76)	-0.27 (0.23)	-0.41* (0.22)	1.76*** (0.61)	-0.31 (0.21)	-0.60*** (0.19)	0.77 (0.47)	-0.39** (0.18)	-0.46*** (0.17)				
Treat at cycle 4 × Cycle 5	0.66*** (0.17)	1.21*** (0.23)	-0.04 (0.15)	1.25 (0.82)	0.48 (0.33)	0.17 (0.29)	0.26 (0.65)	0.49* (0.25)	0.39* (0.22)	0.69 (0.61)	0.50** (0.22)	0.26 (0.18)	-0.31 (0.48)	0.44** (0.18)	0.41** (0.17)				
Treat at cycle 4 × Cycle 6	0.27* (0.15)	1.03*** (0.22)	0.16 (0.14)	-0.46 (0.68)	0.70* (0.37)	0.66** (0.32)	0.45 (0.63)	0.78*** (0.29)	0.62** (0.27)	0.91 (0.61)	0.71** (0.28)	0.38* (0.22)	0.02 (0.45)	0.46* (0.24)	0.37* (0.21)				
Treat at cycle 4 × Cycle 7	0.22 (0.16)	1.28*** (0.22)	0.27* (0.15)	0.34 (0.69)	1.29*** (0.44)	1.00** (0.40)	-0.59 (0.65)	1.17*** (0.32)	1.10*** (0.31)	0.19 (0.57)	0.93*** (0.30)	0.71*** (0.27)	-0.15 (0.46)	0.69*** (0.25)	0.58** (0.23)				
Running variable form	council 865	council 865	council 865	ward 865	ward 865	ward 865	ward 864	ward 864	ward 864	ward 864	ward 864	ward 864	ward 864	ward 864	ward 864				
N	No	No	Yes	No	No	Yes	No	No	Yes	No	No	Yes	No	No	Yes				
Fix (male+female) total																			
p-value: Cycle 4=Cycle 5	0.079	0.228	0.041	0.000	0.000	0.000	0.027	0.000	0.000	0.002	0.000	0.000	0.001	0.000	0.000				
p-value: Cycle 4=Cycle 6	0.000	0.739	0.001	0.000	0.002	0.000	0.076	0.000	0.000	0.014	0.000	0.000	0.027	0.000	0.000				
p-value: Cycle 4=Cycle 7	0.000	0.101	0.000	0.000	0.000	0.000	0.003	0.000	0.000	0.000	0.000	0.000	0.005	0.000	0.000				

Note: This table reports the effect of being above the threshold on the gender composition of candidates across election cycles. The regression specification is given by equation (1). The sample includes bins 1 and 2. Columns 1-6 include candidates from all parties (excluding independent candidates), and in columns 7-15 the sample is restricted to the two main parties. The outcome variable is the number of candidates separately in the two arms—ward and PR—by gender in each municipality. “Useful positions” refer to candidates in high-up positions on the ballot for the party in a ward (position 1 if the ward elects 1-2 councilors, and positions 1 and 2 if the ward elects 3-4 councilors). “Rank 1” candidates refer to the first or only candidate for the party in each ward. The number of observations when the sample is restricted to the two main parties is 864 instead of 865, because in one municipality × election cycle, main parties only had PR candidates. In columns 3, 6, 9, 12, and 15 we control for the total number (male+female) of relevant candidates (either ward or PR) in the relevant positions (either all, useful, or rank 1). Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

V MECHANISMS

The evidence so far shows that parties respond to more stringent PR quota requirements by initially shifting the unregulated ward arm toward men: they nominate more male ward candidates, thereby reducing the female share, especially in ballot positions with higher chances of election. From cycle 5 onward, however, this response reverses. Parties in treated municipalities gradually increase female ward nominations and, by the last cycle, nominate *more* female ward candidates than parties in control municipalities.

In this section, we show that the quota changes parties' nomination strategies in favor of women along two dimensions. First, it increases the number of women with incumbency advantage within the existing candidate pool. Second, it increases the nomination of rookie women with no prior council experience. We focus on explaining the second margin.

We hypothesize that parties initially select a suboptimally low number of women because they hold limited information and downward-biased beliefs about women's competence. In this context, by exposing parties to new female politicians, the quota can improve parties' beliefs about women's competence as a group, including women parties have not previously observed in office.

Table IV organizes several pieces of evidence supporting the party-learning mechanism. After presenting this evidence, we use an Oaxaca-Blinder-style decomposition to quantify how much party learning contributes to the total increase in female ward candidates.

Table IV - Summary of Evidence on the Main Mechanisms

What might explain the change in parties' candidate nomination strategies?		
• More females with incumbency advantage ✓		
- Strong ↑ in incumbent female ward candidates		Tab V
• Beyond incumbent women: weakening discrimination against women as a group ✓		
- ↑ rookie female ward candidates, not only incumbents		Tab V
- Overall trend: women's selection threshold higher than men's. Gap ↓ over time		Fig III
<u>Sources</u>		
▶ Taste-based discrimination ✗		
- ↑ female cand. in competitive wards, contrary to model prediction for ↓ distaste		Tab VI
▶ Statistical discrimination: information matters ✓		
- Quota women are rookies, who bring new info about quality of women		Tab VII
- Counteraction & reversal stronger where info is scarce (no women pre-quota)		Tab VIII
▶ Statistical discrimination: biased beliefs on women's competence ✓		
- ↑ in female ward cand. stronger where first elected women show stronger council performance		Tab IX

Note: Summary of the evidence supporting the main mechanisms proposed in the current paper. The check (cross) marks a mechanism that we find (do not find) support for. We provide brief explanations and indicate where the evidence can be found.

This section concludes by exploring alternative mechanisms that could have generated a similar pattern of initial counteraction and gradual reversal. We address two non-party drivers—a shift in voter preferences and in the pool of potential candidates—and alternative party-side explanations, such as a lack of available women and an increase in women’s influence within the party.

V.A Parties’ strategies: beyond incumbency advantage

Stronger quota requirements shift parties’ nomination strategies in favor of women in two ways: first through female incumbents from cycle 5, and more gradually through female rookies.

Incumbency advantage We find that more women gaining incumbency advantage—a direct effect of the treatment—is an important channel driving the increase in female candidates in the ward arm. The quota gets PR women elected, and, fueled by a no-re-election norm in the PR arm (Shin, 2014), grants them access to the ward arm. Table V separately estimates treatment effects for incumbent and rookie ward candidates,³² showing that parties nominate more incumbent women in the ward arm over time, starting in cycle 5 (columns 4-6).

Table V - Treatment Effect on Incumbent and Rookie Ward Candidates

	Main political parties								
	All ward candidates			Incumbent ward candidates			Rookie ward candidates		
	Male (1)	Female (2)	Female (3)	Male (4)	Female (5)	Female (6)	Male (7)	Female (8)	Female (9)
Treat at cycle 4 × Cycle 4	1.31* (0.76)	-0.27 (0.23)	-0.41* (0.22)	0.63 (0.40)	-0.00 (0.13)	-0.09 (0.12)	0.67 (0.62)	-0.27 (0.17)	-0.32** (0.15)
Treat at cycle 4 × Cycle 5	0.26 (0.65)	0.49* (0.25)	0.39* (0.22)	1.06*** (0.38)	0.47*** (0.17)	0.25 (0.16)	-0.80 (0.59)	0.02 (0.16)	0.11 (0.14)
Treat at cycle 4 × Cycle 6	0.45 (0.63)	0.78*** (0.29)	0.62** (0.27)	0.84** (0.38)	0.62*** (0.21)	0.41** (0.19)	-0.38 (0.49)	0.16 (0.18)	0.19 (0.16)
Treat at cycle 4 × Cycle 7	-0.59 (0.65)	1.17*** (0.32)	1.10*** (0.31)	-0.08 (0.40)	0.79*** (0.20)	0.69*** (0.19)	-0.50 (0.52)	0.38* (0.21)	0.40* (0.20)
Running variable form	ward	ward	ward	ward	ward	ward	ward	ward	ward
N	864	864	864	864	864	864	864	864	864
Fix (male+female) total	No	No	Yes	No	No	Yes	No	No	Yes
p-value: Cycle 4=Cycle 5	0.027	0.000	0.000	0.288	0.000	0.007	0.003	0.044	0.002
p-value: Cycle 4=Cycle 6	0.076	0.000	0.000	0.635	0.001	0.004	0.027	0.011	0.002
p-value: Cycle 4=Cycle 7	0.003	0.000	0.000	0.080	0.000	0.000	0.034	0.001	0.001

Note: This table reports the effect of being above the threshold on the gender composition of ward candidates. The sample includes bins 1 and 2 and is restricted to the two main parties. The outcome variable is the number of ward candidates in each municipality, by gender and incumbency status—whether they were elected in at least one previous election. Column 1 equals the sum of columns 4 and 7, and column 2 equals the sum of columns 5 and 8. In columns 3, 6, and 9, we control for the total number (male+female) of incumbent and rookie ward candidates, respectively. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

³²A candidate or councilor is an incumbent if they have been elected in at least one previous election.

Beyond incumbency advantage However, the treatment effects extend beyond incumbent women. Parties initially nominate significantly fewer women among rookie candidates (column 9). Nonetheless, they slowly nominate more rookie women alongside incumbent women in the ward arm over time. This is most clearly seen in column 9, which captures the extent to which rookie women are substituting for rookie men in a given candidate position: the coefficient becomes statistically significant by the last election cycle (p-value=0.053 for cycle 7).

Complementary party-level analysis exploiting close races The aggregate municipality-level patterns in Table V are confirmed when we estimate the effect of experiencing a quota woman in action more directly. Specifically, we compare the subsequent nomination strategies of parties that marginally won or lost the election of their first—and therefore female—PR candidate in the previous election cycle. Under the standard close-election RD assumption that parties marginally winning and losing the PR seat would have followed similar nomination strategies absent the victory, this comparison identifies the causal effect of having a woman from one’s own party elected. Appendix F provides details on the estimation strategy and results.

The party-level analysis shows that exposure to an additional woman elected through the quota raises female ward candidacies not only by giving that woman an incumbency advantage, but also by gradually increasing rookie women’s access to ward elections. The probability that a party’s number-1 PR candidate in cycle 4 is renominated as a ward candidate in cycle 5 is 34 percentage points higher if she was elected in cycle 4. However, this direct incumbency channel does not persist: the renomination effect weakens and becomes statistically insignificant in cycle 6, falling to zero in cycle 7.³³ Crucially, on the other hand, winning the first PR candidate in cycle 4 has long-term effects, significantly increasing the number of female *rookie* candidates in cycles 6-7.

We show in Section V.E that these positive spillovers on rookie females cannot be explained by more and better women becoming newly available. In the following sections, we propose that they are better explained by an improvement in parties’ perceptions of women.

V.B A framework for interpreting party learning

Before the quota, South Korean municipal councils had almost no female councilors, and 60% of the population believed men make better political leaders than women (Appendix Figure A.1). In such an environment, the quota can disrupt the status quo: by exposing parties to female politicians, it could update beliefs about women’s competence, including women not yet observed in office.

³³This aligns with the limited re-candidacy pattern in the overall sample. 85% of individuals run in a maximum of three elections, three-quarters of whom run only once or twice. There is no gender difference. After all, the average age of candidates is around 50 for both men and women, even for rookies, and 2 terms is already 8 years.

We formalize this mechanism in a model of electoral competition with discrimination, which helps to interpret our empirical results. We build on Le Barbanchon and Sauvagnat (2022) by enriching the candidate-selection process and incorporating the allocation of candidates across wards, a key change that tightens the link between the model’s predictions and our empirical tests.

We also allow for two forms of discrimination: taste-based and statistical. Statistical discrimination is absent from Le Barbanchon and Sauvagnat (2022), who assume perfect information about potential candidates’ ability. To model statistical discrimination, we extend the standard framework of Aigner and Cain (1977) by introducing a dynamic process of Bayesian belief updating. Unlike standard models, beliefs are updated not only about individual ability but also about the average ability of the group the candidates belong to, an idea often discussed in the discrimination literature but rarely formalized, with the exception of Lepage (2024).

Appendix E.1 presents the full model; here, we summarize the key elements. Candidates differ by gender and competence. Parties choose candidates to maximize seats won, but may also care about the gender composition of candidates. Competence matters because it increases electability, an assumption supported by the data.³⁴ Parties do not perfectly observe competence at the time of nomination. Instead, they observe a noisy signal and form expectations based on the signal and their beliefs about the average competence of male and female candidates. After the election, parties observe the true ability of elected councilors and update their beliefs about average competence by gender.

The model has two main implications for the empirical exercise. First, if parties initially underestimate women’s average ability, exposure to female councilors through the quota leads them to update these beliefs. This makes parties more willing over time to nominate women in the ward arm, including female rookies with no prior council experience. Moreover, updating is faster when parties observe more women,³⁵ although with diminishing returns. As a result, belief updating occurs faster in treated municipalities than in control municipalities, even when control municipalities also experience some female councilors. Updating is also faster when the gap between elected women’s revealed ability and parties’ prior expectations is larger, when parties’ initial beliefs are more downward-biased, and when parties have less prior information about women.

Second, the model distinguishes statistical from taste-based discrimination. Statistical discrimination means that parties assign lower expected competence to female than to male candidates with the same ability signal. Taste-based discrimination means that parties have an intrinsic pref-

³⁴Appendix Table E.1 shows that candidates with stronger proxies for competence, such as education and political experience, receive higher vote shares.

³⁵When parties observe more women, the maximum likelihood estimator of women’s average competence—formed from observing elected women’s true abilities—becomes more precise. It therefore receives greater weight in posterior beliefs, moving the posterior more quickly away from the biased prior.

erence against nominating women. Both reduce the female share of candidates by raising the signal threshold required for women relative to men. Hence, in both cases, weaker discrimination lowers this threshold and leads parties to nominate more women. The two mechanisms, however, generate different predictions for how parties allocate candidates across more and less competitive wards. The key assumption is that candidate competence is more electorally valuable in more competitive wards, which is strongly met in the data (see Appendix E.5).³⁶ If taste-based discrimination weakens, parties nominate more women without revising their beliefs about women’s competence, so the additional women nominated are expected to have lower ability and would be placed in less competitive wards. If statistical discrimination weakens, parties raise their assessment of women’s competence more broadly, so both existing and additionally nominated women would be placed in more competitive wards.

V.C Empirical corroboration: parties statistically discriminate based on biased beliefs about women’s competence

The model formalizes how changes in candidate selection could arise from parties learning about women after the quota exposes them to female councilors. This section presents the empirical evidence that supports this hypothesis.

Our empirical analysis focuses on the two main parties.³⁷ Party composition changes over time: new parties emerge and dissolve, and existing parties merge. Restricting the sample to the two parties present throughout all election cycles ensures that any estimated change in nomination strategies reflects genuine behavioral change rather than compositional turnover. These two parties’ strategies are also by far the most consequential, as they alone account for at least 74% of ward councilors and 82% of PR councilors in every election (Appendix Table B.1).

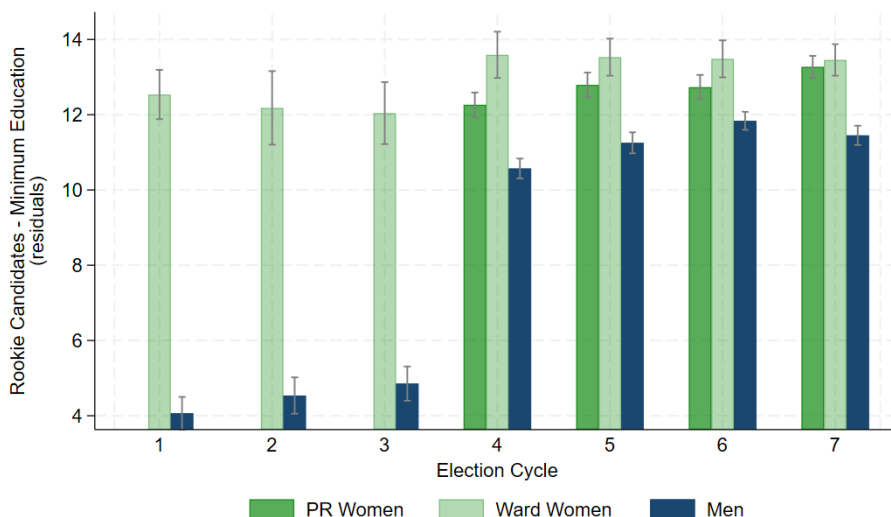
The discrimination framework We begin by using nationwide descriptive trends to motivate the discrimination framework as a useful lens for studying parties’ nomination strategies. Note that these nationwide trends are distinct from, and cannot speak to, the local treatment effects of stronger quota requirements at the discontinuity threshold.

³⁶Esteve-Volart and Bagues (2012) also exploit the premise that placing strong candidates matters more in competitive races. They show that, in a setting where female candidates attract more votes than male candidates, parties nevertheless place women more favorably only in more competitive races. Through the lens of our model, this pattern is consistent with taste-based discrimination under accurate beliefs: parties already recognize that women are strong candidates, but favor men unless electoral competition forces them to do otherwise. This interpretation differs from the statistical-discrimination interpretation in our setting because the information environment is different.

³⁷Due to frequent party rebranding (name changes), mergers, and splits, we follow the largest faction in each of the two dominant blocs (Progressives and Conservatives).

The model predicts that if parties discriminate, either due to taste-based or statistical discrimination, the lowest signal of ability observed among female candidates is higher than that among male candidates. Empirically, we take candidates' education as a signal of ability observed by parties at candidate selection.³⁸

Figure III - Minimum Education of Rookie Female and Male Candidates



Note: This figure illustrates the evolution of the education of the “marginal” (meaning last to be selected, in the lens of the model) female and male candidates over time. It plots, for each election cycle, the average years of schooling of the least educated female rookie PR candidate, female rookie ward candidate, and male rookie candidate selected by each party in each municipality. The years of schooling variable is residualized for year of birth and municipality fixed effects. The figure includes only bins 1 and 2. All candidates are plotted for cycles 1-3 as no party affiliation existed, while only candidates from the two main parties are plotted for cycles 4-7.

Figure III shows that this prediction is empirically met in the data. We plot, for each election cycle, the average years of schooling of the least educated female PR candidate, female ward candidate, and male candidate in each municipality. To focus on parties' selection strategies based on signals of quality rather than incumbency advantage, we restrict the sample to rookies.³⁹ The years of schooling variable is residualized for year-of-birth (cohort) fixed effects, to account for rising educational attainment over time, and for municipality fixed effects, to control for local culture, norms, etc., that might affect educational disparities between men and women.

Throughout the sample period, and especially before the quota was introduced in cycle 4, the minimum level of education among women is higher than among men, implying that women are

³⁸We use education because, apart from it being a common measure of ability in the literature (e.g. Bagues and Campa, 2021; Baltrunaite et al., 2014), it is a good proxy of competence in our setting. Voters value education, as candidates with high education levels win higher vote shares in the data. Furthermore, parties allocate candidates with higher education to the most competitive wards, where the cost of withholding the highest-ability candidates is greater.

³⁹Incumbents are on average older, less educated, and mostly male.

more positively selected ($\bar{s}_f > \bar{s}_m$). In cycle 4, the marginal female *ward* candidate has the highest education, consistent with the initial counteraction we observe in the ward arm: in the model, when parties remove women from the ward arm, they remove the lowest-signal women first. After cycle 4, the gender gap generally continues to narrow.⁴⁰ Through the lens of the model, a decline in $(\bar{s}_f - \bar{s}_m)$ suggests that discrimination against women is weakening. To the extent that these women displace, rather than simply supplement, existing candidates, the least competent men are replaced by more competent women. In this sense, weakening discrimination provides one explanation for why quotas can lead to the replacement of “mediocre men,” also documented in other contexts (e.g. Baltrunaite et al., 2014; Besley et al., 2017).

Taste-based vs. statistical discrimination Having motivated the discrimination framework, we now return to the local treatment effects of stronger quota requirements. What might be the source of the discrimination explaining our main results? Distinguishing between statistical discrimination and taste-based discrimination is challenging because both deliver observationally equivalent predictions in the first cycle post-quota: a reduction in the share of female ward candidates and an increase in the minimum signal of women compared to men.⁴¹ It is the subsequent *dynamic* response that allows us to distinguish which mechanism better fits the evidence.

Recall the model’s predictions for women’s ward placement over time. If the quota weakens parties’ taste-based discrimination, the additional women will be of lower ability than the existing women and therefore would get placed in less competitive wards—where the electoral outcome is virtually predetermined. On the other hand, if the quota weakens parties’ statistical discrimination, i.e. women’s perceived ability rises, then women would get increasingly placed in competitive wards. These are wards where withholding the best candidates is electorally more costly (Esteve-Volart and Bagues, 2012; Folke and Rickne, 2016). We find the latter in the data.

Table VI displays the evolution in the gender composition of candidates across different ward types—stronghold, competitive, and weakhold—defined based on the party’s margin of victory in

⁴⁰The narrowing gender gap primarily reflects rising education among marginal male candidates, although education among marginal female ward candidates also shows a very weak decline. Both patterns are consistent with weakening discrimination. In the background, the education of the overall candidate pool (both men and women) rises over time nationwide, even net of cohort effects (Appendix Figure A.7). This force and weakening discrimination operate in the same direction for men, explaining the strong rise in marginal male candidates’ education, but work against each other for women. For female PR candidates, the supply channel likely dominates, consistent with PR seat numbers being largely fixed across cycles. Thus, female PR candidates’ minimum education rises, though by less than the rise for men. For female ward candidates, by contrast, the statistical discrimination channel is stronger, producing the very weak decline.

⁴¹This observational equivalence is common in the gender-quota literature. For example, findings that quotas increase the number of female candidates while leaving women concentrated in less favorable positions (Bagues and Campa, 2021; Lippmann, 2021), or that quotas lead parties to select women who are more qualified than the men they replace (Baltrunaite et al., 2014), can be consistent with both taste-based and statistical discrimination.

Table VI - Treatment Effect on the Number of Female Ward Candidates by Competitiveness of Ward

Ward type:	Position on ballot:	Main political parties											
		Stronghold			Competitive			Weakhold			Less competitive (Stronghold + Weakhold)		
		All Female (1)	Useful Female (2)	Rank 1 Female (3)	All Female (4)	Useful Female (5)	Rank 1 Female (6)	All Female (7)	Useful Female (8)	Rank 1 Female (9)	All Female (10)	Useful Female (11)	Rank 1 Female (12)
Treat at cycle 4 × Cycle 4		-0.00 (0.06)	-0.07 (0.05)	-0.05 (0.04)	-0.03 (0.07)	-0.05 (0.07)	-0.03 (0.06)	-0.06 (0.06)	-0.09 (0.06)	-0.09* (0.05)	-0.08 (0.08)	-0.17** (0.07)	-0.15** (0.06)
Treat at cycle 4 × Cycle 5		0.05 (0.05)	0.02 (0.04)	0.05 (0.04)	0.19** (0.09)	0.15* (0.08)	0.14** (0.07)	0.03 (0.05)	0.03 (0.05)	0.05 (0.05)	0.06 (0.08)	0.04 (0.07)	0.09 (0.07)
Treat at cycle 4 × Cycle 6		0.06 (0.05)	0.00 (0.04)	0.01 (0.03)	0.23** (0.10)	0.21** (0.09)	0.16* (0.08)	0.04 (0.05)	0.03 (0.05)	0.02 (0.04)	0.08 (0.07)	0.02 (0.07)	0.03 (0.06)
Treat at cycle 4 × Cycle 7		0.08 (0.05)	0.02 (0.04)	0.01 (0.04)	0.41*** (0.12)	0.36*** (0.11)	0.32*** (0.10)	0.03 (0.06)	-0.00 (0.06)	-0.02 (0.05)	0.11 (0.09)	0.01 (0.08)	-0.01 (0.07)
Running variable form		ward	ward	ward	ward	ward	ward	ward	ward	ward	ward	ward	ward
N		1551	1551	1551	1551	1551	1551	1551	1551	1551	1551	1551	1551
Party dummy		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of wards of each type		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Average margin of victory		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fix (male+female) total		Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No
p-value: Cycle 4=Cycle 5		0.317	0.074	0.057	0.004	0.003	0.006	0.178	0.061	0.019	0.069	0.006	0.001
p-value: Cycle 4=Cycle 6		0.304	0.181	0.268	0.011	0.004	0.015	0.144	0.054	0.027	0.051	0.011	0.009
p-value: Cycle 4=Cycle 7		0.144	0.121	0.274	0.001	0.001	0.001	0.231	0.244	0.240	0.034	0.036	0.077

Note: This table reports the effect of being above the threshold on the female share of ward candidates in different wards. The regression specification is given by equation (1). Wards are divided into stronghold, competitive, and weakhold wards based on the party's margin of victory in the PR arm of the immediately preceding election (see notes under Appendix Figure A.8 for how it is calculated). Stronghold, competitive, and weakhold wards correspond to margins of victory of ≥ 0.05 , $[-0.05, 0.05]$, and < -0.05 , respectively, closely approximating the top quarter, middle half, and the bottom quarter. The results are qualitatively similar when we use the cutoffs of ± 0.10 . The sample includes bins 1 and 2 and is restricted to the two main parties. The level of observation is municipality $\times$ party. The outcome variable is the number of female ward candidates in each type of ward. The columns for "Useful" and "Rank 1" represent increasingly high-up positions on the ballot paper, implying higher likelihoods of winning; "Useful" denotes position 1 if the ward elects 1-2 councilors and positions 1 and 2 if the ward elects 3-4 councilors, while "Rank 1" denotes the first candidate of the party in the ward. We control for the number of wards of each type, party dummies, and the average margin of victory for the party in the municipality. We also control for the total number (male+female) of relevant ward candidates, except in columns 3, 6, 9, and 12, since the total number of rank 1 candidates is perfectly multicollinear with the number of wards. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

the PR arm in the immediately preceding election (the National Assembly election, 2 years before each municipal council election). The three ward types closely approximate the top quarter, middle half, and bottom quarter of the margin of victory.⁴² We refer to stronghold and weakhold wards jointly as *less competitive wards*, in contrast to *competitive wards*. As the ward categorization depends on the party at hand, we estimate equation (1) at the municipality $\times$ party level. We also control for variables relevant to the party when allocating candidates across wards: the number of wards of each type, and the party’s average margin of victory in the municipality.

We confirm that parties’ counteraction to the quota in cycle 4 conforms with the model’s prediction: women removed from the ward arm are of low competence and tend to be placed in less competitive wards (columns 10-12).⁴³ The subsequent increase in women in cycles 5-7, by contrast, is concentrated in competitive wards (columns 4-6). Focusing on candidates in useful positions, columns 5 and 11 show that parties in treated municipalities initially nominate 0.17 fewer women in less competitive wards, and by the last cycle, 0.32 more women in competitive wards. These patterns hold when we restrict attention to rookie candidates only (Appendix Table B.11), ruling out an alternative story in which incumbency advantage, rather than candidate quality, matters more in competitive wards.

Our results demonstrate that a reduction in taste-based discrimination is not able to reconcile the reversal in party strategies across election cycles. Instead, they are consistent with a reduction in statistical discrimination.⁴⁴ In the following sections, we present additional evidence supporting this hypothesis. To be clear, our findings do not rule out taste-based discrimination as a barrier against female nomination in South Korea altogether—only that it does not appear to drive the quota’s effect on party nominations over time.

Statistical discrimination: lack of information Statistical discrimination is directly tied to information; it would disappear under perfect information on ability. Here, we show that the reversal of parties’ strategies in treated municipalities can be traced back to quotas bringing in new information. This idea corresponds to the model prediction that the reversal in strategies happens faster the greater the number of newly elected female councilors.

First, we show that the quota allows rookie women to get elected in the PR arm, representing new signals about women’s average competence. Table VII reports the treatment effect on the num-

⁴²Stronghold, competitive, and weakhold wards correspond to victory margins of ≥ 0.05 , $[-0.05, 0.05)$, and < -0.05 , respectively. The distribution of the victory margins (and how it is constructed) is in Appendix Figure A.8.

⁴³This prediction is also consistent with the initial counteraction being concentrated among rookie women, a result reported in Table V. If parties hold downward-biased beliefs about women’s ability, the expected ability of rookie women would tend to be lower than the revealed true ability of previously elected incumbent women.

⁴⁴Beaman et al. (2009) find a similar lack of change in (dis)taste for women as a result of reserving seats for women in India. Deep preferences and social norms remain difficult to erode, while beliefs on effectiveness may be much more malleable.

ber of rookie and incumbent women elected in the municipality.⁴⁵ We can see that the additional women elected through the PR arm in treated municipalities are predominantly rookie women. In fact, up until cycle 5, the additional women are solely rookie women. As a consequence, despite parties' counteraction resulting in a lower, albeit not significant, number of female rookie ward councilors in cycle 4, the *total* number of female rookie councilors in treated municipalities is higher overall (column 6). Hence, the quota exposed parties to new female politicians in action.

Table VII - Treatment Effect on Incumbent and Rookie Female Councilors

Election arm:	All political parties					
	Incumbent councilors			Rookie councilors		
	Ward	PR	All	Ward	PR	All
	Female	Female	Female	Female	Female	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Treat at cycle 4 $\times$ Cycle 4	-0.15 (0.12)	-0.01 (0.02)	-0.23 (0.15)	-0.20 (0.15)	0.76*** (0.09)	0.52** (0.20)
Treat at cycle 4 $\times$ Cycle 5	0.10 (0.15)	-0.00 (0.03)	0.02 (0.17)	0.19 (0.15)	0.71*** (0.08)	0.86*** (0.19)
Treat at cycle 4 $\times$ Cycle 6	0.44** (0.19)	-0.01 (0.02)	0.35* (0.20)	0.04 (0.16)	0.71*** (0.08)	0.72*** (0.20)
Treat at cycle 4 $\times$ Cycle 7	0.43** (0.19)	-0.01 (0.02)	0.34* (0.20)	0.30 (0.19)	0.70*** (0.08)	0.99*** (0.22)
Running variable form	ward	council	council	ward	council	council
<i>N</i>	865	865	865	865	865	865
Fix (male+female) total	No	No	No	No	No	No
<i>p</i> -value: Cycle 4=Cycle 5	0.032	0.956	0.039	0.004	0.463	0.042
<i>p</i> -value: Cycle 4=Cycle 6	0.000	0.776	0.000	0.106	0.399	0.235
<i>p</i> -value: Cycle 4=Cycle 7	0.001	0.815	0.001	0.006	0.414	0.021

Note: This table reports the effect of being above the threshold on the number of female incumbent and rookie councilors. The regression specification is given by equation (1). The sample includes bins 1 and 2 and all the parties participating in elections. The outcome variable is the number of female ward and PR councilors in each municipality, by incumbency status—whether they were elected in at least one previous election. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Second, we show that municipalities with greater informational scarcity about female politicians exhibit both stronger initial counteraction and larger subsequent updating of candidate selection strategies. Table VIII examines heterogeneous treatment effects by whether female councilors were ever present before the quota. Comparing columns 1 and 2, we find that parties substitute away from female candidates in the unregulated arm during cycle 4 only in municipalities with no female councilors before the quota. Moreover, only these municipalities exhibit a statistically significant and steady shift in candidate selection from cycle 5 onward. As indicated by the reported

⁴⁵In this table, we do not fix the male+female total because we are interested in the raw number of new signals about women the treatment brings in, as opposed to how parties select candidates between men and women. We examine the overall number of women elected in the municipality, not just from the main parties, since women from other parties or independents also provide information about women's competence.

p-values, the coefficients for cycle 7 are statistically significantly different between municipalities with and without female councilors pre-quota.

Table VIII - Treatment Effect on the Number of Female Ward Candidates by Presence of Women Before the Quota

Had female councilors before quota?	Main political parties			
	No	Yes	No	Yes
	Female (1)	Female (2)	Female (3)	Female (4)
Treat at cycle 4	-0.44* (0.23)	0.59 (0.58)	-0.52** (0.24)	0.28 (0.51)
Treat at cycle 4 × Cycle 5	0.81*** (0.20)	0.57 (0.34)	0.80*** (0.20)	0.47 (0.35)
Treat at cycle 4 × Cycle 6	0.85*** (0.29)	1.10** (0.52)	0.87*** (0.29)	1.07** (0.51)
Treat at cycle 4 × Cycle 7	1.79*** (0.39)	0.27 (0.55)	1.77*** (0.38)	0.21 (0.54)
Running variable form	ward	ward	ward	ward
N	649	215	649	215
Pre-quota gender environment controls	No	No	Yes	Yes
Fix (male+female) total	Yes	Yes	Yes	Yes
p-value: No=Yes, cycle 4	0.099		0.152	
p-value: No=Yes, cycle 5	0.539		0.396	
p-value: No=Yes, cycle 6	0.676		0.737	
p-value: No=Yes, cycle 7	0.024		0.017	

Note: This table reports the effect of being above the threshold on the number of female ward candidates. The regression specification is given by equation (1), except that we replace (Treat at cycle 4)×(Cycle 4) with (Treat at cycle 4) to highlight the change in the treatment effect in cycles 5-7 relative to the effect in cycle 4. The sample includes bins 1 and 2 and is restricted to the two main parties. We divide municipalities into two groups depending on whether at least one woman was elected in the municipality before the introduction of the quota. The pre-quota gender environment controls are described in the main text. We control for the total number (male+female) of ward candidates. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A natural concern to this heterogeneity analysis is that pre-existing differences in the overall gender environment may have jointly affected whether municipalities had female councilors before the quota and how strongly they responded to the quota afterward. Columns 3 and 4 show our results are robust to controlling for these differences directly. We control for: (i) the supply of women, proxied by the number of female ward candidates in cycle 3, when all candidates ran without party affiliation; (ii) party favorability toward nominating women, proxied by the number of female candidates affiliated with the main parties who ran to represent the municipality in the 2004 National Assembly election, two years before the quota; and (iii) local gender characteristics, measured by the female share of the voting-age population, gender gaps in labor force participation and tertiary education rates, and the female share of total housework and family care time.⁴⁶

⁴⁶These variables include all gender-related factors from the balance tests in Appendix Table B.2. As noted there,

Statistical discrimination: biased beliefs New information about women can reduce statistical discrimination in two ways. First, it can correct downward-biased beliefs about women’s average competence. Second, even if beliefs are accurate, information can weaken statistical discrimination by reducing the noisiness of signals of ability.⁴⁷

In this section, we present evidence consistent with the presence of downward-biased beliefs: not only the number of women experienced but also their revealed performance matters. If exposure to women only reduced signal noise, the response would not depend so directly on whether the first women in office performed better than expected. By contrast, when beliefs are downward-biased, parties should update faster toward the truth where elected women’s revealed ability exceeds prior expectations by a larger margin.

Table IX reports a heterogeneity analysis of the post-cycle-4 treatment effects by the revealed quality of the first female PR councilors elected in cycle 4. Revealed quality is proxied by the number of bills proposed as lead proposer during the cycle-4 term.⁴⁸ This measure captures a salient dimension of councilor performance: proposing bills is one of the main observable ways through which councilors exercise legislative initiative. It is also used in public accountability efforts; for example, a long-standing Korean civic watchdog has criticized councilors with no bill proposals over extended periods (CCEJ, 2024).

We find that the shift toward female ward candidates emerges earlier and is stronger when the first elected PR women in the municipality show higher revealed performance. Columns 1 and 2 divide municipalities into above- and below-median groups based on the average number of bills proposed by these women. The treatment effect is statistically significant from cycle 6 onward, but only among the above-median group. For example, column 1 shows that, by cycle 7, treated municipalities have 1.18 more female ward candidates than control municipalities. Since the total (male+female) number of ward candidates is held fixed, this corresponds to 1.18 fewer male ward candidates.

While columns 1-2 provide a transparent introduction to the performance measure, columns 3-6 are more tightly linked to the model by capturing the gap between revealed ability and parties’ prior expectations. If a woman performs well, but that performance was fully anticipated based on prior signals of ability, there is little scope for belief updating. Using the sample of all elected PR women, we first estimate an individual-level regression of the number of bills proposed on

labor force participation, tertiary education, and housework and family care time are observed only at the province level.

⁴⁷In the model, a lower σ_s^2 increases perceived ability for candidates with positive signals ($s_i > 0$). Positive signals are the relevant case for candidates, because they are positively selected.

⁴⁸We focus on lead-proposer activity because proposer information often appears in the form “Councilor [name] and [number] others”—making the full set of co-proposers inconsistently observed.

education-level dummies and a dummy for political experience (having served as a party member), while controlling for municipality and cycle fixed effects.⁴⁹ We then take the municipality-level average of the residuals for the first PR women, and divide municipalities into above- and below-median groups. This residualized measure, therefore, captures the surprise component of the first PR women's revealed quality. Columns 5-6 additionally control for the same pre-quota gender-environment controls as in Table VIII, to account for pre-existing differences in the supply of women, parties' willingness to nominate women, and local gender characteristics.

Table IX - Treatment Effect on the Number of Female Ward Candidates by Revealed Quality of the First Female PR Councilors

First PR women's quality in terms of:	Main political parties					
	Raw no. of bills		Residualized no. of bills			
	Above	Below	Above	Below	Above	Below
	Female	Female	Female	Female	Female	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Treat at cycle 4 $\times$ Cycle 5	0.23 (0.39)	0.32 (0.34)	0.25 (0.32)	0.43 (0.46)	0.46 (0.33)	0.13 (0.38)
Treat at cycle 4 $\times$ Cycle 6	0.79* (0.45)	0.00 (0.37)	0.71* (0.38)	0.30 (0.51)	0.92** (0.39)	0.05 (0.40)
Treat at cycle 4 $\times$ Cycle 7	1.18** (0.50)	0.59 (0.43)	1.28*** (0.45)	0.62 (0.53)	1.47*** (0.45)	0.31 (0.42)
Running variable form	ward	ward	ward	ward	ward	ward
<i>N</i>	310	286	292	292	292	289
Pre-quota gender environment controls	No	No	No	No	Yes	Yes
Fix (male+female) total	Yes	Yes	Yes	Yes	Yes	Yes
<i>p</i> -value: Above=Below, cycle 5	0.874		0.754		0.522	
<i>p</i> -value: Above=Below, cycle 6	0.175		0.520		0.120	
<i>p</i> -value: Above=Below, cycle 7	0.371		0.337		0.059	

Note: This table reports the treatment effects on the number of female ward candidates, dividing municipalities into two groups by whether the revealed quality of the PR women elected in cycle 4 is below or above the median. Quality is proxied by the number of bills proposed as the lead proposer during the cycle-4 term. Columns 1-2 divide municipalities by the raw number of bills. Columns 3-6 divide municipalities by the number of bills residualized by education and political experience to capture only the surprise component of revealed quality. The pre-quota gender environment controls are the same as in Table VIII. The sample includes bins 1 and 2 and cycles 5-7 and is restricted to municipalities with female PR councilors from main parties in cycle 4. We control for the total number (male+female) of ward candidates. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Across all three above-below comparisons, the estimated treatment effects are generally larger in the above-median group than in the below-median group, and are statistically significant only in the above-median group. In the final comparison (columns 5 and 6), the reported *p*-value indicates that the cycle-7 treatment effects among the above- and below-median groups are statistically significantly different at the 10% level.

⁴⁹Having a high school degree, undergraduate degree, and graduate degree is increasingly associated with the number of bills proposed, supporting education as a signal of ability.

In sum, the results in Table IX support the interpretation that parties held downward-biased beliefs about women’s ability and updated more strongly when early female councilors revealed higher ability than expected.

V.D Decomposition: quantifying party learning

Section V.C has presented evidence consistent with the view that the increase in female ward candidates in cycles 5-7 reflects party learning. Section V.E discusses why alternative mechanisms are less likely to drive this pattern. Before turning to alternative mechanisms, we use an Oaxaca-Blinder-style decomposition to quantify how much party learning accounts for the total increase in female ward candidates. Given the role of incumbency advantage documented in Section V.A, we also quantify this channel as a benchmark. Appendix G provides the details.

Because the increase in female ward candidates comes from both incumbents and rookies (Table V), we decompose the total effect into incumbent and rookie contributions. For incumbents, we separate a *female incumbent stock effect*, which captures the mechanical increase in female incumbent candidates due to the larger pool of previously elected women, from a *renomination effect*, which captures whether previously elected women are more likely to reappear as ward candidates. The former provides our incumbency benchmark. The latter may partly reflect learning, but may also capture individual-specific advantages such as name recognition or local networks.

For rookies, we separate a *rookie number effect*, which captures changes in the total number of rookie candidates regardless of gender,⁵⁰ from a *rookie female share effect*, which captures whether women make up a larger share of rookie candidates. Conditional on the relative supply of male and female rookies remaining unchanged, which we show in Section V.E, the rookie female share effect is the margin most closely tied to learning about women’s average competence.

The results in Appendix Table G.1 show that mechanical incumbency explains a large share of the increase in female ward candidates, but that the rookie female share effect also grows over time. The female incumbent stock effect accounts for 27% of the total increase in cycle 5, 56% in cycle 6, and 50% in cycle 7. The total rookie contribution rises from 0.02 in cycle 5 to 0.38 in cycle 7. Notably, this increase is driven by the rookie female share effect—the margin most closely tied to party learning—indicating that the increase in female rookies does not come from an expansion of rookies in general, but from women making up a larger share of rookie candidates. The rookie female share effect accounts for 17% of the total effect in cycle 5, 24% in cycle 6, and 34% in

⁵⁰Incumbents and rookies differ in many ways unrelated to gender. For example, incumbents have an electoral advantage; incumbents’ ability is better known, so parties face less uncertainty; and incumbents may have stronger bargaining power inside the party.

cycle 7. Thus, relative to the mechanical incumbency benchmark, this margin is economically meaningful and increasingly important over time.

V.E Alternative mechanisms

In this section, we discuss alternative mechanisms that can deliver a similar pattern of initial counteraction and subsequent reversal in women’s nominations. We consider two non-party drivers: shift in voter preferences or in the supply of candidates, and two alternative within-party drivers: difficulty in finding qualified women and increase in women’s power within the party.

We test whether these mechanisms change discontinuously at the council-size threshold. Although the nationwide quota system may affect these margins more broadly, our goal is to assess whether they can explain the discontinuous change in female nominations identified by our research design.

Change in the supply of women The supply or competence of potential female candidates may have evolved differently in treatment municipalities, prompting parties to adjust the gender composition of their candidate slates. For instance, female councilors introduced by the quota may have served as role models, increasing women’s political ambition, lowering perceived entry costs, or improving expectations of electoral success. This role-modeling effect could have been stronger in treatment municipalities, where exposure to women in office was greater. Because we do not observe the full pool of potential candidates, we use two indirect tests.

First, if the pool of female candidates improved, or merely expanded in size, parties would have access to a larger number of qualified women from the upper end of the distribution, which should be reflected in the characteristics of nominated candidates. Appendix Figure A.7 shows that the education distribution of both male and female candidates shifts rightward from cycle 4 to cycle 7, suggesting an overall improvement in candidate quality. Crucially, however, this improvement is common to both treatment and control municipalities, providing no evidence of a differential change in the pool of candidates at the threshold.⁵¹

Second, a differential expansion in the pool of potential candidates should also be visible among independent candidates—individuals with no party affiliation who run on their own initiative in the ward arm. Supply-side mechanisms such as role modelling should be particularly strong for independent candidates, since they bypass party screening, potential bias, and strategic

⁵¹A formal estimation of the treatment effect on the gender gap in characteristics related to candidate quality—multifaceted measures of education and political experience—can be found in Appendix Table B.12. A weak difference (statistically significant at the 10% level) between treatment and control is present in the gender gap in political experience at cycle 4. Crucially, though, there is no evidence of a differential evolution over time. This is therefore unlikely to explain the reversal dynamics in cycles 5-7.

limits on candidate entry to avoid vote-splitting. Appendix Table B.13 shows, however, that the treatment effect on the number of female independent ward candidates is statistically insignificant across all election cycles, with no consistent directional pattern.

The evidence suggests that candidate supply may have changed broadly, but there is little indication that it changed differentially above and below the threshold.

Change in voter preferences for women A second alternative centers on voters. Repeated exposure to female councilors may lead voters in treated municipalities to update favorably about women's competence or develop a stronger taste for female candidates. Parties could then adjust nominations toward women in response to these electoral incentives.

To examine this possibility, we estimate the treatment effect on the gender gap in vote shares among ward candidates over time. Specifically, we estimate an individual-level version of equation (1), where the outcome is a ward candidate's vote share and the regressor of interest is the treatment dummy interacted with the female dummy.

Vote shares are a conditional equilibrium outcome: we observe them only for candidates whom parties choose to nominate, and treatment may itself affect candidate nomination, ward placement, and ballot position. The estimates, therefore, do not isolate voter responses separately from parties' candidate-selection and placement decisions.

Isolating voter demand is also complicated by the party-centered nature of ward voting. A simple prediction of a candidate's vote share based on their party's popularity in the PR arm and their ballot position has a correlation of 0.90 with actual vote share. Therefore, party nomination and placement decisions are central to candidates' electoral performance, limiting how directly vote shares can be interpreted as voter demand. Still, party-centered voting does not rule out candidate-level differences in vote shares: we do detect a baseline gender gap in vote shares through the statistically significant female dummy.

For our purposes, conditional vote shares remain informative. If voters were leading the process—becoming more favorable toward female candidates before, or more strongly than, parties adjusted nominations—we would expect a positive treatment effect on women's vote shares relative to comparable men. If parties moved ahead of voters, we would expect a negative effect.

We find no evidence of a treatment effect on women's relative vote shares, either immediately after the quota or over time. Columns 1-5 of Appendix Table B.14 show that the coefficients on the interaction between the treatment dummy and the female dummy are statistically insignificant, even as we gradually add controls for individual characteristics (age, education, political experience, incumbency status) and electoral conditions (position on the ballot, number of competitors).

These results suggest that the reversal in parties' candidate nominations at the threshold is un-

likely to be explained by an additional voter-side shift in support for women at that same threshold. As with candidate supply, this does not imply that voter beliefs or preferences remained stagnant as the quota system unfolded. Voter-side and party-side changes may have moved broadly in parallel, but our main findings are consistent with a party-learning interpretation in which party beliefs and party selection are the key margins.

Difficulty in finding qualified women Another explanation for the initial counteraction to quotas that is unrelated to statistical discrimination is a shortage of qualified women. A party that is short of women to choose candidates from might have to move women from the ward arm to the PR arm just to fulfill the quota requirements.

If this were true, we should find evidence that parties in treatment municipalities have greater trouble finding female candidates. To test this hypothesis, we define a party as “unconstrained” in female candidate choice if it nominates strictly more female candidates in the PR arm than the quota requirement.⁵² The statistically insignificant coefficients in Appendix Table B.15 show that parties above the threshold are not more constrained in finding women, particularly in cycle 4.

Women becoming more powerful Lastly, the faster growth of women in treatment municipalities may also reflect a compounding effect: an initial, even modest, increase in women strengthens their influence in the candidate nomination process and further brings in more women. This dynamic might operate through several channels. For example, a growing number of lab and field experiments provide evidence that group composition affects women’s perceived expertise, influence, and willingness to take on leadership roles.⁵³ Alternatively, gender quotas can expand the supply of qualified women available for leadership positions (O’Brien and Rickne, 2016).

Although we cannot completely rule out this mechanism, the evidence does not support it as the *sole* driver of the observed changes in party strategies over time. Three pieces of evidence point in this direction.

First, we find that individual women are unlikely to gain persistent power over time. Columns 3-5 of Appendix Table F.1 show that the quota women who entered councils in cycle 4 do not remain for long: marginal winners in close PR races were more likely than marginal losers to re-run in cycle 5, but not in cycles 6 or 7. In fact, re-candidacy is generally limited to one or two times across the full sample, regardless of gender. It is therefore unlikely that these women consolidate influence by rising through party ranks over time.

Second, we test whether women’s growing numbers raise their collective influence within par-

⁵²For example, the minimum number of women a party needs to include in the party list is 1 if the number of PR seats for the municipality is 1 or 2, and 2 if the number of PR seats is 3 or 4.

⁵³Examples include Coffman (2014); Bordalo et al. (2019); Born et al. (2020); Karpowitz et al. (2021); Dupas et al. (2021); Miller and Sutherland (2021).

ties. In a suggestive exercise (Appendix Table B.16), we find that the increase in female rookie candidates is stronger among parties with more female councilors elected from their own party in the previous cycle, consistent with a role for women’s within-party influence. However, a similar, although smaller, increase in female rookies appears when parties are exposed to more women elected from *other parties* in the same municipality.⁵⁴ Female councilors from other parties should not directly increase women’s influence within the focal party, but may still provide information about women’s competence. The similarity between the own-party and other-party patterns, therefore, suggests that broader exposure or learning also plays an important role. Further details and caveats are provided in Appendix H.1.

Third, as discussed in Section V.C, the increase in female candidates is concentrated in competitive wards where electoral outcomes are uncertain, rather than in party strongholds. This suggests that, even if women have gained some influence across election cycles, it has not been sufficient to redirect party strategy toward securing more female seats. These findings align with Spaziani (2022), who shows that gender quotas in Italian municipal elections had limited success in bringing women into positions of power.

To sum up, stronger quota requirements in the PR arm trigger changes in party nomination strategies in the unregulated ward arm. These results are consistent with statistical discrimination that weakens over time, whereas we do not find support for a discontinuous change in voter preferences, candidate supply, or women’s within-party influence at the threshold. This explanation matches interpretations provided by South Korean political scientists. According to Shin (2014) and Yoon and Shin (2017), parties viewed the quotas merely as a rule to comply, without any intention of making long-term commitments to women. However, the quota ultimately fostered positive impacts on female representation in the long run. Beyond giving women the incumbency advantage, confidence, and resources to run again, the quota also allowed parties to revise their perceptions of women’s capabilities as legislators. Consistent with this interpretation, by 2012 only half of the women elected in the ward arm had previously held PR seats.

VI CONCLUSION

This paper highlights that in countries where women are significantly underrepresented, a lack of information and biased beliefs about women’s competence can lead parties to engage in statistical discrimination during candidate selection. This creates a vicious cycle that perpetuates the initial

⁵⁴Parties with more women elected from their own ranks are not simply the same parties that are exposed to more women elected from other parties. In the data, the number of own-party female councilors and the number of other-party female councilors are negatively correlated.

gender imbalance.

In such contexts, gender quotas are needed not only to ensure equal representation but also from an efficiency standpoint. If informational failures cause parties to nominate a suboptimally low number of female politicians, quotas can aid in rectifying this inefficiency. Provided that they are designed to successfully bring in qualified women, quotas offer parties an opportunity to acquire new information about women's abilities. Once the learning process takes off, the policy itself might not be needed.

By comparing municipalities on either side of council-size cutoffs, our estimates capture the causal effect of *stronger quota requirements* on party nomination strategies, but they may not capture the full range of the effects of the nationwide quota system. Female representation also grew substantially in municipalities subject to weaker quota requirements, suggesting that the quota may have induced broader changes that our local comparison cannot capture. Consistent with this, suggestive evidence—though not causal—points to an overall improvement in candidate quality, with less qualified men replaced by more qualified women.

The phenomena discussed in this paper are not unique to the context studied. Although parliamentary gender quotas are now widespread, many countries—such as Egypt, Liberia, Mauritius, Sao Tome and Principe, Sierra Leone, and Sri Lanka—still have no such measures in place and continue to have low levels of female representation. The findings also inform quota design in non-political settings, such as company boards, where incumbents are similarly, if not more, male-dominated and attitudes often male-biased (Appendix Figure A.3). More broadly, the implications extend beyond women to other underrepresented groups facing similar informational frictions.

What remains to be crystallized is precisely which aspect of women's competence parties learn about. Two pieces of evidence offer tentative guidance: the fact that parties increasingly place women in competitive wards suggests learning about women's electability, while the stronger reversal where women propose more bills suggests learning about women's ability to participate actively in the council. Yet the evidence does not allow us to go further. Are parties learning about women's campaigning skills, their ability to meet the demands of the electorate, their loyalty to the party, or their effectiveness as legislators? This paper is part of a broader research agenda aimed at understanding how gender quotas can initiate a gradual process of learning. In future work, we will examine interactions among councilors, as recorded in council meeting transcripts, to study both the mechanisms behind this process and its consequences for local policymaking.

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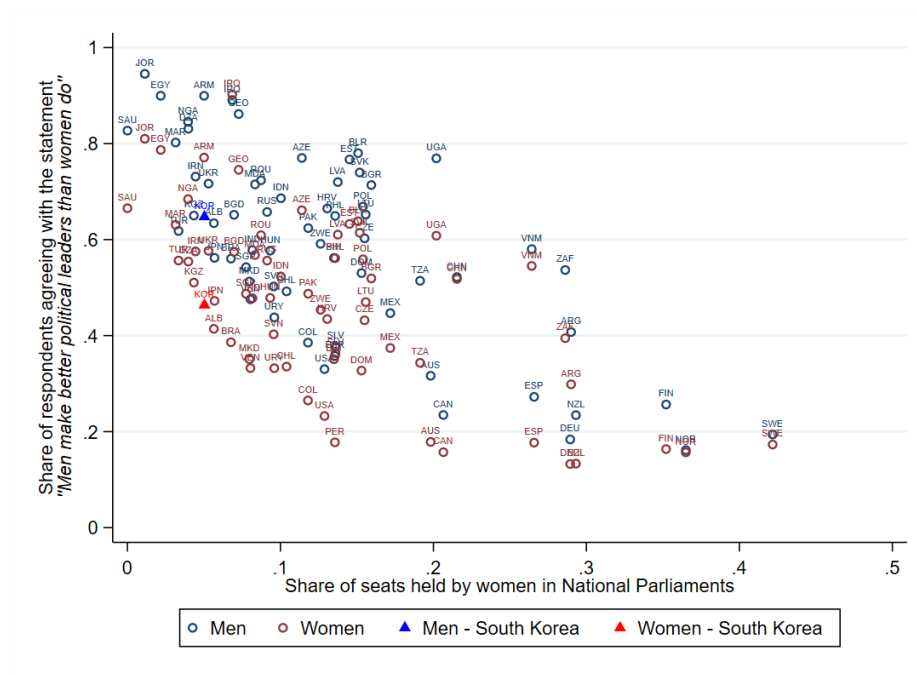
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Online Appendix

Appendix A ADDITIONAL FIGURES

Figure A.1 - Female Share in National Parliaments and Attitudes Towards Women (1995-2004)



Note: The graph depicts the cross-country correlation between attitudes towards women as political leaders and female representation in National Parliaments in the period just before the introduction of the quota in South Korea (2005). On the y-axis, attitudes towards women are measured as the share of respondents (men in blue, women in red) that agree with the statement “Men make better political leaders than women do” in the World Values Survey waves 3 (1995-1998) and 4 (1999-2004) (Inglehart et al., 2018). Higher values indicate attitudes more favorable towards men. The x-axis displays the share of seats held by women in national parliaments according to the Inter-Parliamentary Union (2004), average years 1995-2004. South Korean respondents are indicated with a triangle.

Figure A.2 - Ballot Papers in Municipal Council Elections

BALLOT PAPER
"Ward councilors"
(Multi-member plurality vote)

Municipality A - Ward X

1-a	Party 1	Cand. i	
1-b	Party 1	Cand. ii	
2-a	Party 2	Cand. iii	✓
2-b	Party 2	Cand. iv	
3	Party 3	Cand. v	

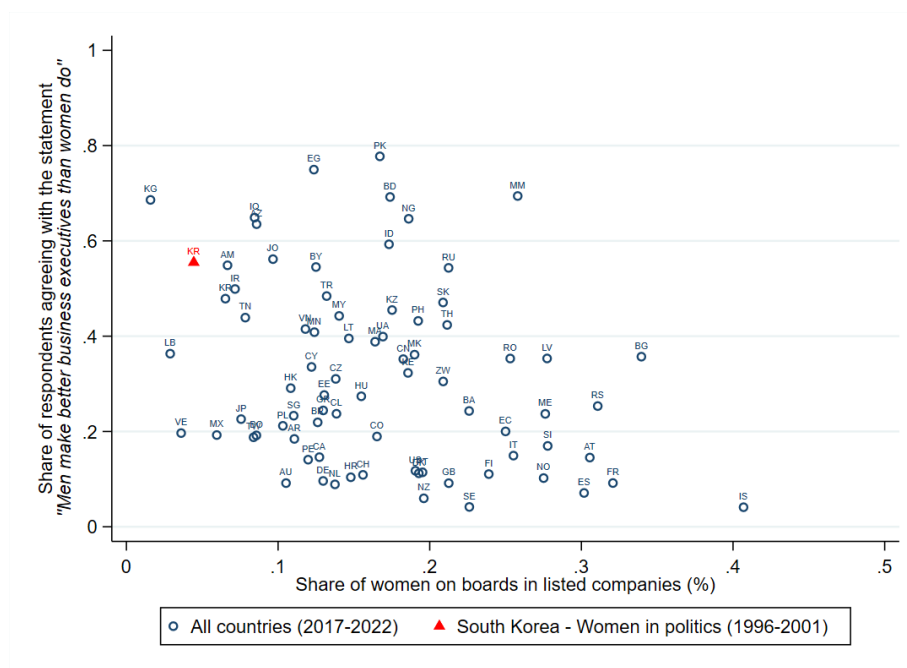
BALLOT PAPER
"PR councilors"
(Party-list proportional representation)

Municipality A

1	Party 1	✓
2	Party 2	
3	Party 3	
4	Party 4	
5	Party 5	

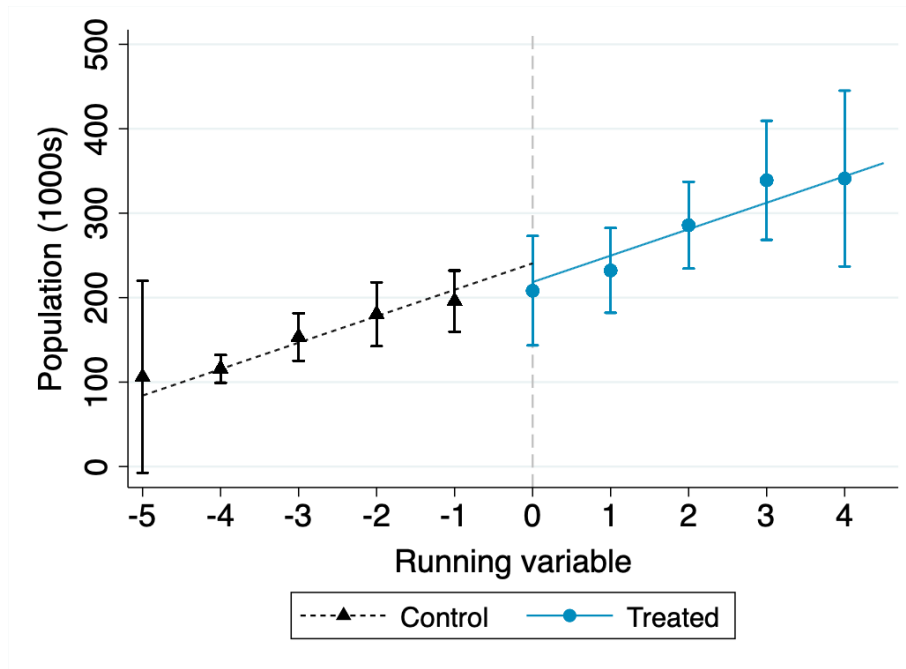
Note: This figure illustrates the ballot papers for a voter residing in ward X of municipality A. The left is used to vote for ward councilors and the right for PR councilors. The ticks indicate how a voter might vote. For the ward arm, a voter votes for an individual candidate. Votes are aggregated at the ward level, and if there are n seats allotted for a ward, the top n candidates winning the most votes get elected. For the PR arm, a voter votes for a party, and the party vote share aggregated at the municipality level determines the number of PR seats a party wins. Then, the PR candidates of the party get elected according to their order on the pre-registered party list.

Figure A.3 - Female Representation on Boards in Listed Companies and Attitudes Towards Women (2017-2022)



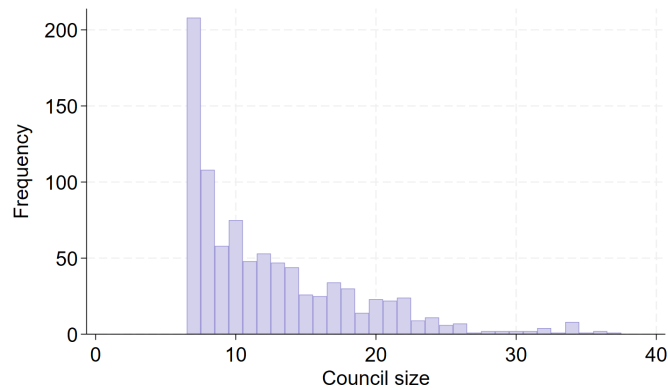
Note: The graph depicts the cross-country relationship between attitudes towards women as business executives and female representation in listed companies' boards. On the y-axis, attitudes towards women are measured as the share of respondents agreeing with the statement "Men make better business executives than women do" in the World Values Survey, wave 7 (2017-2022) (Haerpfer et al., 2020). Higher values indicate attitudes more favorable towards men. The x-axis displays the share of women on the boards of listed companies in 2022, calculated using the ORBIS database (Bureau van Dijk, 2022). Red triangle (▲): for comparison, we plot the situation of women in political bodies in South Korea just before the introduction of the quota. Attitudes towards women as politicians in South Korea are measured as the share of respondents agreeing with the statement "Men make better political leaders than women do" in the World Values Survey, waves 3 (1996) and 4 (2001) (Inglehart et al., 2018). Female representation in politics is measured as the share of seats held by women in the South Korean National Assembly (1997-2001 average) according to the Inter-Parliamentary Union (2004) .

Figure A.4 - Balance Test: Treatment Effect on Population Size



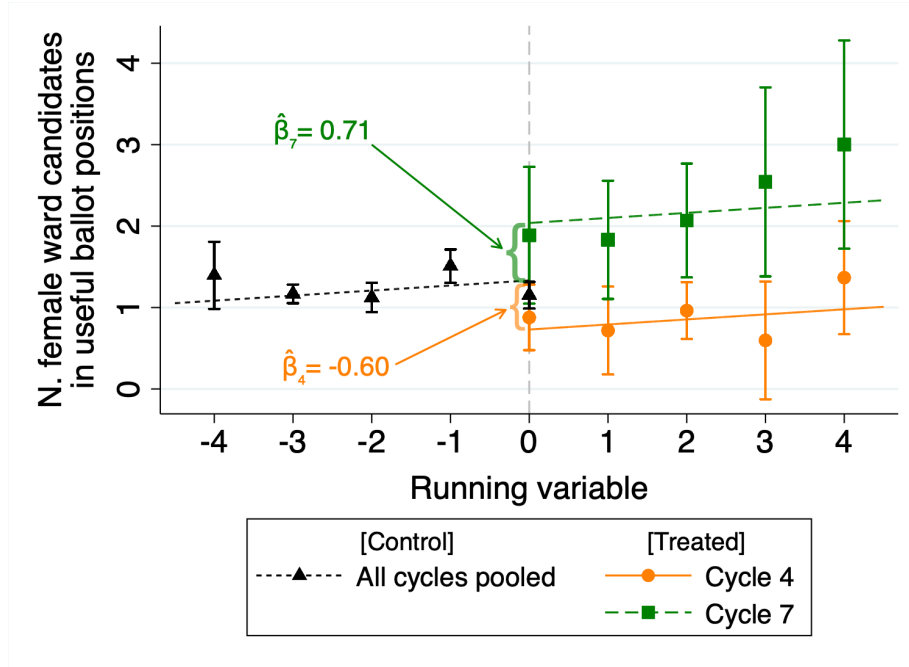
Note: This figure is the graphical counterpart to the results in Panel A, column 1 of Table B.2. It plots the average population size of municipalities in cycle 4, residualized and re-centered for bin effects (α_b), for each value of the running variable (x_{cb}). The error bars indicate 95% confidence intervals of each mean value. The short-dashed and solid lines represent $f(x_{cb})$. Visually, there is no discontinuous jump at $x_{cb} = 0$, indicating that population size evolves smoothly with council size, with no discontinuity at the cutoff.

Figure A.5 - Histogram of Council Size



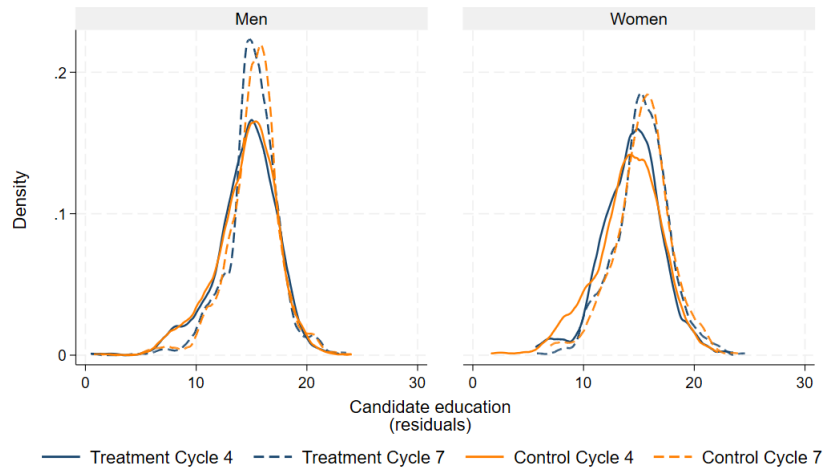
Note: The figure displays the number of councils for each council size, in election cycles 4, 5, 6, and 7.

Figure A.6 - Treatment Effect on the Number of Female Ward Candidates in Useful Ballot Positions



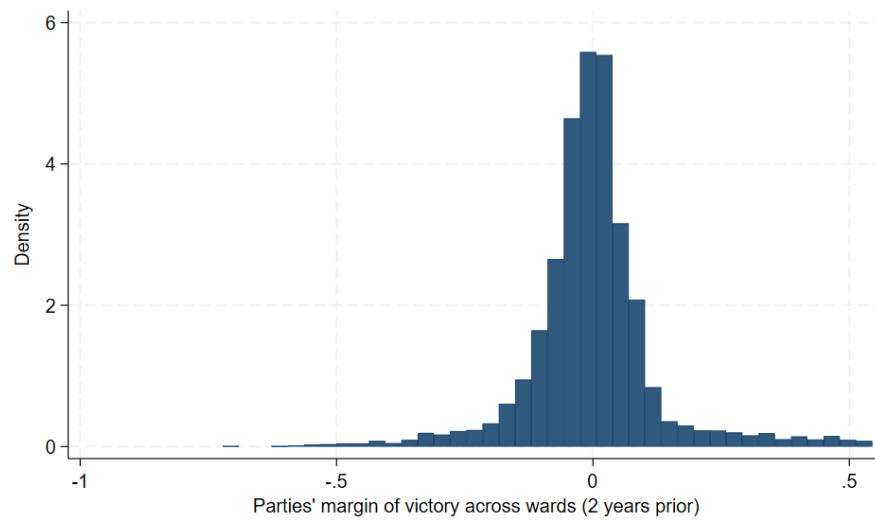
Note: This figure is the graphical counterpart to the results in Table III, column 12, of estimating equation (1). It plots, for each value of the running variable $\tilde{x}_{cb}$, the average number of female ward candidates from the two main parties placed in “useful” ballot positions: position 1 in wards electing 1-2 councilors and positions 1-2 in wards electing 3-4 councilors. The outcome is residualized and re-centered for bin and cycle effects (α_b and α_t), the contemporaneous number of ward seats, and the total (male+female) number of main-party ward candidates in useful positions. The error bars indicate 95% confidence intervals of each mean value. Only the treatment effects for cycles 4 and 7 are shown for presentational clarity. The lines (short-dashed, dashed, and solid) represent $f(\tilde{x}_{cb})$. Because the running variable is based on the ward arm ($\tilde{x}_{cb}$), it ranges from -4 to 0 for control municipalities, in contrast to x_{cb} , the running variable in the RD plot in Figure II Panel B, which ranges from -5 to -1.

Figure A.7 - Evolution of Candidate Education



Note: This figure illustrates the evolution of the education of female and male candidates over time. It plots, separately for treatment and control municipalities and for election cycles 4 and 7, the years of schooling of female and male candidates, residualized for year of birth, municipality fixed effects (capturing the effect of bin fixed effects and the baseline running variable), and the total number of MPs as in equation (1). Comparing treatment and control municipalities therefore, yields an estimate of the treatment effect in each election cycle. The residuals are re-centered using the overall average years of schooling. The figure includes only bins 1 and 2 and only candidates from the two main parties. P-values for tests of Treatment = Control (equality of means/distributions): Cycle 4 men, 0.94/0.67; Cycle 4 women, 0.26/0.29; Cycle 7 men, 0.58/0.13; Cycle 7 women, 0.68/0.65. P-values for tests of Cycle 4 = Cycle 7 (equality of means/distributions): 0.00/0.00.

Figure A.8 - Parties' Margin of Victory in Previous Elections Across Wards



Note: This figure plots the party's popularity in the ward. The party's margin of victory is calculated as the difference in the party's vote share with that of the most popular competitor in the ward in the immediately preceding election (the National Assembly election, 2 years before each municipal council election). We demean the party's margin of victory by election $\times$ party, to address electoral swings occurring at the national level. The sample includes bins 1 and 2.

Appendix B ADDITIONAL TABLES

Table B.1 - Shares of Candidates and Councilors by Party Affiliation

	Candidates		Councilors	
	Ward	PR	Ward	PR
Election Cycle 1				
Independent	100%	-	100%	-
Election Cycle 2				
Independent	100%	-	100%	-
Election Cycle 3				
Independent	100%	-	100%	-
Election Cycle 4				
Independent	41%	-	11%	-
Progressive Party	16%	29%	20%	18%
Conservative Party	24%	47%	54%	64%
Average N. parties	4.42	2.84	2.45	1.60
Election Cycle 5				
Independent	32%	-	14%	-
Progressive Party	21%	30%	32%	41%
Conservative Party	33%	49%	43%	44%
Average N. parties	4.50	2.88	2.63	1.67
Election Cycle 6				
Independent	34%	-	13%	-
Progressive Party	24%	38%	37%	40%
Conservative Party	35%	55%	48%	59%
Average N. parties	3.67	1.96	2.32	1.44
Election Cycle 7				
Independent	20%	-	9%	-
Progressive Party	32%	43%	54%	66%
Conservative Party	29%	38%	34%	32%
Average N. parties	4.68	2.67	2.45	1.43

Note: This table illustrates the shares of candidates and councilors by party affiliation in each municipal council and election cycle. The affiliation is reported separately for candidates/councilors nominated in the two different arms—ward and PR. Three affiliations are reported: the two main parties—Progressives and Conservatives (>70% of party-nominated candidates)—and independents, i.e. candidates that run with no party affiliation. The residual category (omitted) includes all other parties. “Average N. parties” indicate the average number of distinct party affiliations represented, where the group of independents are counted as one party.

Table B.2 - Balance Tests on Pre-Determined Characteristics

Panel A: Population characteristics						
	Population	Voting-age population				Households
	Total (1,000s) (1)	Total (1,000s) (2)	Male (1,000s) (3)	Female (1,000s) (4)	Female share (%) (5)	Total (1,000s) (6)
Treat at cycle 4	-22.2 (31.1)	-15.9 (22.7)	-7.2 (11.3)	-8.7 (11.4)	-0.56 (0.35)	-5.0 (11.1)
Running variable form	council	council	council	council	council	council
Mean dependent variable	189.14	144.25	71.01	73.24	50.71	69.85
Panel B: Time use and employment						
	Housework + family care (min./day)		Labor force participation (%)		Unemployment rate (%)	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)
Treat at cycle 4	1.40 (1.40)	0.07 (3.22)	0.74 (0.81)	0.68 (1.07)	-0.99 (0.62)	-1.02 (0.84)
Running variable form	council	council	council	council	council	council
Mean dependent variable	41.11	228.25	71.83	45.32	8.00	13.03
Panel C: Education attainment						
	≤ Elementary school (%)		Middle school (%)		High school (%)	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)
Treat at cycle 4	1.51 (1.81)	1.58 (2.89)	0.06 (0.46)	-0.15 (0.28)	0.01 (1.09)	0.26 (1.33)
Running variable form	council	council	council	council	council	council
Mean dependent variable	12.94	26.05	10.74	11.46	38.00	35.48
	Technical university (%)		University (%)		Graduate studies (%)	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)
Treat at cycle 4	-0.20 (0.31)	-0.48 (0.46)	-1.30 (1.58)	-1.11 (1.48)	-0.09 (0.49)	-0.10 (0.26)
Running variable form	council	council	council	council	council	council
Mean dependent variable	11.46	9.98	23.21	15.48	3.66	1.55
Panel D: Political leaning, economic, and ward division characteristics						
	Past vote share by party (p.p.)		Budget (billion KRW)		Ward characteristics	
	Progressive (1)	Conservative (2)	Total (3)	Council expenses (4)	Num of wards (5)	Seats per ward (6)
Treat at cycle 4	-0.45 (6.77)	0.45 (6.77)	50.75 (86.78)	0.02 (0.05)	-0.24 (0.18)	0.17 (0.11)
Running variable form	council	council	council	council	ward	ward
Mean dependent variable	55.02	44.98	450.33	0.88	5.17	2.64

Note: This table shows the absence of discontinuities in pre-determined characteristics at the threshold in cycle 4. The regression specification follows equation (1). The sample consists of bins 1 and 2 in election cycle 4. $N = 218$ for all columns. Panel A: residents, residents of voting age, and number of households by gender (source: National Election Commission, 2018b). Panel B: province-level information on time spent on unpaid domestic or care-giving services (Statistics Korea, 2004) and labor force participation and unemployment rates (Statistics Korea, 2005). Panel C: province-level shares of men and women by education attainment (Statistics Korea, 2005). Panel D: vote share of main parties in the PR arm of the National Assembly Election of 2004, municipal government expenditure (Local Finance Disclosure System, 2005), and number and size of wards in each municipality (National Election Commission, 2018b). Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.3 - Robustness: Treatment Effect on the Number of Candidates and Councilors, for Various Bandwidths

	Ward						PR						All					
	Candidates			Councilors			Candidates			Councilors			Candidates			Councilors		
	Male (1)	Female (2)		Male (3)	Female (4)		Male (5)	Female (6)		Male (7)	Female (8)		Male (9)	Female (10)		Male (11)	Female (12)	
Panel A: distance ≤ 4 (baseline specification)																		
Treat	1.17*	0.53		-0.28	0.28		0.53***	1.12***		0.08	0.71***		0.66	1.50***		-0.89***	0.89***	
	(0.68)	(0.32)		(0.22)	(0.22)		(0.15)	(0.18)		(0.06)	(0.07)		(0.80)	(0.45)		(0.27)	(0.27)	
N	865	865		865	865		865	865		865	865		865	865		865	865	
Panel B: distance ≤ 3																		
Treat	1.05	0.56*		-0.29	0.29		0.45***	1.13***		0.07	0.71***		0.42	1.58***		-0.92***	0.92***	
	(0.69)	(0.32)		(0.22)	(0.22)		(0.15)	(0.19)		(0.06)	(0.07)		(0.82)	(0.46)		(0.28)	(0.28)	
N	821	821		821	821		821	821		821	821		821	821		821	821	
Panel C: distance ≤ 2																		
Treat	1.77**	0.60*		-0.20	0.20		0.43**	1.16***		0.01	0.72***		1.62	1.73***		-0.83**	0.83**	
	(0.79)	(0.34)		(0.24)	(0.24)		(0.19)	(0.23)		(0.07)	(0.09)		(1.02)	(0.51)		(0.33)	(0.33)	
N	524	524		524	524		524	524		524	524		524	524		524	524	
Panel D: distance ≤ 1																		
Treat	1.35	0.86**		-0.29	0.29		0.62**	0.89***		-0.00	0.64***		1.01	2.29***		-1.15**	1.15**	
	(0.88)	(0.39)		(0.27)	(0.27)		(0.28)	(0.31)		(0.11)	(0.13)		(1.36)	(0.70)		(0.46)	(0.46)	
N	324	324		324	324		324	324		324	324		324	324		324	324	
Panel E: distance = 0																		
Treat	1.01	0.69		-0.21	0.21		0.59**	0.70***		0.01	0.59***		0.60	1.03**		-0.59	0.59	
	(0.97)	(0.41)		(0.29)	(0.29)		(0.23)	(0.21)		(0.08)	(0.10)		(1.05)	(0.45)		(0.35)	(0.35)	

Note: This table tests the robustness of the estimated treatment effect using regression equation (1) for different bandwidths. Bandwidths refer to the distance to the threshold. To illustrate, the council sizes for which distance equals 0 are 10, 11, 20, and 21, while the council sizes for which distance equals 1 are 9, 12, 19, and 22. The sample includes bins 1 and 2 and includes all election cycles. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.4 - Robustness: Treatment Effect ($f(x)$): Linear Interaction)

	All political parties							
	Candidates				Councilors			
	Ward		PR		Ward		PR	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Treat at cycle 4 $\times$ Cycle 4	3.50*** (0.99)	-0.50 (0.36)	0.97*** (0.23)	0.94*** (0.22)	0.64** (0.28)	-0.48** (0.24)	0.10 (0.08)	0.76*** (0.09)
Treat at cycle 4 $\times$ Cycle 5	1.30 (0.87)	0.28 (0.35)	0.67*** (0.17)	1.19*** (0.22)	0.05 (0.31)	0.17 (0.25)	0.10 (0.07)	0.71*** (0.08)
Treat at cycle 4 $\times$ Cycle 6	-0.40 (0.75)	0.51 (0.37)	0.28* (0.15)	1.01*** (0.22)	-0.05 (0.35)	0.39 (0.28)	0.09 (0.06)	0.71*** (0.08)
Treat at cycle 4 $\times$ Cycle 7	0.39 (0.76)	1.10** (0.46)	0.24 (0.16)	1.25*** (0.21)	-0.11 (0.37)	0.69** (0.31)	0.05 (0.06)	0.72*** (0.08)
Running variable form	ward	ward	council	council	ward	ward	council	council
N	865	865	865	865	865	865	865	865

Note: This table tests the robustness of the estimated treatment effect of equation (1) to the choice of the functional form for the relationship between council size and the outcome. In this table, $f(x) = \pi_1 x + \pi_2 x \cdot Treat$. The baseline results are, in order, in columns 4-5 and 1-2 of Table III, and 1-2 and 3-4 of Table II. The sample includes bins 1 and 2. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.5 - Robustness: Treatment Effect ($f(x)$): Quadratic)

	All political parties							
	Candidates				Councilors			
	Ward		PR		Ward		PR	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Treat at cycle 4 $\times$ Cycle 4	3.53*** (0.97)	-0.43 (0.35)	1.00*** (0.23)	0.89*** (0.22)	0.58** (0.27)	-0.43* (0.23)	0.10 (0.08)	0.77*** (0.09)
Treat at cycle 4 $\times$ Cycle 5	1.34 (0.85)	0.35 (0.35)	0.70*** (0.18)	1.14*** (0.23)	-0.01 (0.30)	0.22 (0.24)	0.11 (0.07)	0.71*** (0.09)
Treat at cycle 4 $\times$ Cycle 6	-0.37 (0.73)	0.57 (0.37)	0.31* (0.16)	0.95*** (0.22)	-0.11 (0.34)	0.43 (0.28)	0.09 (0.07)	0.71*** (0.09)
Treat at cycle 4 $\times$ Cycle 7	0.42 (0.74)	1.16** (0.45)	0.26 (0.17)	1.20*** (0.21)	-0.17 (0.37)	0.73** (0.30)	0.05 (0.06)	0.73*** (0.09)
Running variable form	ward	ward	council	council	ward	ward	council	council
N	865	865	865	865	865	865	865	865

Note: This table tests the robustness of the estimated treatment effect of equation (1) to the choice of the functional form for the relationship between council size and the outcome. In this table, $f(x) = \pi_1 x + \pi_2 x^2$. The baseline results are, in order, in columns 4-5 and 1-2 of Table III, and 1-2 and 3-4 of Table II. The sample includes bins 1 and 2. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.6 - Robustness: Treatment Effect ($f(x)$): Quadratic Interaction)

	All political parties							
	Candidates				Councilors			
	Ward		PR		Ward		PR	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Treat at cycle 4 $\times$ Cycle 4	3.50*** (1.14)	-0.25 (0.41)	0.75** (0.30)	0.86** (0.33)	0.69* (0.37)	-0.43 (0.30)	0.00 (0.11)	0.71*** (0.14)
Treat at cycle 4 $\times$ Cycle 5	1.31 (0.99)	0.53 (0.39)	0.45* (0.26)	1.10*** (0.33)	0.10 (0.39)	0.23 (0.32)	0.01 (0.10)	0.66*** (0.14)
Treat at cycle 4 $\times$ Cycle 6	-0.40 (0.92)	0.76* (0.43)	0.06 (0.24)	0.92*** (0.33)	-0.00 (0.42)	0.44 (0.33)	-0.01 (0.10)	0.66*** (0.14)
Treat at cycle 4 $\times$ Cycle 7	0.40 (0.93)	1.35*** (0.50)	0.02 (0.26)	1.17*** (0.32)	-0.06 (0.43)	0.74** (0.36)	-0.05 (0.10)	0.68*** (0.14)
Running variable form	ward	ward	council	council	ward	ward	council	council
<i>N</i>	865	865	865	865	865	865	865	865

Note: This table tests the robustness of the estimated treatment effect of equation (1) to the choice of the functional form for the relationship between council size and the outcome. In this table, $f(x) = \pi_1 x + \pi_2 x^2 + (\pi_3 x + \pi_4 x^2) \cdot Treat$. The baseline results are, in order, in columns 4-5 and 1-2 of Table III, and 1-2 and 3-4 of Table II. The sample includes bins 1 and 2. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.7 - Robustness: Treatment Effect (All Bins)

	All political parties							
	Candidates				Councilors			
	Ward		PR		Ward		PR	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Treat at cycle 4 $\times$ Cycle 4	3.26*** (0.97)	-0.45 (0.36)	1.04*** (0.22)	0.85*** (0.23)	0.51* (0.26)	-0.48** (0.23)	0.13* (0.08)	0.71*** (0.09)
Treat at cycle 4 $\times$ Cycle 5	1.03 (0.82)	0.24 (0.39)	0.65*** (0.16)	1.15*** (0.22)	-0.06 (0.31)	0.22 (0.27)	0.11 (0.07)	0.67*** (0.08)
Treat at cycle 4 $\times$ Cycle 6	-0.76 (0.69)	0.44 (0.42)	0.30** (0.15)	0.94*** (0.22)	-0.15 (0.34)	0.38 (0.30)	0.11* (0.06)	0.66*** (0.09)
Treat at cycle 4 $\times$ Cycle 7	0.34 (0.67)	1.10** (0.50)	0.30* (0.16)	1.21*** (0.22)	-0.21 (0.38)	0.66* (0.34)	0.06 (0.06)	0.68*** (0.09)
Running variable form	ward	ward	council	council	ward	ward	council	council
N	898	898	898	898	898	898	898	898

Note: This table tests the robustness of the estimated treatment effect using regression equation (1) to the sample selection criteria. Here we include all bins. The baseline results are, in order, in columns 4-5 and 1-2 of Table III, and 1-2 and 3-4 of Table II. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.8 - Robustness: Treatment Effect (All Municipalities in Bins 1 and 2)

	All political parties							
	Candidates				Councilors			
	Ward		PR		Ward		PR	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Treat at cycle 4 $\times$ Cycle 4	3.27*** (0.92)	-0.37 (0.33)	0.92*** (0.22)	0.95*** (0.22)	0.47* (0.25)	-0.35* (0.21)	0.08 (0.08)	0.77*** (0.09)
Treat at cycle 4 $\times$ Cycle 5	1.25 (0.81)	0.42 (0.33)	0.60*** (0.17)	1.08*** (0.23)	-0.14 (0.28)	0.30 (0.23)	0.09 (0.06)	0.64*** (0.09)
Treat at cycle 4 $\times$ Cycle 6	-0.35 (0.68)	0.67* (0.36)	0.26* (0.15)	1.00*** (0.22)	-0.23 (0.33)	0.52* (0.27)	0.08 (0.06)	0.70*** (0.08)
Treat at cycle 4 $\times$ Cycle 7	0.36 (0.69)	1.29*** (0.44)	0.21 (0.16)	1.24*** (0.22)	-0.29 (0.36)	0.83*** (0.29)	0.04 (0.06)	0.72*** (0.08)
Running variable form	ward	ward	council	council	ward	ward	council	council
N	873	873	873	873	873	873	873	873

Note: This table tests the robustness of the estimated treatment effect using regression equation (1) to the sample selection criteria. Here we include all existing municipalities. The baseline results are, in order, in columns 4-5 and 1-2 of Table III, and 1-2 and 3-4 of Table II. The sample includes bins 1 and 2. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.9 - Robustness: Treatment Effect (Balanced Panel in Bins 1 and 2)

	All political parties							
	Candidates				Councilors			
	Ward		PR		Ward		PR	
	Male (1)	Female (2)	Male (3)	Female (4)	Male (5)	Female (6)	Male (7)	Female (8)
Treat at cycle 4 $\times$ Cycle 4	3.52*** (0.94)	-0.31 (0.33)	0.94*** (0.22)	0.96*** (0.22)	0.46* (0.26)	-0.34 (0.22)	0.09 (0.08)	0.76*** (0.09)
Treat at cycle 4 $\times$ Cycle 5	1.38* (0.81)	0.47 (0.34)	0.65*** (0.17)	1.21*** (0.23)	-0.12 (0.29)	0.31 (0.24)	0.10 (0.06)	0.70*** (0.08)
Treat at cycle 4 $\times$ Cycle 6	-0.39 (0.68)	0.69* (0.37)	0.26* (0.15)	1.02*** (0.22)	-0.22 (0.33)	0.52* (0.27)	0.08 (0.06)	0.70*** (0.09)
Treat at cycle 4 $\times$ Cycle 7	0.40 (0.69)	1.29*** (0.44)	0.21 (0.16)	1.27*** (0.22)	-0.28 (0.36)	0.82*** (0.29)	0.04 (0.06)	0.72*** (0.08)
Running variable form	ward	ward	council	council	ward	ward	council	council
<i>N</i>	863	863	863	863	863	863	863	863

Note: This table tests the robustness of the estimated treatment effect using regression equation (1) to the sample selection criteria. Here we include only municipalities for which we can observe all election cycles (balanced panel), excluding 1 municipality that becomes a provincial council in election cycle 6. The baseline results are, in order, in columns 4-5 and 1-2 of Table III, and 1-2 and 3-4 of Table II. The sample includes bins 1 and 2. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.10 - Placebo Test—Treatment Effects Before the Reform

	Number of male ward candidates	
	Treatment fixed at Cycle 4 (1)	Contemporaneous treatment (2)
Treat × Cycle 1	-1.47 (3.38)	0.82 (1.30)
Treat × Cycle 2	-0.16 (1.74)	0.42 (1.04)
Treat × Cycle 3	-1.60 (1.71)	0.89 (0.92)
Treat × Cycle 4	3.47*** (1.21)	3.52*** (1.12)
Treat × Cycle 5	-0.02 (1.08)	0.07 (0.85)
Treat × Cycle 6	-2.01* (1.04)	-1.11 (0.82)
Treat × Cycle 7	-2.66** (1.16)	-1.76* (0.97)
Running variable form	ward	ward
<i>N</i>	1304	1517
Sample	Contemp. bin = cycle-4 bin & cycle-4 bin ∈ {1, 2}	Cycle-4 bin ∈ {1, 2}

Note: This table reports a placebo check estimating the effect of being above the threshold on the number of male ward candidates in the pre-quota period (cycles 1-3). The regression specification is given by equation (1). Column 1 uses treatment status defined in cycle 4 (our main treatment variable) to test for pre-existing differences in candidacy patterns between treatment and control municipalities. Because treatment status is fixed at cycle 4, this specification compares treatment and control municipalities within their cycle-4 bins. This creates a potential issue in the pre-quota period because electoral district boundaries changed substantially during the first three cycles: municipalities that fall in the same bin in cycle 4 may have had very different council sizes in earlier cycles. Such comparisons would be inconsistent with the logic of a regression discontinuity design, which relies on comparing observations within narrow bandwidths. We address this issue by restricting the sample to municipality×cycle observations whose contemporaneous bin matches their bin in cycle 4. This restriction ensures that treatment and control municipalities compared within a cycle-4 bin are also similar in size in the pre-quota cycle being analyzed. Note that council sizes remained largely stable from cycle 4 onward. Column 2 instead uses treatment status defined contemporaneously, to test whether the thresholds themselves (e.g., 10 vs. 11, 20 vs. 21) had any particular significance before the reform. The sample includes municipalities in bins 1 or 2 in cycle 4, for consistency with the rest of the paper. No further restriction is needed because, under contemporaneous treatment, treatment and control municipalities are always compared within the same contemporaneous bin. Lastly, the sample throughout includes candidates with no party affiliation, since no candidates were affiliated with parties before cycle 4. Standard errors, reported in parentheses, are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.11 - Treatment Effect on the Number of Female Rookie Ward Candidates by Competitiveness of Ward

Ward type:	Position on ballot:	Main political parties											
		Stronghold			Competitive			Weakhold			Less competitive (Stronghold + Weakhold)		
		All Female (1)	Useful Female (2)	Rank 1 Female (3)	All Female (4)	Useful Female (5)	Rank 1 Female (6)	All Female (7)	Useful Female (8)	Rank 1 Female (9)	All Female (10)	Useful Female (11)	Rank 1 Female (12)
Treat Cycle 4 × Cycle 4		-0.02 (0.04)	-0.03 (0.03)	-0.03 (0.03)	-0.05 (0.05)	-0.05 (0.05)	-0.02 (0.04)	-0.06 (0.04)	-0.06 (0.04)	-0.07* (0.04)	-0.08 (0.06)	-0.09* (0.05)	-0.10** (0.05)
Treat Cycle 4 × Cycle 5		0.03 (0.03)	0.04 (0.03)	0.05* (0.03)	0.08 (0.06)	0.09* (0.05)	0.10** (0.04)	-0.02 (0.04)	-0.02 (0.04)	-0.01 (0.04)	-0.00 (0.05)	0.01 (0.04)	0.04 (0.04)
Treat Cycle 4 × Cycle 6		0.03 (0.04)	0.01 (0.03)	0.02 (0.03)	0.13** (0.06)	0.12** (0.05)	0.07 (0.05)	-0.07* (0.04)	-0.05 (0.03)	-0.03 (0.03)	-0.04 (0.05)	-0.04 (0.05)	-0.02 (0.04)
Treat Cycle 4 × Cycle 7		0.02 (0.03)	0.00 (0.03)	0.01 (0.02)	0.17** (0.08)	0.18** (0.08)	0.22*** (0.06)	-0.01 (0.05)	-0.02 (0.04)	-0.03 (0.04)	0.01 (0.06)	-0.02 (0.05)	-0.02 (0.05)
Running variable form		ward	ward	ward	ward	ward	ward	ward	ward	ward	ward	ward	ward
N		1551	1551	1551	1551	1551	1551	1551	1551	1551	1551	1551	1551
Party dummy		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N. of wards of each type		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Average margin of victory		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fix (male+female) total		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports the effect of being above the threshold on the number of female rookie ward candidates in different wards. The regression specification is given by equation (1). Wards are divided into stronghold, competitive, and weakhold wards based on the party's margin of victory in the PR arm of the immediately preceding election (see notes under Figure E.1 for how it is calculated). Stronghold, competitive, and weakhold wards correspond to margins of victory of ≥ 0.05 , $[-0.05, 0.05]$, and < -0.05 , respectively, closely approximating the top quarter, middle half, and the bottom quarter. The results are qualitatively similar when we use the cutoffs of ± 0.10 . The sample includes bins 1 and 2 and is restricted to the two main parties. The level of observation is municipality $\times$ party. The outcome variable is the number of rookie female ward candidates in each type of ward. The columns for "Useful" and "Rank 1" represent increasingly high-up positions on the ballot paper, implying higher likelihoods of winning; "Useful" denotes position 1 if the ward elects 1-2 councilors and positions 1 and 2 if the ward elects 3-4 councilors, while "Rank 1" denotes the first candidate of the party in the ward. We control for the number of wards of each type, party dummies, the average margin of victory for the party in the municipality, and the total number (male+female) of relevant ward candidates. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.12 - Treatment Effect on the Gender Gap in Candidate Characteristics

	Main political parties				
	Education level (1)	Years of schooling (2)	1(Bachelor or more) (3)	1(Attended top 20 uni) (4)	1(Political experience) (5)
Female	-0.13 (0.11)	-0.24 (0.18)	0.00 (0.03)	0.00 (0.02)	0.10*** (0.03)
Female × Cycle 5	0.09 (0.12)	0.14 (0.19)	-0.02 (0.03)	0.02 (0.02)	-0.06* (0.03)
Female × Cycle 6	0.00 (0.14)	0.04 (0.23)	-0.00 (0.04)	-0.02 (0.02)	-0.04 (0.04)
Female × Cycle 7	0.39*** (0.13)	0.60*** (0.21)	0.00 (0.03)	-0.01 (0.02)	-0.02 (0.03)
Treat at cycle 4	-0.15 (0.15)	-0.26 (0.25)	-0.02 (0.04)	0.06** (0.03)	0.02 (0.03)
Treat at cycle 4 × Cycle 5	-0.04 (0.09)	-0.07 (0.14)	-0.03 (0.02)	-0.01 (0.01)	-0.05** (0.02)
Treat at cycle 4 × Cycle 6	-0.08 (0.09)	-0.13 (0.15)	-0.01 (0.03)	-0.03** (0.01)	-0.00 (0.03)
Treat at cycle 4 × Cycle 7	-0.02 (0.10)	-0.04 (0.16)	-0.02 (0.03)	-0.02 (0.02)	0.01 (0.03)
Female × Treat at cycle 4	0.16 (0.17)	0.29 (0.28)	-0.02 (0.04)	-0.03 (0.02)	0.08* (0.04)
Female × Treat at cycle 4 × Cycle 5	0.05 (0.17)	0.03 (0.28)	0.06 (0.05)	-0.02 (0.03)	-0.03 (0.05)
Female × Treat at cycle 4 × Cycle 6	0.05 (0.20)	0.02 (0.32)	0.01 (0.05)	0.03 (0.03)	-0.08 (0.06)
Female × Treat at cycle 4 × Cycle 7	-0.22 (0.20)	-0.39 (0.33)	-0.00 (0.06)	0.03 (0.03)	-0.03 (0.05)
Cycle 5	0.30*** (0.05)	0.50*** (0.08)	0.10*** (0.02)	-0.02** (0.01)	0.11*** (0.01)
Cycle 6	0.48*** (0.06)	0.78*** (0.10)	0.13*** (0.02)	-0.00 (0.01)	0.12*** (0.02)
Cycle 7	0.38*** (0.06)	0.64*** (0.10)	0.15*** (0.02)	-0.03*** (0.01)	0.10*** (0.02)
<i>N</i>	13177	13177	13177	13177	13724
Running variable form	council	council	council	council	council
Year of birth fixed effects	Yes	Yes	Yes	Yes	Yes
Party affiliation	Yes	Yes	Yes	Yes	Yes
Rookie or incumbent	Yes	Yes	Yes	Yes	Yes

Note: This table reports the effect of being above the threshold on the gender gap in education and political experience of candidates across election cycles. The outcome variables are the candidates' level of education (0-12), years of schooling (0-22), and dummies for having a bachelor's degree or more, for receiving tertiary education from a top 20 university in Korea—as ranked in any year between 1995 and 2019 by the JoongAng Newspaper Annual University Rankings, and for having an occupation related to politics. The regression specification is given by equation (3). The analysis is performed at the individual level. The sample includes bins 1 and 2, and only candidates from the two main parties for whom we have education/occupation information. “Party affiliation” is a dummy equal to one if the candidate is affiliated with the Progressive Party. “Rookie or incumbent” refers to a dummy equal to one if the candidate is an incumbent—they have been elected in at least one previous election. Standard errors (in parentheses) are clustered by municipality.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

**Table B.13 - Treatment Effect on the Number of Female Ward Candidates
with No Party Affiliation**

	Independent candidates
	Female (1)
Treat at cycle 4 $\times$ Cycle 4	0.09 (0.11)
Treat at cycle 4 $\times$ Cycle 5	0.04 (0.10)
Treat at cycle 4 $\times$ Cycle 6	0.24 (0.15)
Treat at cycle 4 $\times$ Cycle 7	-0.18 (0.12)
Running variable form	ward
N	865

Note: This table reports the effect of being above the threshold on the number of women running for elections without party affiliation in the ward arm. The regression specification is given by equation (1). The sample includes bins 1 and 2. The outcome variable is the number of female candidates with no party affiliation in each election cycle and municipality. Table B.1 provides details regarding the size of this group of candidates, which remains substantial over the election cycles. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.14 - Treatment Effect on the Gender Gap in Voter Preferences

Main political parties						
	Candidate's vote share in the ward election arm (%)					
	Controls:	None	Ballot position	Individual characteristics	N. ward candidates	Party×muni×cycle fixed effects
		(1)	(2)	(3)	(4)	(5)
Female		0.48 (1.06)	-0.71 (0.97)	-0.04 (1.02)	0.46 (1.02)	-2.00** (0.87)
Female × Cycle 5		-0.68 (1.32)	-1.59 (1.13)	-2.65** (1.19)	-3.09*** (1.17)	-1.00 (0.99)
Female × Cycle 6		-1.46 (1.17)	-1.75* (1.00)	-2.94*** (1.03)	-3.47*** (1.04)	-1.46* (0.87)
Female × Cycle 7		1.68 (1.22)	1.61 (1.15)	0.14 (1.19)	-0.68 (1.16)	-0.10 (0.97)
Treat at cycle 4		-2.11** (1.02)	-1.99** (0.80)	-2.21*** (0.81)	-0.81 (0.68)	
Treat at cycle 4 × Cycle 5		0.14 (0.61)	0.29 (0.54)	0.32 (0.54)	-0.48 (0.56)	
Treat at cycle 4 × Cycle 6		0.84 (0.73)	0.91 (0.63)	0.94 (0.63)	-0.48 (0.68)	
Treat at cycle 4 × Cycle 7		1.26 (0.79)	1.17* (0.64)	1.22* (0.63)	-0.38 (0.63)	
Female × Treat at cycle 4		-1.21 (1.87)	0.45 (1.54)	-0.14 (1.61)	-0.47 (1.52)	0.43 (1.15)
Female × Treat at cycle 4 × Cycle 5		2.09 (2.09)	-0.68 (1.65)	-0.40 (1.74)	-0.20 (1.67)	-1.18 (1.37)
Female × Treat at cycle 4 × Cycle 6		1.56 (1.88)	-0.46 (1.53)	0.17 (1.60)	0.60 (1.53)	-0.09 (1.30)
Female × Treat at cycle 4 × Cycle 7		0.60 (2.09)	-0.96 (1.72)	-0.20 (1.75)	0.12 (1.68)	0.16 (1.42)
N		11192	11192	10737	10737	10665
Running variable form		ward	ward	ward	ward	ward
Ballot position		No	Yes	Yes	Yes	Yes
Age polynomials		No	No	Yes	Yes	Yes
Political experience		No	No	Yes	Yes	Yes
Rookie or incumbent		No	No	Yes	Yes	Yes
Years of schooling		No	No	Yes	Yes	Yes
Fix (male+female) total		No	No	No	Yes	Yes
Party×muni×cycle fixed effects		No	No	No	No	Yes

Note: This table reports the effect of being above the threshold on the gender gap in the vote share (in percentages) obtained by candidates across election cycles. The regression specification is given by equation (3). The analysis is performed at the individual level. The sample includes all candidates nominated in municipalities included in bins 1 and 2 by the two main parties. Each column to the right includes additional controls compared to the preceding column. The controls ensure that we are comparing the gender gap in vote shares for candidates with similar characteristics and under similar election conditions. “Ballot position” refers to dummies for each position in the ballot list. For the definition of variables on individual characteristics (“Age polynomials,” “Party affiliation,” “Rookie or incumbent,” “Years of Schooling”), refer to notes of Table B.12. “N. ward candidates” refers to the total number of candidates competing in the municipality, independently of their party affiliation. “Party×muni×cycle fixed effects” refers to party×municipality×election cycle fixed effects—allowing us to compare men and women of the same party in the same municipality, in the same election cycle. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.15 - Probability of Being Constrained in the Number of Female Candidates

	Main political parties (party $\times$ municipality)	
	Pr(unconstrained)	
	All main parties (1)	Participates in ward arm (2)
Treat at cycle 4 $\times$ Cycle 4	-0.08 (0.08)	-0.09 (0.07)
Treat at cycle 4 $\times$ Cycle 5	0.08 (0.07)	0.12 (0.07)
Treat at cycle 4 $\times$ Cycle 6	0.02 (0.07)	0.05 (0.06)
Treat at cycle 4 $\times$ Cycle 7	0.07 (0.07)	0.09 (0.07)
Running variable from	council	council
<i>N</i>	1551	1514

Note: This table reports the effect of being above the threshold (defined based on cycle 4 council size) on the probability that the party is unconstrained in the selection of candidates. A party is defined as not constrained if the number of female candidates in the party's list is strictly greater than the number of women the party must place in its list due to quotas. All odd-numbered candidates in the party list for the PR arm need to be female due to the quota, so the required number of women is 1 if 1 or 2 seats are elected through the PR arm, and 2 if 3 seats are elected through the PR arm. The regression specification is given by equation (1). The unit of analysis is party by municipality. The sample includes bins 1 and 2, and only the two main parties. In column 2, the sample is restricted to only the main parties that participate in the ward arm, i.e. have at least one ward candidate. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.16 - Party Learning versus Women's Growing Power Within Parties

Sample selection:	Main political parties						
	All parties	Parties with N. own-party female councilors in $t - 1$			Parties with N. other-party female councilors in $t - 1$		
		≥ 1	≥ 2	≥ 3	≥ 1	≥ 2	≥ 3
	Rookie Female (1)	Rookie Female (2)	Rookie Female (3)	Rookie Female (4)	Rookie Female (5)	Rookie Female (6)	Rookie Female (7)
Treat at cycle 4 $\times$ Cycle 4	-0.15* (0.08)						
Treat at cycle 4 $\times$ Cycle 5	0.09 (0.08)	0.08 (0.10)	0.16 (0.12)	0.34** (0.17)	0.10 (0.09)	0.21* (0.12)	-0.01 (0.16)
Treat at cycle 4 $\times$ Cycle 6	0.11 (0.09)	0.17 (0.11)	0.26** (0.13)	0.47*** (0.17)	0.12 (0.10)	0.18 (0.11)	0.32** (0.15)
Treat at cycle 4 $\times$ Cycle 7	0.22** (0.11)	0.24** (0.12)	0.33** (0.13)	0.62*** (0.20)	0.25** (0.12)	0.41*** (0.13)	0.55*** (0.18)
Running variable form	ward	ward	ward	ward	ward	ward	ward
N	1551	1084	934	830	1128	986	859
Fix (male+female) total	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N in Treatment group							
Cycle 5	128	100	24	9	118	49	15
Cycle 6	139	108	69	27	135	99	49
Cycle 7	132	125	90	43	124	87	44

Note: This table investigates whether the rise in female rookie nominations reflects women's growing influence as more women are elected within parties, or whether women elected from other parties also predict similar changes. We estimate a party-level version of equation (1). The outcome variable is the number of female rookie ward candidates nominated by the party. Column 1 replicates column 9 of Table V at the party level, showing a similar initial reduction in female rookies and a positive effect by cycle 7. In columns 2-4, we restrict the treated group to parties in treatment municipalities with at least one, two, or three female councilors elected from within the party in the previous cycle. In columns 5-7, we instead restrict the treated group to parties in treatment municipalities with at least one, two, or three female councilors elected from other parties in the previous cycle. In all columns, the comparison group consists of all parties in control municipalities. The sample includes bins 1 and 2 and is restricted to the two main parties. Standard errors (in parentheses) are clustered by municipality.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C INSTITUTIONAL SETTING

C.1 Other reforms introduced simultaneously

Table C.1 summarizes the reforms to municipal council elections. We describe these reforms and discuss their implications for our analysis of the quota's effects.

Besides the gender quota on the PR arm, the 2006 electoral reform included two additional measures intended to increase the share of women among ward candidates. First, subsidies were offered to parties based on the female share among their ward candidates nationwide. Second,

another clause of the reform stated that parties “shall endeavor to nominate” at least 30% women among ward candidates nationwide.

Could these measures account for our findings? Because they do not vary discontinuously at the council-size thresholds that underlie our regression discontinuity design, they cannot by themselves generate the treatment effects we estimate. Moreover, as we discuss below, neither measure appears to have been quantitatively important in shaping party nomination strategies in practice. That said, it is plausible that they may have amplified the quota’s effects at the margin.

Because both measures targeted female shares at the national level, parties may have strategically decided where to place female candidates across municipalities. If the quota increased the supply of experienced female politicians, or changed parties’ perceptions of women’s electoral viability, treatment municipalities may have become relatively more attractive places to nominate women to ward seats. In this sense, these provisions could have amplified the quota’s effects through the incumbency and learning channels discussed in Section V.A. They would not, however, generate a treatment-control discontinuity absent their interaction with the quota.

Crucially, neither measure appears to have strongly bound parties’ nomination strategies in practice. The scale of the subsidies has been criticized as too small to meaningfully expand female nominations (Lee, 2003; Kim et al., 2003), accounting for only 5-6% of the total value of election subsidies (National Election Commission, 2018a). Furthermore, the 30% target clause was aspirational rather than binding (shall “endeavor” to nominate) and appears to have been largely disregarded in practice. No party with national presence⁵⁵ ever nominated more than 30% women, and the female share in the two main parties fell well short of 30% in every election.⁵⁶

To sum up, the other measures did not independently generate the treatment-control discontinuities we estimate. To the extent they mattered, they may have amplified parties’ responses to the stronger quota requirements just above the council-size thresholds. Our estimates can therefore be interpreted as treatment effects at the threshold, estimated in an environment that also included weak nationwide incentives to nominate women in the ward arm.

Amendments to electoral rules continued between the 2006 and 2010 elections. One provision stipulated that in either the municipal council elections or the higher-level provincial council elections, there must be at least one female candidate in each National Assembly election district. As

⁵⁵Among the 11 party×election cycle observations with at least 300 ward candidates nationwide, the average female share of ward candidates was 6.6% in cycle 4 (2006) and 20.5% in cycle 7 (2018). These figures include relatively small parties, as the two main parties fielded 1,300-2,000 ward candidates per cycle. Only a few very small parties exceeded the 30% threshold: none in cycle 4, one each in cycles 5 and 6, and three in cycle 7, accounting for 3.8, 4.6, and 3.1% of ward candidates nationwide, respectively.

⁵⁶The Progressive Party’s female share of ward candidates nationwide in cycles 4-7 was 4.1%, 11.6%, 16.3%, and 23.3%, respectively; the corresponding shares for the Conservative Party were 4.6%, 10.0%, 15.0%, and 18.7%.

there are around 250 such districts, compared to 226 municipalities, a National Assembly election district is roughly comparable in size to a municipality.⁵⁷ Legislative Impact Analysis Reports indicate that most parties chose to satisfy this rule in the municipal council elections, likely because these elections involve a larger number of candidates (Lee, 2019). Because this provision effectively set a *uniform* minimum number of female candidates across both treatment and control municipalities, we do not expect it to have driven our treatment effects.

Table C.1 - Amendments to Legislation on Municipal Council Elections

First applicable election	Amendment
Election cycle 4 (2006)	[PR] Proportional representation introduced
	[W] Single-member plurality vote → Multi-member plurality vote
	[PR] odd-numbered candidates in party lists must be female
	[W] Subsidies to parties for nominating female candidates
	[W] Parties encouraged to nominate $\geq 30\%$ women among ward candidates nationwide
Election cycle 5 (2010)	[PR] odd-numbered candidates in party lists must be female (enforced)
	[W] At least one female candidate per National Assembly Election district

Note: The table summarizes the amendments to the legislation on South Korean municipal council elections. [PR] indicates rules relating to the PR arm and [W] to the ward arm. Adapted from Lim (2018).

C.2 Background behind the adoption of gender quotas

Before gender quotas were adopted in the municipal council elections in 2006, they were adopted first in the National Assembly Election in 2004. The adoption was influenced by increasing demands by women’s organizations to raise female representation in politics, which at the time was dramatically behind the international average.⁵⁸ As females constitute half the voters, it was in the interest of political parties to put gender quotas forward among their election pledges. Moreover, Jeon (2013) argues that the adoption of the quota was also a political tactic. Political parties wanted to increase the size of the National Assembly back to what it was before the size cut during the Asian Financial Crisis, and the fact that the majority of the added seats will go to females, with the quota, made for a good excuse to expand the Assembly.

Once the quota was adopted in the National Assembly Election, it became the natural next step to introduce it in the regional elections. The gender quota in the municipal council election was passed in the National Assembly, led by both major parties. Some argue that there was political motivation behind it, too (Kim, 2005). One new element in the reform was the party nomination system—a ward candidate must be nominated by a party in order to run with the party affilia-

⁵⁷National Assembly election districts are defined based on population size and local representativeness. A large municipality may contain five National Assembly election districts, while up to five small municipalities may comprise a single National Assembly election district.

⁵⁸See Cho and Kim (2010) for a summary of the major activities of women’s organizations.

tion—and it has been disputed as a ploy to deepen party influence. Political parties used the quota to justify the party nomination system since the gender quota was embedded in the PR arm, where party nomination was essential.

To sum up, it is unlikely there was a major division among political parties in their support of the gender quota when it was passed.

Appendix D EMPIRICAL STRATEGY

D.1 Configuration of the running variable

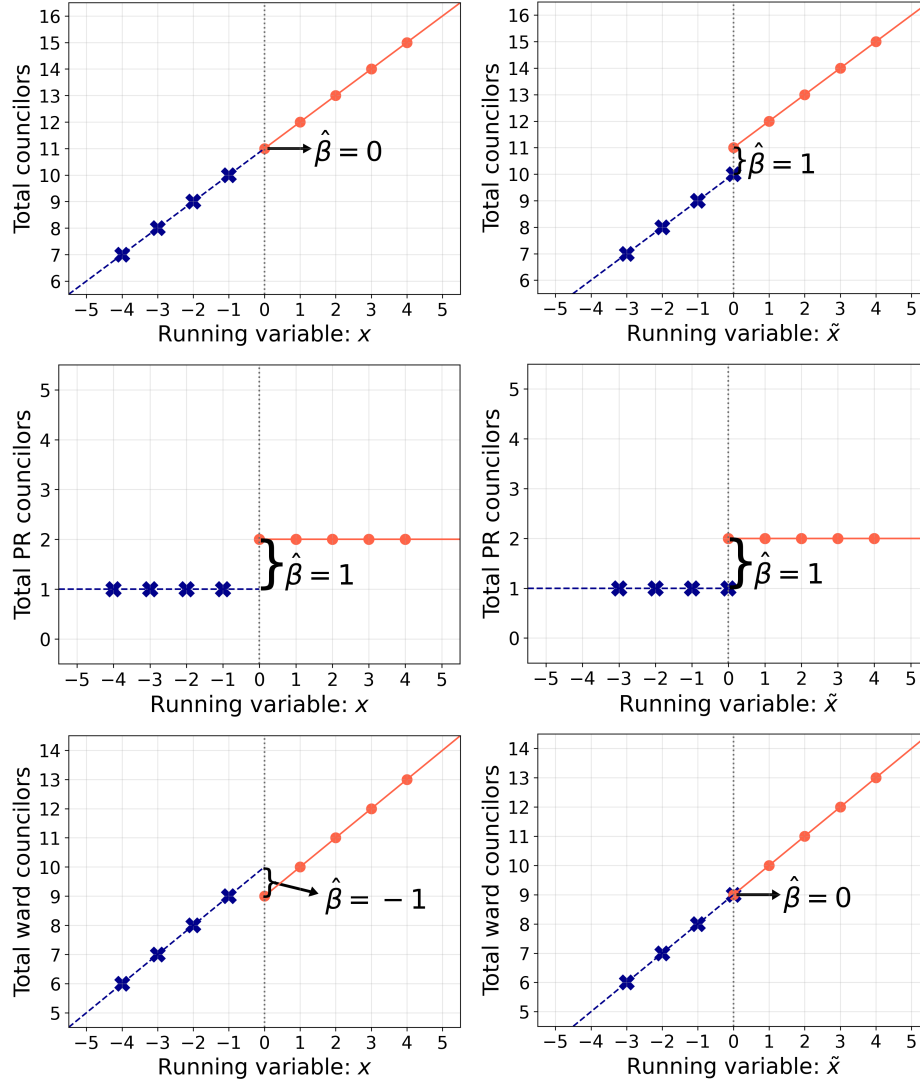
Figure D.1 illustrates why, in estimating equation (1), we use a different running variable for outcomes in the ward arm than for outcomes in the PR arm or for outcomes combining both arms.

The figure shows the treatment effect $\hat{\beta}$ for three outcomes: the total number of councilors, the total number of PR councilors, and the total number of ward councilors. These outcomes correspond to the number of total seats (council size), PR seats, and ward seats. In each case, the treatment effect is identified at the council-size cutoff.

The left panels use $x_{cb} \equiv (\text{council size})_{cb} - \text{threshold}_b$ as the running variable. Since no control municipality lies exactly at the threshold, the control outcome at $x_{cb} = 0$ is extrapolated from the linear trend below the cutoff. For the total number of councilors, this produces no discontinuity at the threshold, so $\hat{\beta} = 0$. For the total number of PR councilors, the threshold adds one PR seat, so $\hat{\beta} = 1$. For the total number of ward councilors, however, the implied discontinuity is negative: the treated municipality at the threshold has *one fewer* ward seat than the fitted control municipality at $x_{cb} = 0$, yielding $\hat{\beta} = -1$.

This creates an interpretation problem for ward outcomes, such as the number of female ward councilors, by producing a discrepancy in ward-seat capacity. Even if treatment increases the number of female ward councilors, the estimated treatment effect could still be negative as the treated council is effectively compared to a control-side benchmark with one more ward seat. To resolve this issue, for ward outcomes we redefine the running variable as $\tilde{x}_{cb} \equiv (\text{number of ward seats})_{cb} - (\text{number of ward seats at the threshold})_b$. Using $\tilde{x}_{cb}$ means comparing a treated council with the *same* number of ward seats as a control council at the threshold ($\hat{\beta} = 0$ in the bottom-right panel), which makes the treatment effect easier to interpret. Importantly, this change in the running variable does not otherwise affect the estimation: $\hat{\alpha}_b$, $\hat{\alpha}_t$, $\hat{\gamma}$, and the R^2 remain unchanged.

Figure D.1 - Running-Variable Choice in the Regression Discontinuity Design



Note: This figure shows how the treatment effect coefficient in equation (1) differs when the running variable is based on council size (x_{cb}) rather than the number of ward seats ($\tilde{x}_{cb}$). For illustrative purposes, we focus on bin 1 only. The three panels on the left estimate a simplified version of equation (1)—abstracting from multiple bins, multiple cycles, and additional controls—using x as the running variable: $Y_c = \beta \times \mathbb{1}(x_c \geq 0) + x_c$. The three panels on the right instead use $\tilde{x}$ as the running variable: $Y_c = \beta \times \mathbb{1}(x_c \geq 0) + \tilde{x}_c$. From left to right in each panel, the dark blue crosses correspond to control councils of sizes 7 through 10, and the red dots correspond to treated councils of sizes 11 through 15. Using $\tilde{x}$ shifts the dark blue crosses one unit to the right. In each case, the treatment effect is identified from the discontinuity when the running variable is 0. The implied treatment effect $\hat{\beta}$ is shown for three outcomes: the total number of councilors (top), the total number of PR councilors (middle), and the total number of ward councilors (bottom).

D.2 Factors that bind a candidate to a municipality

A candidate is legally required to have been a resident of the municipality they are running in for at least 60 days prior to the election. In addition, as municipal councilors deal with local grass-roots matters, a candidate familiar with the municipality will win more votes *ceteris paribus*. Hence, a candidate usually runs in the municipality they have a connection with, such as their birthplace, long-term residence, or place of education. Moreover, the final say of a party's nomination lies on the head of the municipal branch of the party, so a candidate typically serves the local activities of the party in the municipality they desire to run in for a long time before getting nominated. Finally, once a candidate is nominated in a municipality, they put on a campaign and become known to the residents. So if they were to run again, they would not start over at a new location. For all these factors, rarely do parties move around candidates across municipalities for strategic reasons.

D.3 Why an additional PR seat increases female PR representation

The treatment—an additional PR councilor—strongly tends to increase the number of female PR councilors because the additional PR seat is usually allocated to an odd-numbered position on a party list, whose gender is restricted to be female.

This is a result of the PR seat allocation rule (Hare-Niemeyer method):

1. Exclude parties receiving less than 5% of the total PR votes in the municipality.
2. For each qualifying party, compute the allocation score $X = \text{PR seats} \times \text{vote share among qualifying parties}$.
3. Allocate to each party the number of seats equal to the integer part of X .
4. Allocate remaining seats by the ranking of the decimal part of X .

Example allocation Suppose council A has three parties competing for three PR seats. Parties 1, 2, and 3 receive 60%, 38%, and 2% of the votes, respectively. Because party 3 receives less than 5% of the votes, it does not qualify for seat allocation. Among qualifying parties, the normalized vote shares are $60/(60 + 38) \approx 61.2\%$ for party 1 and $38/(60 + 38) \approx 38.8\%$ for party 2. The corresponding allocation scores are $3 \times 0.612 \approx 1.84$ for party 1 and $3 \times 0.388 \approx 1.16$ for party 2. Since the integer part of each score is 1, each party first receives one PR seat. The remaining PR seat is then allocated to party 1, because its fractional remainder is larger: $0.84 > 0.16$.

Implications for our empirical setting Our empirical analysis restricts the sample to municipalities in bins 1 and 2, with either 1, 2, or 3 PR seats.

Consider first the treatment threshold for bin 1, where the additional PR seat is the second seat. This seat can either go to the number-1 candidate of the second-ranked party or to the number-2 candidate of the first-ranked party. For the first-ranked party to win both PR seats, however, its normalized vote share must exceed that of the second-ranked party by more than 50 percentage points: $2v_1 - 1 > 2v_2 \Leftrightarrow v_1 > v_2 + \frac{1}{2}$. This is a large margin. Therefore, the second PR seat is far more likely to go to the number-1 candidate of the second-ranked party than to the number-2 candidate of the first-ranked party.

The same logic applies when there are 3 PR seats. Seats can be allocated in three ways: one seat each to the first-, second-, and third-ranked parties; two seats to the first-ranked party and one seat to the second-ranked party; or all three seats to the first-ranked party. The only case in which the additional third seat may go to an even-numbered position is the comparison between the first two seat allocations: that is, whether the seat goes to the number-2 candidate of the first-ranked party or to the number-1 candidate of the third-ranked party. For the number-2 candidate of the first-ranked party to win this seat, the first-ranked party's normalized vote share must exceed that of the third-ranked party by more than 33 percentage points: $3v_1 - 1 > 3v_3 \Leftrightarrow v_1 > v_3 + \frac{1}{3}$. This is again a large margin. In the remaining comparison—between the first-ranked party winning two seats or all three seats—the marginal seat goes either to the number-1 candidate of the second-ranked party or to the number-3 candidate of the first-ranked party. Both positions are odd-numbered and hence restricted to female candidates.

D.4 The composition of PR councilors changes only at the thresholds

In order to buttress the regression discontinuity design, we test whether there is a change in the number of female PR councilors as council size increases, at points *other* than the thresholds. We estimate the following equation:

$$(\text{N. of female PR councilors})_{cbt} = \beta \times (\text{Larger by one})_{cbt} + \delta_b + \gamma_t + \epsilon_{cbt} \quad (2)$$

$$\text{where } (\text{Larger by one})_{cbt} = \begin{cases} 1, & \text{if } (\text{council size})_{cbt} - \text{threshold}_{bt} = x \\ 0, & \text{if } (\text{council size})_{cbt} - \text{threshold}_{bt} = x - 1 \end{cases}$$

for each value of $x \in \{-4, -3, \dots, 3, 4\}$, i.e. distance from the threshold. The threshold is council $\times$ election cycle specific, as it depends on the bin the council belongs to.

Equation (2), therefore, estimates the change in the number of female PR councilors when the

council size increases by 1, for all points around the threshold. Table D.1 reports the results. It confirms that there is a positive effect only at the threshold.

Table D.1 - The Effect of Council Size on the Number of Female PR Councilors

	<i>x</i> value (distance from the threshold)								
	-4	-3	-2	-1	0	1	2	3	4
Coefficient ($\hat{\beta}$)	-0.03	0.03	-0.03	-0.01	0.92***	-0.02	-0.02	-0.03	0.09
Standard error	(0.09)	(0.02)	(0.03)	(0.04)	(0.06)	(0.08)	(0.08)	(0.08)	(0.09)
<i>N</i>	267	380	210	170	168	147	133	111	87

Note: This table reports the results of the coefficient of $(Larger\ by\ one)_{cbt}$ in regression equation (2), for different values of x . For each x , the sample is councils that are x and $x-1$ away from the threshold ($Larger\ by\ one = 1$ and $= 0$, respectively.) When $x = 0$, $(Larger\ by\ one)_{cbt}$ corresponds to the treatment definition used in the main identification strategy (with the exception that the sample is restricted to councils just one seat above and below the threshold). The table shows that the number of female PR councilors increases discontinuously only at the treatment thresholds. The sample includes bins 1 and 2. Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

D.5 Individual-level version of the main regression specification

The individual-level version of equation (1) is:

$$Y_{icbt} = \alpha_b + \alpha_t + \sum_{s=4}^7 \beta_s \cdot (Treat\ in\ cycle\ 4)_{cb} + \sum_{s=4}^7 \pi_s \cdot (Treat\ in\ cycle\ 4)_{cb} \times Female_{icbt} + \sum_{s=4}^7 \kappa_s \cdot Female_{icbt} + f() + X'_{cbs} \gamma + X'_{icbs} \delta + \epsilon_{cbs} \quad (3)$$

where $(Treat\ in\ cycle\ 4)_{cb}$ is treatment status in election cycle 4, X_{cbs} denotes municipality-level control variables such as the contemporaneous number of seats, α_b and α_t are bin and cycle fixed effects, and X_{icbs} indicates individual-level controls.

Appendix E MODEL OF PARTY LEARNING

E.1 Parties' strategies: a dynamic model of discrimination

Section V.C's empirical evidence on the party learning mechanism is guided by the predictions of a model of electoral competition featuring discrimination. We innovate from Le Barbanchon and Sauvagnat (2022)'s model of electoral competition by enriching the candidate selection process and adding in the aspect of candidate allocation to different election wards, a key change that tightens the link between the model predictions and the empirics. Moreover, we allow for two

types of discrimination, taste-based and statistical. There is no room for statistical discrimination in Le Barbanchon and Sauvagnat (2022), as they assume perfect information about the ability of potential candidates. In modeling statistical discrimination, we extend standard models (Aigner and Cain, 1977) with a dynamic process of belief updating, not just about the individual but also about the group, similarly to Lepage (2024).

Setup Consider party $p \in \{L, R\}$ in a municipality with constituent wards $1, \dots, W$. Party ideology I_p is fixed. The two parties participate in the ward and the PR election arms in election cycle t . For simplicity, we assume that each party has only one ward candidate position in each ward and one PR candidate position on the PR party list. A party selects candidates from a group of potential candidates and allocates them to different candidate positions. Potential candidate i is characterized by their gender $g \in \{m, f\}$ and their competence a_i :

$$a_i \sim N(\mu_g, \sigma^2)$$

where μ_g is the mean ability of gender g .

Parties have imperfect information regarding the competence of a new potential candidate. Their true competence is only revealed if they get elected and serve as councilors. However, at the time of candidate selection, parties observe a signal of competence:

$$s_i = a_i - \mu_g + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma_s^2)$$

$E(s_i) = 0$, so the signal is informative about the *relative* competence of i within gender.⁵⁹

Party's beliefs at candidate selection Parties have imperfect information not only on the ability of individual candidates but also on the group mean ability, μ_g .⁶⁰ A party's prior belief about the value of μ_g , before the election in election cycle t , follows a normal distribution with mean $\tilde{\mu}_{g,t}$, variance $\tilde{\sigma}_{g,t}^2$.

For a potential candidate i with signal s_i , the party expects i 's ability to be

$$\tilde{E}(a_i | s_i, g, t) = \tilde{\mu}_{g,t} + \frac{\sigma^2}{\sigma_s^2 + \sigma^2} s_i$$

where $\tilde{E}$ indicates expectation taken over the prior distribution. Statistical discrimination against women implies that for a man and a woman with the same signal s_i , his perceived ability is higher

⁵⁹Hence, the signals do not contain information on the value of μ_g .

⁶⁰We assume for simplicity that the variances of ability and signal (σ^2 and σ_s^2) are the same across gender and that parties know these values, which allows us to focus on their perceptions of μ_g . Learning about productivity variance or individual signal precision affects the weight attached to individual signals versus group membership but leaves substantive implications unchanged.

than hers: $\tilde{E}(a_i|s_i, m, t) > \tilde{E}(a_i|s_i, f, t)$.

Party's problem In election cycle t , party p nominates candidates to win as many seats as possible, but it also cares about the gender ratio among its candidates. From what follows, we denote the PR arm as ward “0” for notational convenience. Party p maximizes⁶¹

$$U_p = \mathbb{E} \left(\sum_{w=0}^W V_{p,w} \right) - b \left| \frac{1}{W+1} \sum_{w=0}^W F_{p,w} - f_p^* \right|$$

where $V_{p,w}$ is an indicator for party p winning the seat in ward w , $F_{p,w}$ is an indicator for party p 's candidate in ward w being female, f_p^* is party p 's desired female share among its candidates, and b regulates how much the gender ratio matters for the party relative to winning seats. Taste-based discrimination against women implies a low f_p^* . If $b = 0$, then gender does not factor into the candidate nomination strategy.

Voters vote according to party ideology and councilor quality. We assume that voters have single-peaked preferences for candidates such that the Median Voter Theorem holds. To fix ideas, consider party R . The median voter in ward w with ideology I^w gets the following utility if party R 's candidate with ability a_i wins:

$$U_{R,w} = a_i - |I^w - I_R| - \delta_w$$

where $\delta_w \sim N(0, 1)$ is the relative voter preference shock for party L , unforeseen at the time of candidate nomination. For the PR arm, I^0 is the ideology of the median voter in the whole municipality.

We assume party R takes the probability of winning in ward w with candidate i to be^{62,63}

$$\mathbb{E}(V_{R,w}) = \Phi \left(\tilde{E}(a_i|s_i, g, t) - A_{L,w} - \underbrace{|I^w - I_R| + |I^w - I_L|}_{\equiv R_w(\text{popularity of party } R)} \right)$$

where Φ denotes the cumulative distribution function of the standard normal distribution, and $A_{L,w}$ is party R 's expectation of the ability of the party- L candidate in ward w . High R_w indicates that

⁶¹Subscript t is omitted for notational convenience because candidate nomination is a static problem.

⁶²We assume the party believes that voters assess the expected competence of candidate i in the same way as it does. However, it suffices for all our model implications that the party believes that voter beliefs on the competence of candidates are *increasing* in party beliefs.

⁶³For simplicity, we assume that the party assesses the likelihood of winning by taking the perceived ability of candidate i as fixed. Model predictions are qualitatively the same if we instead had $\mathbb{E}(V_{R,w}) = \tilde{E}[\Phi(a_i - A_{L,w} - |I^w - I_R| + |I^w - I_L|) | s_i, g, t]$.

w is party R 's stronghold.

Note here that $\frac{\partial^2 \mathbb{E}(V_{R,w})}{\partial \tilde{E}(a_i | s_i, g, t) \partial |R_w|} < 0$. Candidate competence increases the likelihood of victory, but more so in competitive wards and less so in strongholds or weakholds.

Party learning Once candidate i is elected and serves as councilor, the party learns about their competence. We assume for simplicity that learning is complete and the true competence of the candidate $\mathbf{a} = \{a_i\}$ is revealed.⁶⁴

Given the observed abilities, the party also learns about the average ability of each gender μ_g . The party makes an maximum-likelihood inference about the value of μ_g , considering that these councilors were positively selected with signals $\mathbf{s} = \{s_i\}$. Put simply, the party solves, “What must μ_f be for females with signals $\mathbf{s}$ to have true abilities $\mathbf{a}$?”

The maximum likelihood estimator, derived in Appendix E.2, is

$$\hat{\mu}_g = \frac{1}{n} \sum_{i=1}^n \left(a_i - \frac{\sigma^2}{\sigma_s^2 + \sigma^2} s_i \right) \sim N \left(\mu_g, \frac{1}{n} \left(\frac{\sigma^2 \sigma_s^2}{\sigma_s^2 + \sigma^2} \right)^2 \right) \quad (4)$$

Call $V = Var(\hat{\mu}_g)$. Then, the posterior distribution about the value of μ_g is normal with mean $\tilde{\mu}_{g,t+1}$ and variance $\tilde{\sigma}_{g,t+1}^2$, which are weighted averages of the prior and the maximum likelihood estimator:

$$\tilde{\mu}_{g,t+1} = \frac{V \tilde{\mu}_{g,t} + \tilde{\sigma}_{g,t}^2 \hat{\mu}_g}{V + \tilde{\sigma}_{g,t}^2}, \quad \tilde{\sigma}_{g,t+1}^2 = \frac{V \tilde{\sigma}_{g,t}^2}{V + \tilde{\sigma}_{g,t}^2}$$

The updating speed of party beliefs about the value of μ_g is given by

$$\tilde{\mu}_{g,t+1} - \tilde{\mu}_{g,t} = \frac{\tilde{\sigma}_{g,t}^2}{V + \tilde{\sigma}_{g,t}^2} (\hat{\mu}_g - \tilde{\mu}_{g,t})$$

Timing in election cycle t , nature first determines the group of potential candidates available to each party, and parties only know about their own groups. Then the two parties play a simultaneous game: to select candidates, based on their beliefs on the mean ability by gender ($\tilde{\mu}_{g,t}$) and the within-gender competence signals ($\{s_i\}$), and to allocate them to wards $1, \dots, W$ and the PR arm. Next, the relative voter preference shock δ_w is realized, and voters vote. The true competence of each elected councilor ($\{a_i\}$) is revealed during their term. Based on this, the parties update their beliefs on the mean group ability to $\tilde{\mu}_{g,t+1}$.

Party's selection of candidates The party nominates candidates taking into account the value of

⁶⁴This assumption could be relaxed. An extension of the model with imperfect belief updating regarding a councilor's competence can be found in Appendix E.3. In this extension, learning about women's mean ability is faster if a new female councilor is from your own party. This extension explains the result in Table F.1 where parties that marginally won a quota woman nominate more rookie females in later elections than parties that marginally lost.

$\tilde{E}(a_i|s_i, g, t)$. Within gender, it chooses the potential candidate with the highest value of s_i first, then the one with the second-highest value of s_i , etc.

For simplicity of exposition, we consider the scenario where the Nash equilibrium allocation of candidates is for each party to place the most competent candidate in the most competitive ward, the second-most competent candidate in the second-most competitive ward, etc.⁶⁵ The intuition is that candidate competence increases the likelihood of victory disproportionately more in more competitive wards.

To describe the candidate selection strategy more clearly, we make an innocuous tweak that circumvents the randomness of the draws of potential candidates. Instead of selecting $W + 1$ candidates from the group of potential candidates already drawn, parties choose the minimum signal threshold for each gender $\bar{s}_g$, and $W + 1$ candidates are drawn randomly from the parts of the signal distribution above $\bar{s}_g$. The female share of candidates is then $\frac{1 - \Phi(\bar{s}_f/(\sigma^2 + \sigma_s^2))}{2 - \Phi(\bar{s}_f/(\sigma^2 + \sigma_s^2)) - \Phi(\bar{s}_m/(\sigma^2 + \sigma_s^2))}$.⁶⁶

Here we formalize how parties select candidates, and illustrate it diagrammatically in Appendix E.6:

- **[Only statistical discrimination]** If $b = 0$, a party selects the most competent $W + 1$ individuals, regardless of gender. Hence, the marginal male and marginal female candidates have the same perceived ability: $\tilde{E}(a_i|\bar{s}_m, m, t) = \tilde{E}(a_i|\bar{s}_f, f, t)$. If men are perceived to have a higher mean ability than women, i.e. $\tilde{\mu}_{m,t} > \tilde{\mu}_{f,t}$, then the marginal female candidate has a higher signal than the marginal male candidate: $\bar{s}_f > \bar{s}_m$. Call the female share in this benchmark case $f^{benchmark}$.
- **[Both taste-based and statistical discrimination]** If $b > 0$ and $f^{benchmark} > f_p^*$, a party selects a smaller female share than $f^{benchmark}$. There is a trade-off between gender preference and candidate competence. A party nominates some male candidates even if they have a lower perceived ability than the marginal female candidate: $\tilde{E}(a_i|\bar{s}_m, m, t) < \tilde{E}(a_i|\bar{s}_f, f, t)$. The gap in the signals of the marginal female and male candidates ($\bar{s}_f - \bar{s}_m$) is higher than in the benchmark case.

Evidence on model assumptions Two features of the model are critical. The first is that voters care about councilor competence. This assumption is important because parties do not have an incentive to select competent candidates otherwise. We show in Table E.1 that indeed, candidates with high education levels and prior political experience get higher vote shares in practice. The second is that parties place competent candidates in more competitive wards—a model implication

⁶⁵The conditions underlying this Nash equilibrium are formally described in Appendix E.4.

⁶⁶The probability a potential female candidate's signal exceeds $\bar{s}_f$ is $\Pr(s_i > \bar{s}_f) = 1 - \Phi(\bar{s}_f/(\sigma^2 + \sigma_s^2))$.

that follows from parties caring about winning seats as opposed to, say, ensuring that competent candidates get elected. This assumption is important as it allows us to reject taste-based discrimination as a driver of party responses to the quota. Figure E.1 confirms that candidate education level exhibits a clear inverted V-shaped relationship with party popularity in the ward, peaking in the most competitive wards, for both main parties. For more details, see Appendix E.5.

Implications—Taste-based vs. statistical discrimination Both sources of discrimination decrease the share of female candidates and raise the minimum signal of women compared to men's. Taste-based discrimination does this by raising the lowest perceived ability of women a party would accept above that of men's, i.e. $\tilde{E}(a_i|\bar{s}_f, f, t) > \tilde{E}(a_i|\bar{s}_m, m, t)$. So even if the group mean abilities were perceived to be the same, $\bar{s}_f > \bar{s}_m$. Statistical discrimination does this by raising the female signal required to equate the perceived ability of the marginal man and woman; $\bar{s}_f > \bar{s}_m$ when $\tilde{E}(a_i|\bar{s}_f, f, t) = \tilde{E}(a_i|\bar{s}_m, m, t)$.

However, they do have different implications for how a party *allocates* candidates. If taste-based discrimination weakens, the party does not alter its perception of women's competence but merely gains recognition that women are “not so bad to work with,” i.e., f_p^* increases. Then, a party nominates more women overall, willing to accept women with lower perceived ability than before, i.e., $\bar{s}_f$ decreases. However, the new women will be less competent than the existing women and, therefore, get placed in less competitive wards where candidate quality has little impact on electoral outcomes.

On the other hand, if the quota leads statistical discrimination to weaken as parties gain new information about women's abilities, parties realize that female candidates are more competent than previously thought, i.e., $\tilde{\mu}_{f,t+1} > \tilde{\mu}_{f,t} \Rightarrow \tilde{E}(a_i|s_i, f, t+1) > \tilde{E}(a_i|s_i, f, t) \forall s_i$. Hence, all women (existing and additional) are increasingly placed in more competitive wards where candidate quality is crucial for electoral success.

Comparative statistics—Speed of belief updating Two comparative statistics are relevant to our empirical exercise. When the prior belief about the group mean ability is biased down ($\tilde{\mu}_{g,t} < \mu_g$), updating toward the truth is faster

1. The larger the number of female councilors: $\frac{\partial(\tilde{\mu}_{g,t+1}-\tilde{\mu}_{g,t})}{\partial n} > 0$
2. The higher the ability of the female councilor encountered: $\frac{\partial(\tilde{\mu}_{g,t+1}-\tilde{\mu}_{g,t})}{\partial a_i} > 0$
3. The more biased the prior was: $\frac{\partial(\tilde{\mu}_{g,t+1}-\tilde{\mu}_{g,t})}{\partial \tilde{\mu}_{g,t}} < 0$
4. The less certain the prior belief was: $\frac{\partial(\tilde{\mu}_{g,t+1}-\tilde{\mu}_{g,t})}{\partial \tilde{\sigma}_{g,t}^2} > 0$

E.2 Derivation: maximum likelihood estimator of μ_g

Conditional on s_i , the distribution of a_i is:

$$a_i|s_i \sim N\left(\mu_g + \frac{\sigma^2}{\sigma_s^2 + \sigma^2}s_i, \frac{\sigma^2\sigma_s^2}{\sigma_s^2 + \sigma^2}\right) \quad (5)$$

Let's define $c = \frac{\sigma^2}{\sigma_s^2 + \sigma^2}$, $m(s_i) = \mu_g + \frac{\sigma^2}{\sigma_s^2 + \sigma^2}s_i = \mu_g + cs_i$, $\bar{\sigma}^2 = \frac{\sigma^2\sigma_s^2}{\sigma_s^2 + \sigma^2}$.

Then the likelihood function is

$$\begin{aligned} \mathcal{L}(\mu_g) &= P(\mathbf{a}|\mathbf{s}; \mu_g) \\ &= \prod_{i=1}^n \phi(a_i|s_i; \mu_g) \quad \text{where } \phi : \text{Gaussian probability density function} \\ &= \frac{1}{(\sqrt{2\pi}\bar{\sigma})^n} \exp\left(-\frac{1}{2\bar{\sigma}^2} \sum_{i=1}^n (a_i - m(s_i))^2\right) \end{aligned}$$

Thus, the maximum likelihood estimator is

$$\hat{\mu}_g = \frac{1}{n} \sum_{i=1}^n (a_i - cs_i) \sim N\left(\mu_g, \frac{1}{n}\bar{\sigma}^2\right) \quad (6)$$

E.3 Extension: if the exact ability of councilors is not revealed

What if the exact ability of councilors is not revealed while they serve their term? Rather, for councilor i , a party receives a second signal of ability that is highly informative about the *absolute* ability of i :

$$v_i \sim N(a_i, \sigma_v^2)$$

where σ_v^2 is a small number. Moreover, say the precision of v_i is inversely related to the closeness of the interaction between councilor i and a party. For instance,

$$\sigma_v^2 = \begin{cases} \sigma_1^2 & \text{if } i \text{ belongs to own party} \\ \sigma_2^2 & \text{else} \end{cases}$$

with $\sigma_1^2 < \sigma_2^2$.

Once the values of the second signals of the ability of councilors, $\mathbf{v} = \{v_i\}$, are revealed, the party makes an inference about the value of μ_g via maximum likelihood as before.

Conditional on s_i , the distribution of v_i is

$$v_i|s_i \sim N\left(\mu_g + \frac{\sigma^2}{\sigma_s^2 + \sigma^2}s_i, \sigma_v^2 + \frac{\sigma^2\sigma_s^2}{\sigma_s^2 + \sigma^2}\right) \quad (7)$$

As in section E.2, let's define $c = \frac{\sigma^2}{\sigma_s^2 + \sigma^2}$, $m(s_i) = \mu_g + \frac{\sigma^2}{\sigma_s^2 + \sigma^2}s_i = \mu_g + cs_i$. Additionally, define $\bar{\sigma}_1^2 = \sigma_1^2 + \frac{\sigma^2\sigma_s^2}{\sigma_s^2 + \sigma^2}$ and $\bar{\sigma}_2^2 = \sigma_2^2 + \frac{\sigma^2\sigma_s^2}{\sigma_s^2 + \sigma^2}$. N_1 denotes the set of own-party councilors of size n_1 , and N_2 the set of other councilors of size n_2 .

The likelihood function is

$$\begin{aligned} \mathcal{L}(\mu_g) &= P(\mathbf{v}|\mathbf{s}; \mu_g) \\ &= \Pi_{i \in N_1} f(v_i|s_i; \mu_g, \sigma_1^2) \times \Pi_{i \in N_2} f(v_i|s_i; \mu_g, \sigma_2^2) \\ &= \frac{1}{(\sqrt{2\pi}\bar{\sigma}_1)^{n_1}} \exp\left(-\frac{1}{2\bar{\sigma}_1^2} \sum_{i \in N_1} (v_i - m(s_i))^2\right) \times \frac{1}{(\sqrt{2\pi}\bar{\sigma}_2)^{n_2}} \exp\left(-\frac{1}{2\bar{\sigma}_2^2} \sum_{i \in N_2} (v_i - m(s_i))^2\right) \end{aligned}$$

Then the maximum likelihood estimator is

$$\hat{\mu}_g = \frac{\bar{\sigma}_2^2 \sum_{i \in N_1} (v_i - cs_i) + \bar{\sigma}_1^2 \sum_{i \in N_2} (v_i - cs_i)}{\bar{\sigma}_2^2 n_1 + \bar{\sigma}_1^2 n_2} \quad (8)$$

The distribution of $\hat{\mu}_g$ is

$$\hat{\mu}_g \sim N\left(\mu_g, \frac{\bar{\sigma}_1^2 \bar{\sigma}_2^2}{\bar{\sigma}_2^2 n_1 + \bar{\sigma}_1^2 n_2}\right) \quad (9)$$

The posterior distribution about the value of μ_g follows the same structure as before.

E.4 Conditions for Nash equilibrium allocation of candidates

Because each party does not know about the potential candidates available to the other party, they can only form expectations about the competence of the opponent in each ward. Call the expected competence of the opposing party's best candidate $A^{(1)}$, that of the second best candidate $A^{(2)}$, etc.⁶⁷ Also rank wards by competitiveness. Call $|R|^{(1)}$ the absolute value of party R 's popularity in the most competitive ward, i.e. the smallest absolute value of $-|I^w - I_R| + |I^w - I_L|$. For simplicity of exposition, we assume that there are sufficiently many potential candidates available to each party such that the expected gap in competence between any candidate and the next-best one

⁶⁷For example, say $\tilde{\mu}_{f,t} < \tilde{\mu}_{m,t}$ (men are perceived to have a higher mean ability than women) so that the top candidate is expected to be male, drawn from $N(\tilde{\mu}_{m,t}, \sigma^2)$. Then $A^{(1)} = \tilde{\mu}_{m,t} + \sigma \int_{-\infty}^{\infty} x \frac{d}{dx} \Phi(x)^n dx$, where Φ is the cdf of the standard normal distribution, and n is the total number of male potential candidates.

is small relative to the gap in competitiveness: $\max_k \{A^{(k)} - A^{(k+1)}\} < \min_k \{|R|^{(k+1)} - |R|^{(k)}\}$. Under this assumption, the Nash equilibrium allocation of candidates is for each party to place the most competent candidate in the most competitive ward.⁶⁸

E.5 Empirical support for key model assumptions

Voters value councilor competence We study the election probabilities for the candidates elected in cycles 1-3, before the introduction of the quota. All councilors were directly elected through plurality vote in single-member constituent wards with no party affiliation. Hence, we can study how much voters value different characteristics, free of potential confounders, such as party preference, differences in party strategic behaviors across wards, or the influence of within-party ballot list position. We restrict the analysis to male candidates as women were rare and highly selected.

Table E.1 displays the results of this analysis. The probability of being elected and the candidates' vote share are regressed on municipality $\times$ election cycle fixed effects, and candidates' characteristics—age, education, a dummy for whether the candidate works in politics, and a dummy for whether the candidate was ever elected before. The table demonstrates that higher vote shares go to incumbents, older candidates (likely reflecting their social networks and community influence), and candidates with political experience. More relevantly for our modeling assumption, voters value the education of candidates. In fact, the coefficients on the education degrees show that election outcomes improve monotonically with higher levels of education.

⁶⁸The assumption of $\max_k \{A^{(k)} - A^{(k+1)}\} < \min_k \{|R|^{(k+1)} - |R|^{(k)}\}$ ensures that the expected competence of the opponent does not overturn the competitiveness ranking of wards in equilibrium.

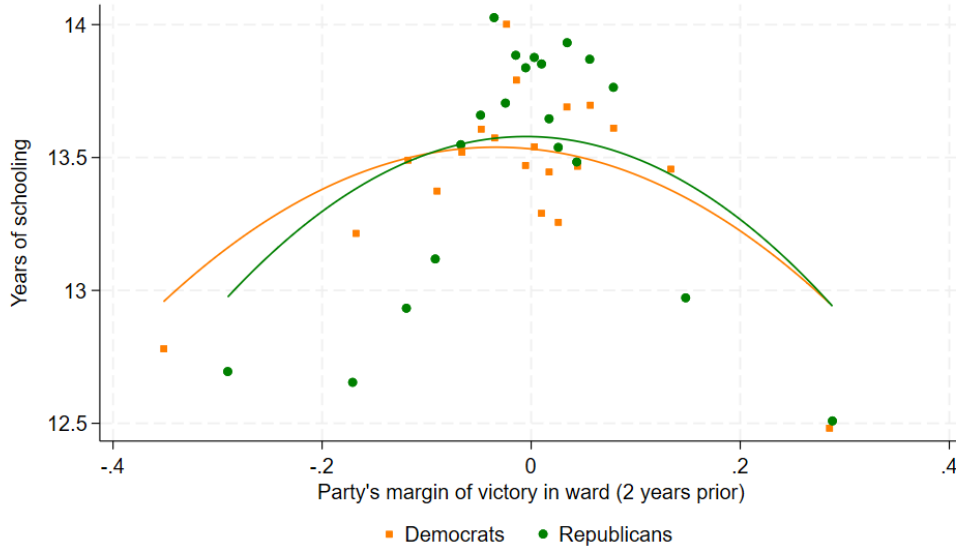
Table E.1 - What Do Voters Care About?

	Pr(Elected) (1)	Vote Share (2)
Middle School	0.01 (0.01)	0.52 (0.40)
High School	0.02* (0.01)	0.88** (0.34)
Undergraduate Degree	0.05*** (0.01)	1.53*** (0.36)
Grduate Degree	0.10*** (0.02)	2.83*** (0.46)
Incumbent	0.16*** (0.01)	6.06*** (0.29)
Age	0.04*** (0.00)	1.65*** (0.11)
Age squared	-0.00*** (0.00)	-0.02*** (0.00)
Political experience	0.04*** (0.01)	1.59*** (0.33)
<i>N</i>	25201	23919
Municipality×cycle fixed effects	Yes	Yes

Note: The probability of being elected (column 1) and the candidate's vote share (column 2) are regressed on municipality × election cycle fixed effects, and candidates' characteristics—age, education (a dummy for different levels of education achieved), a dummy for whether the candidate has political experience, and a dummy for whether the candidate was ever elected before. The sample consists of the universe of male candidates for the three municipal elections before the introduction of the quota in 1995, 1998, and 2002 (26,606 male candidates), for which we could retrieve occupation, education (95%), and vote share information (90%). Standard errors (in parentheses) are clustered by municipality. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Competent candidates are placed in competitive wards To eliminate confounding factors related to gender and incumbency status, we focus on the sample of male rookie candidates. Using education as a proxy for competence, Figure E.1 confirms that candidate education level exhibits an inverted V-shaped relationship with party popularity. It peaks in the most competitive wards with a near-zero margin of victory. Other notable features are the symmetry around zero and the marked similarity between the two parties. These solidly support the idea that parties allocate candidates according to model predictions: placing competent candidates in competitive wards and less competent ones in strongholds or weakholds.

Figure E.1 - Education of Male Rookie Candidates by Ward-Level Party Popularity

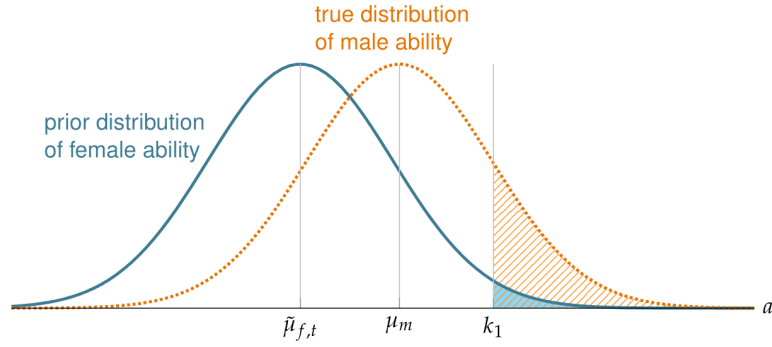


Note: This figure gives a binned scatter plot of the years of schooling of male rookie ward candidates by the party's popularity in the ward. We residualize the years of schooling variable for age, to account for the tendency of increasing educational attainment over time. The party's margin of victory is calculated as the difference in the party's vote share with that of the most popular competitor in the ward in the immediately preceding election (the National Assembly election, 2 years before each municipal council election). We demean the party's margin of victory by election $\times$ party, to address electoral swings occurring at the national level. The curves are quadratic fits. The sample includes bins 1 and 2.

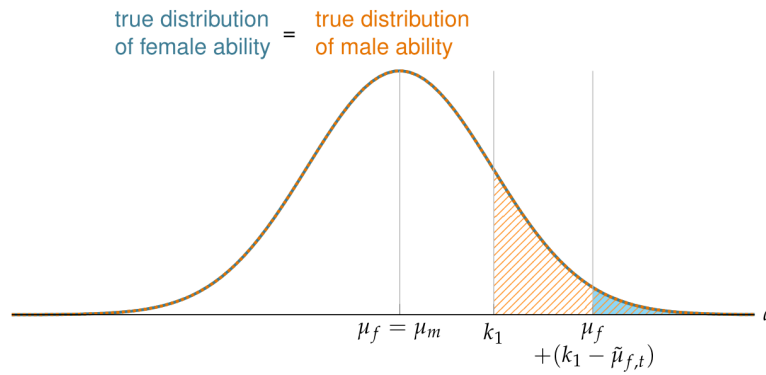
E.6 A diagrammatic illustration of candidates' selection

Figure E.2 illustrates the model for the $b = 0$ case. To simplify matters, suppose the true distribution of male ability is known. If the prior distribution of female ability at election t lies to the left of the male ability distribution, as in diagram (a), then the party will select candidates with perceived ability above a certain cutoff k_1 , determined by the total number of candidates. The lowest signal among females is $(k_1 - \tilde{\mu}_{f,t})$, much higher than the lowest among males, $(k_1 - \mu_m)$. The female share of candidates will be given by the relative size of the shaded areas in (a). Now, if the true ability distribution of females was identical to that of males, as plotted in diagram (b), then the true abilities of women, revealed post-election, would turn out to be much higher than expected. The party thus reconsiders the value of μ_f using maximum likelihood. The posterior distribution of female ability then is a weighted average of the prior and the MLE, and will lie closer to the true female ability distribution, as shown in diagram (c). At election $t + 1$, the new cutoff will be k_2 , and the female share of candidates will be higher than at t .

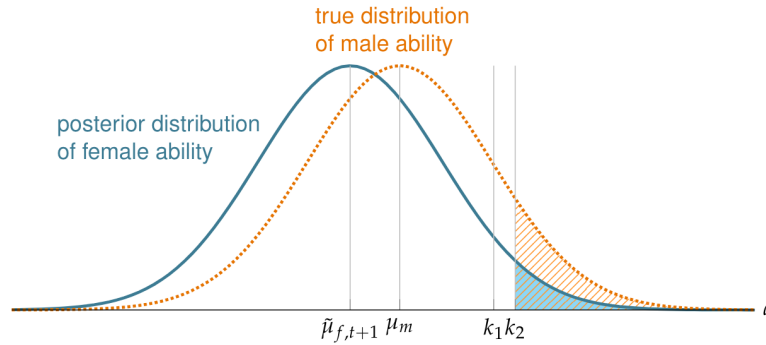
Figure E.2 - Illustration of Model of Belief Updating



(a) Candidate selection at t



(b) Maximum likelihood during the council term of t



(c) Candidate selection at $t + 1$

Note: This figure provides a simplified illustration of the model. A few assumptions are made. First, parties only care about maximizing the average ability of councilors, and not about meeting a particular female ratio. Moreover, we assume that the true distribution of male ability is known. Panel (a) describes the selection of candidates at time t if the prior distribution of female ability at election t lies to the left of the true male ability distribution. Panel (b) illustrates that the true abilities of women revealed post-election would turn out to be much higher than expected if the true ability distribution of females were identical to that of males. Panel (c) describes candidates selection at time $t + 1$, when the posterior distribution of female ability lies closer to the true female ability distribution.

Appendix F PARTY-LEVEL RD DESIGN

We conduct a *party-level* analysis, where we compare the strategies of parties that marginally won or lost the election of their first—and therefore female—PR candidate in the previous election cycle. As PR seats are allocated to parties according to the Hare-Niemeyer method (largest remainder method), the closeness to winning is not straightforward from the vote shares. We construct the vote margin variable (v_{cpt}) following Luechinger et al. (2024). It represents the distance from the party’s actual vote share in the PR arm to the one the party needs to win 1 PR seat.

For the one-period-ahead effects of winning (columns 1-2 of Table F.1), we employ a regression discontinuity design:

$$Y_{cpt} = \alpha_t + \beta \times \text{Winner}_{cp,t-1} + f(v_{cp,t-1}) + X'_{cpt}\gamma + \epsilon_{cpt}$$

where $\text{Winner}_{cp,t-1} \equiv \mathbb{1}(v_{cp,t-1} \geq 0)$. $f(v_{cp,t-1})$ is linear and allows for different slopes to the left and right of the cutoff $v_{cp,t-1} = 0$. X_{cpt} represents the control variables: party dummies, election cycle dummies, the number of ward seats, and council size. The sample includes only the two major parties in South Korea, as these are the parties that can be tracked over time.

For the cumulative effects of winning in cycle 4 on period- t outcomes (columns 3-8 of Table F.1), we estimate for each $t = 5, 6, 7$:

$$Y_{cpt} = \alpha + \beta \times \text{Winner}_{cp,4} + f(v_{cp,4}) + X'_{cpt}\gamma + \epsilon_{cpt}$$

The results are reported in Table F.1, and for key outcomes, accompanying RD plots are in Figure F.1. Columns 1 and 2 show one-period-ahead effects, pooling all cycles. The probability that the number-1 PR candidate in cycle $t - 1$ is renominated as a ward candidate in cycle t is higher by 46 percentage points if she got elected in $t - 1$ (column 1). Hence, parties clearly factor in incumbency advantage during ward candidate selection. The renomination of the elected PR woman raises the overall female share of ward candidates of the winning party in cycle t , which is 14 percentage points higher than that of the losing party (column 2).

How far does the incumbency advantage go in explaining the increase in female ward candidates in the long term? Columns 3-8 report the dynamic cumulative effects of marginally winning the first PR candidate in cycle 4. Columns 3-5 show that the incumbency effect does not persist. The probability of her renomination decreases and becomes statistically insignificant in cycle 6, and drops to zero in cycle 7.⁶⁹ Critically, however, columns 6-8 show that winning the first PR

⁶⁹This aligns with the limited re-candidacy pattern in the overall sample. 85% of individuals run in a maximum of three elections, three-quarters of whom run only once or twice. There is no gender difference. After all, the average age of candidates is around 50 for both men and women, even for rookies, and 2 terms is already 8 years.

candidate in cycle 4 gradually increases the number of female *rookie* candidates, from zero more in cycle 5, to 0.47 more in cycle 6 and 1.33 more in cycle 7.⁷⁰

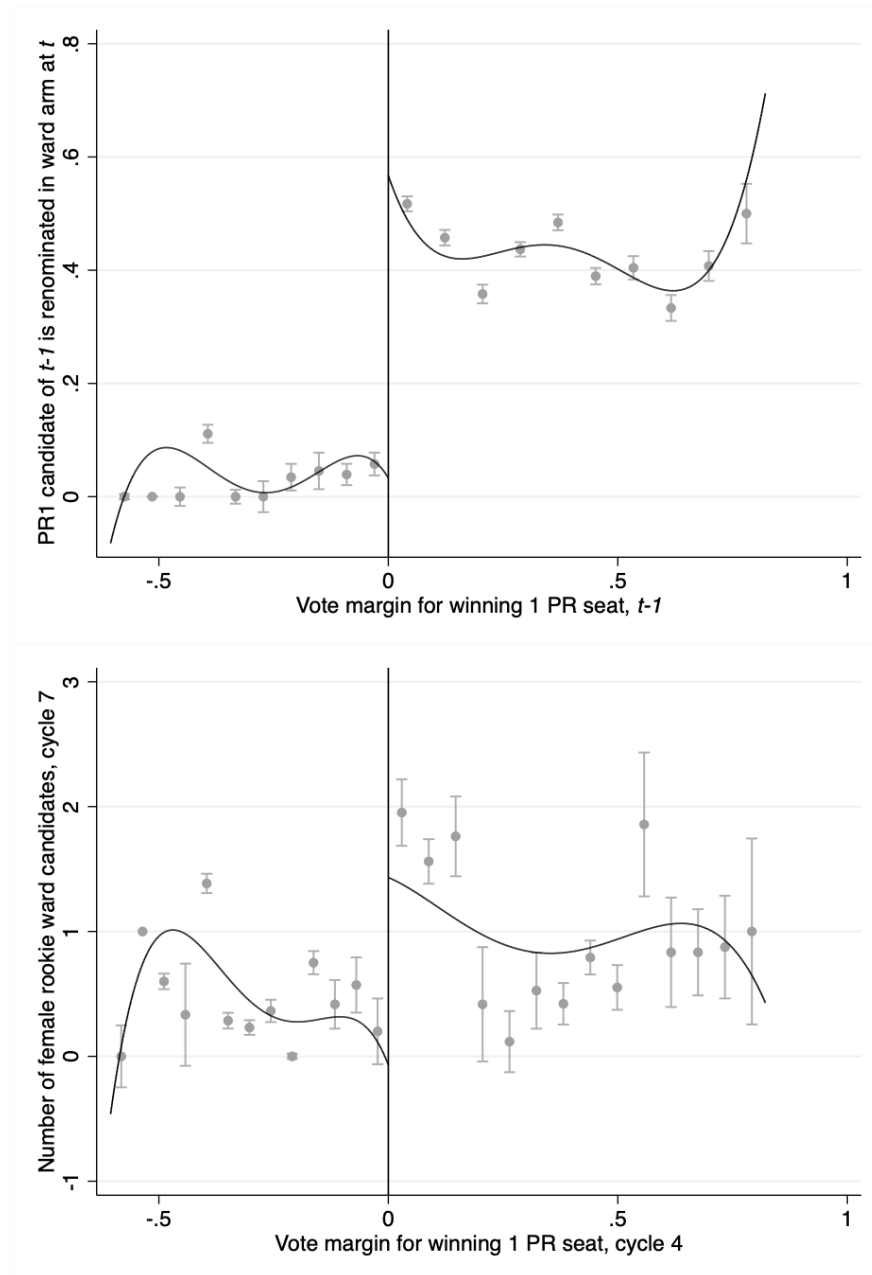
Table F.1 - The Effect of a Party's First PR Candidate Winning in Close Elections on the Party's Future Candidate Composition

	Effect of PR1 candidate at t-1 winning on...		Effect of PR1 candidate in cycle 4 winning on...					
	1(PR1 candidate at t-1 is renominated in ward arm, t)	Female share of ward candidates, t	1(PR1 candidate in cycle 4 is renominated in ward arm, t)			Number of rookie female ward candidates, t		
	t=5,6,7 (1)	t=5,6,7 (2)	t=5 (3)	t=6 (4)	t=7 (5)	t=5 (6)	t=6 (7)	t=7 (8)
Winner _{t-1}	0.46*** (0.09)	0.14*** (0.05)						
Winner _{t=4}			0.34** (0.14)	0.19 (0.17)	-0.01 (0.05)	-0.01 (0.09)	0.47** (0.23)	1.33*** (0.43)
Running var.	Vote margin for winning 1 PR seat, t-1		Vote margin for winning 1 PR seat, cycle 4					
Bandwidth	0.179	0.116	0.182	0.199	0.139	0.122	0.216	0.141
N	311	220	121	133	114	98	141	114

Note: This table reports the effect of the first PR (“PR1”) candidate—a woman—of a party marginally winning in close PR elections on the party’s future candidate composition. The estimation details are in Appendix F. Columns 1-2 report the one-period-ahead effects, pooling all cycles. Columns 3-8 report the cumulative effects of winning in cycle 4 for each future cycle separately. Mean-squared-error-optimal bandwidth proposed by Calonico et al. (2014) is used. The regression discontinuity plots of the binned sample means of the outcome variable for columns 1 and 8 are in Figure F.1. The standard errors (in parentheses) are clustered by municipality×party. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

⁷⁰The patterns are similar when we estimate the effect on the female share of rookie candidates of the party.

Figure F.1 - The Effect of a Party's First PR Candidate Winning in Close Elections on the Party's Future Candidate Composition



Note: Regression discontinuity plots of binned sample means with 95% confidence intervals, corresponding to results in Columns 1 and 8 of Table F.1. The bins are selected to mimic the underlying variability of the data (Calonico et al., 2014), and the curves show the 4th-order global polynomial fit on each side of the cutoff.

Appendix G Decomposition of the treatment effect on female ward candidates

This appendix provides the details on the decomposition exercise discussed in Section V.D.

Let $d \in \{0, 1\}$ denote treatment, following the potential outcomes framework, and let t index election cycles. $F_t(d)$, $F_t^I(d)$ and $F_t^R(d)$ denote the RD-estimated potential outcomes for the total number of female ward candidates, the number of female incumbent and female rookie ward candidates, respectively.⁷¹ Let $\Delta F_t \equiv F_t(1) - F_t(0)$, and define ΔF_t^I and ΔF_t^R analogously. The total treatment effect on female ward candidates is

$$\Delta F_t = \Delta F_t^I + \Delta F_t^R.$$

For the incumbent component, let $C_{t-1}(d)$ denote the number of female councilors elected in the previous election cycle under treatment status d , and let $\rho_t(d)$ denote the probability that a female councilor elected in cycle $t - 1$ appears as an incumbent ward candidate in cycle t .⁷² Then

$$F_t^I(d) = C_{t-1}(d)\rho_t(d).$$

Using a Oaxaca-Blinder-style decomposition, the incumbent component can be written as

$$\Delta F_t^I = \underbrace{[C_{t-1}(1) - C_{t-1}(0)] \bar{\rho}_t}_{\text{female incumbent stock effect}} + \underbrace{[\rho_t(1) - \rho_t(0)] \bar{C}_{t-1}}_{\text{renomination effect}},$$

where

$$\bar{\rho}_t = \frac{\rho_t(1) + \rho_t(0)}{2}, \quad \bar{C}_{t-1} = \frac{C_{t-1}(1) + C_{t-1}(0)}{2}.$$

The first term captures the increase in female incumbent candidates arising from a larger stock of previously elected female councilors, holding fixed the average rate at which they become incumbent ward candidates. The second term captures changes in that renomination rate, holding fixed the female elected stock.

For rookies, we write

$$F_t^R(d) = N_t^R(d)s_t^R(d),$$

⁷¹As an example, Figure G.1 visually illustrates how $F_7^I(0)$ and $F_7^I(1)$ are estimated at the cutoff of the regression discontinuity. $\Delta F_7^I = F_7^I(1) - F_7^I(0)$ corresponds to the treatment effect for cycle 7 in column 5 of Table V.

⁷²Strictly speaking, incumbents in Table V include all previously elected individuals, not only those elected in the previous cycle. In practice, this distinction is quantitatively small: among female incumbent ward candidates, 95% were elected in the previous cycle. We therefore define the relevant stock of incumbents as women elected in the previous cycle.

where $N_t^R(d)$ is the total number of rookie ward candidates and $s_t^R(d)$ is the female share among rookie ward candidates. The rookie component is therefore

$$\Delta F_t^R = \underbrace{[N_t^R(1) - N_t^R(0)] \bar{s}_t^R}_{\text{rookie number effect}} + \underbrace{[s_t^R(1) - s_t^R(0)] \bar{N}_t^R}_{\text{rookie female share effect}},$$

where

$$\bar{s}_t^R = \frac{s_t^R(1) + s_t^R(0)}{2}, \quad \bar{N}_t^R = \frac{N_t^R(1) + N_t^R(0)}{2}.$$

The rookie number effect captures the contribution of changes in the total number of rookie candidate slots, holding fixed the average female share among rookies. The rookie female share effect captures whether women constitute a larger share of rookie candidates.

Combining the incumbent and rookie decompositions gives

$$\Delta F_t = \underbrace{[C_{t-1}(1) - C_{t-1}(0)] \bar{\rho}_t}_{\text{female incumbent stock effect}} + \underbrace{[\rho_t(1) - \rho_t(0)] \bar{C}_{t-1}}_{\text{renomination effect}} + \underbrace{[N_t^R(1) - N_t^R(0)] \bar{s}_t^R}_{\text{rookie number effect}} + \underbrace{[s_t^R(1) - s_t^R(0)] \bar{N}_t^R}_{\text{rookie female share effect}}.$$

$= \Delta F_t^I$ (total incumbent contribution) $= \Delta F_t^R$ (total rookie contribution)

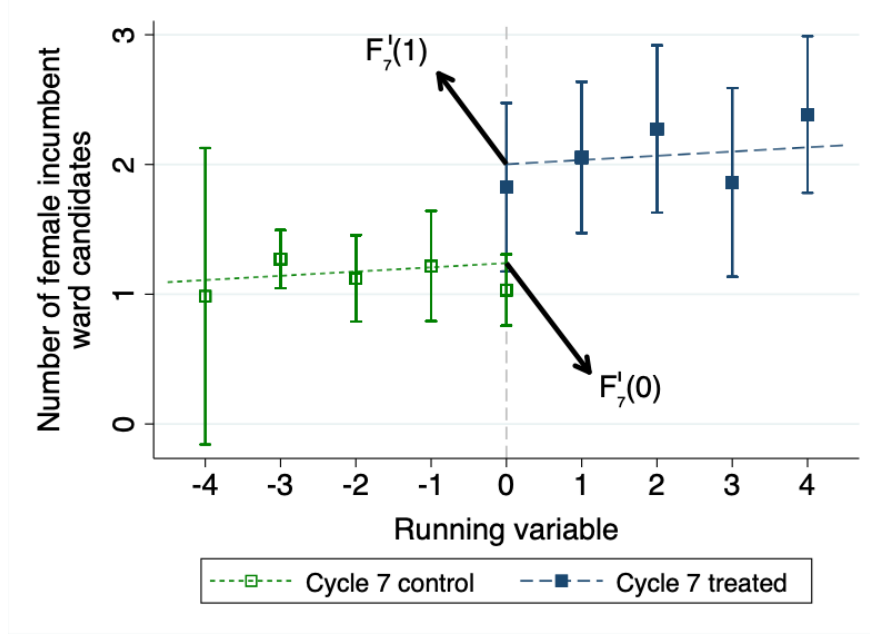
To calculate each part of the decomposition, we use the values in Table G.2. The decomposition results are in Table G.1.

Table G.1 - Decomposition of the Treatment Effect on Female Ward Candidates

Cycle	Incumbents				Rookies			Rookie Fem Share	Incumb Stock
	ΔF	ΔF^I	Stock	Renom	ΔF^R	Number	Fem Share	ΔF	ΔF
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
5	0.486	0.467	0.133	0.333	0.020	-0.063	0.083	17.1%	27.4%
6	0.781	0.618	0.436	0.181	0.163	-0.026	0.189	24.2%	55.9%
7	1.174	0.791	0.578	0.213	0.384	-0.019	0.403	34.3%	49.2%

Note: This table decomposes the RD-estimated treatment effect on the number of female ward candidates, ΔF . ΔF^I and ΔF^R are the incumbent and rookie contributions, such that $\Delta F = \Delta F^I + \Delta F^R$. The values of ΔF , ΔF^I , and ΔF^R correspond to the treatment-effect coefficients for cycles 5-7 in columns 2, 5, and 8 of Table V, respectively. The incumbent contribution is split into a female incumbent stock effect (col. 3), due to the larger stock of incumbent women elected in the previous cycle, and a renomination effect (col. 4), due to changes in the rate at which those women get renominated as ward candidates. The rookie contribution is split into a rookie number effect (col. 6), due to a change in the total number of rookies, and a rookie female share effect (col. 7), due to a change in the female share among rookies. The final two columns report the rookie female share effect (col. 7) and the female incumbent stock effect (col. 3) as percentages of ΔF (col. 1).

Figure G.1 - Potential Outcomes for the Number of Female Incumbent Candidates



Note: This figure plots the average number of female incumbent ward candidates in cycle 7, residualized and re-centered for bin effects (α_b), for each value of the running variable ($\tilde{x}_{cb}$). The error bars indicate 95% confidence intervals of each mean value. The lines (short-dashed and dashed) represent $f(\tilde{x}_{cb})$.

Table G.2 - Ingredients of the Decomposition Exercise

Cycle t	Directly estimated quantities								Imputed quantities					
	$F_t^I(0)$	$F_t^I(1)$	$F_t^R(0)$	$F_t^R(1)$	$M_t^R(0)$	$M_t^R(1)$	$C_{t-1}(0)$	$C_{t-1}(1)$	$\rho_t(0)$	$\rho_t(1)$	$N_t^R(0)$	$N_t^R(1)$	$s_t^R(0)$	$s_t^R(1)$
5	0.592	1.058	0.525	0.545	6.492	5.692	1.610	1.898	0.367	0.557	7.018	6.237	0.075	0.087
6	0.992	1.610	0.662	0.825	5.723	5.340	2.151	3.030	0.461	0.531	6.385	6.165	0.104	0.134
7	1.274	2.064	0.953	1.336	6.349	5.846	2.518	3.586	0.506	0.576	7.302	7.182	0.130	0.186

Note: This table reports the RD-estimated potential-outcome components used in the decomposition. All directly estimated quantities are evaluated at the RD cutoff using equation (1), the same specification as in the main analysis. $F_t^I(d)$ denotes the number of female incumbent ward candidates under treatment status d , where $F_t^I(1) - F_t^I(0)$ equals the treatment effect reported in column 5 of Table V. As an example, see Figure G.1 for a visual illustration of how $F_7^I(0)$ and $F_7^I(1)$ are estimated. $F_t^R(d)$ and $M_t^R(d)$ denote the numbers of female and male rookie ward candidates, corresponding to columns 8 and 7 of Table V. $C_{t-1}(d)$ denotes the number of female councilors elected in the previous cycle, corresponding to column 6 of Table II. The renomination rate is defined as $\rho_t(d) = F_t^I(d)/C_{t-1}(d)$. Finally, $N_t^R(d) = F_t^R(d) + M_t^R(d)$ is the total number of rookie ward candidates, and $s_t^R(d) = F_t^R(d)/N_t^R(d)$ is the female share among rookies.

Appendix H TESTS OF ALTERNATIVE MECHANISMS

H.1 Women becoming more powerful

Estimating the role of women's influence, whether alone or in a group, in isolation is challenging. Indeed, an increase in the number of female councilors that can influence decisions for candidate selection implies an increase in the women the party was able to experience in action during the previous election cycle, potentially learning about their competence. We report in the following the results of an exercise that we believe can help unravel the two mechanisms.

We test whether a similar increase in rookie female candidates arises when parties are exposed to a growing number of elected female councilors who are *not* affiliated with the party. When focusing on the number of elected women within the party, we cannot disentangle the two effects of influence and learning. However, while parties can learn from experiencing women from other parties, these women should not have influence over the parties' decisions.

Table B.16 estimates a party-level version of equation (1). The outcome is the number of female rookie ward candidates nominated by the party. Column 1 replicates column 9 of Table V at the party level, showing a similar initial reduction in female rookies and a positive treatment effect by cycle 7. In columns 2-4, we restrict the treated group to parties with at least one, two, or three own-party female councilors in the previous election cycle. In columns 5-7, we instead restrict the treated group to parties with at least one, two, or three female councilors elected from other parties in the previous election cycle. In all columns, the comparison group consists of all parties in control municipalities. The analysis cannot be performed for cycle 4 because candidates did not run with party affiliation before that cycle. For the own-party vs. other-party comparison to be meaningful, the number of women elected from the party's own ranks and the number of women elected from other parties must capture distinct variation; we check that the two measures are not positively correlated in the data.⁷³

We view the analysis as a suggestive diagnostic rather than a definitive test, because the threshold-based exercise (i.e., at least 1, 2, 3 women) has clear limitations. First, the number of female councilors in the previous election cycle is endogenous, so the treated sample becomes increasingly selected as the threshold rises. Second, we do not impose the same threshold restrictions on parties in control municipalities. Doing so would compare selected treated parties with even more positively selected control parties, making the counterfactual difficult to interpret. Third, the composition of the treated sample changes across election cycles; for example, amid a trend of growing female representation, it is easier to meet a threshold of at least two female councilors in

⁷³Conditional on election cycle and bin fixed effects, the correlation between the number of women elected within the party and the number of women from other parties is equal to -0.15 among the treatment group, and -0.11 among the treatment and control groups combined.

cycle 6 than in cycle 4. Last, as the threshold rises, the treated sample becomes increasingly small, e.g. 15 parties in column 7, cycle 5.

Despite these limitations, the exercise is informative because own-party and other-party female councilors have different implications for women's power growing within the party. Women elected from the party's own ranks may increase women's influence within the party, whereas women elected from other parties should not directly do so. The fact that similar patterns appear for both groups suggests that the reversal is consistent with broader exposure or learning playing an important role.

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